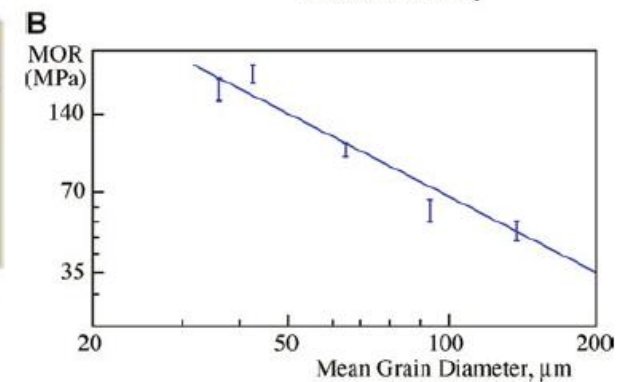
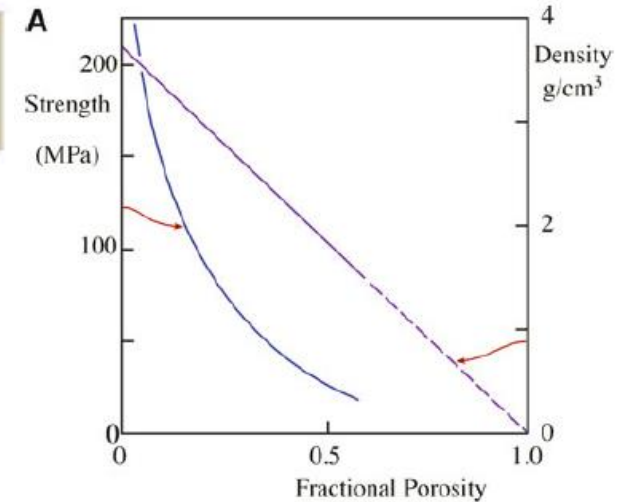
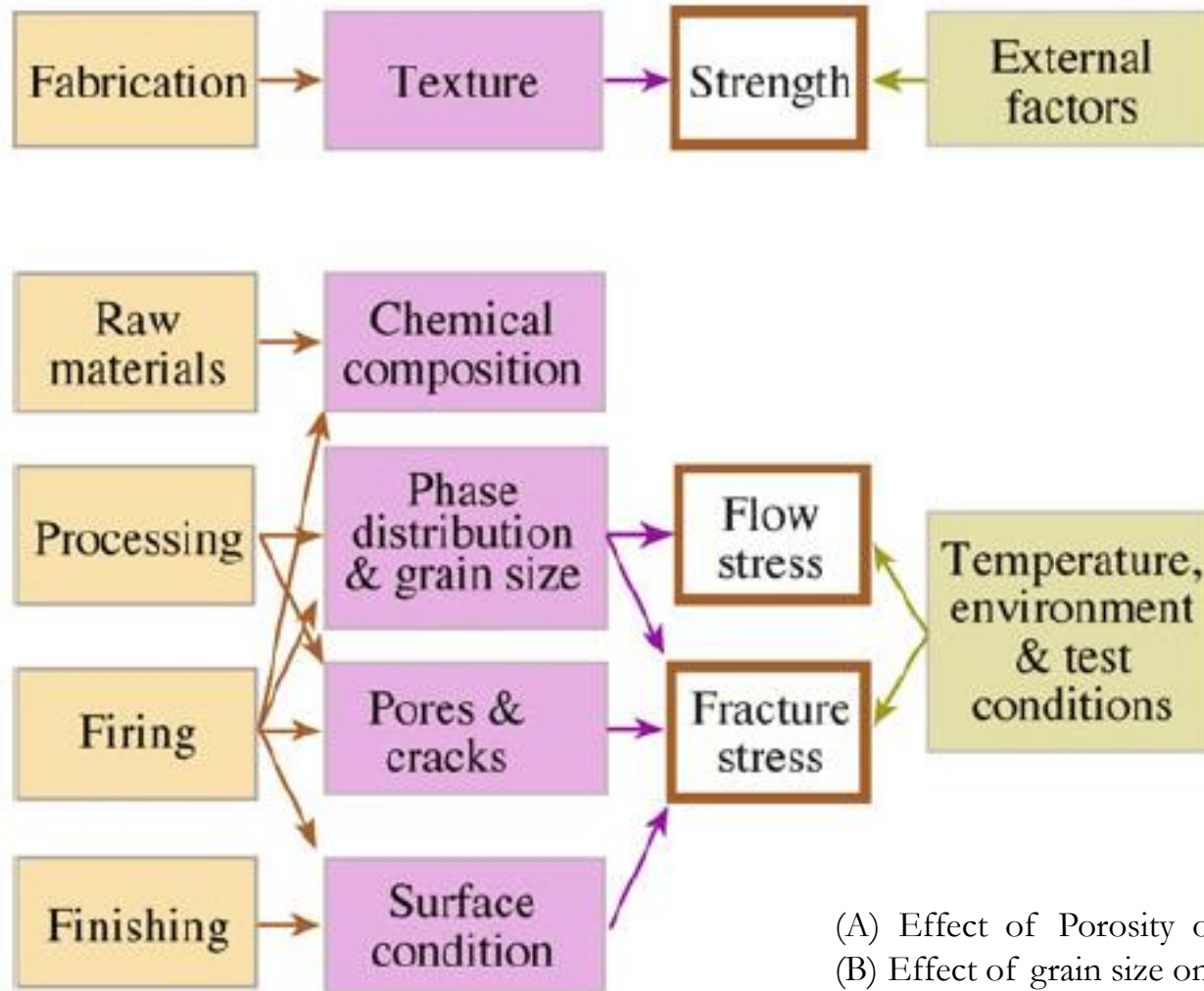


Chapter 8 Mechanical Properties of Ceramic Materials

- Density and Porosity (Physical)
- Elastic Modulus and Hardness
- Strength
- Fracture Toughness
- Fatigue Resistance
- Creep Resistance
- Abrasive and Sliding Wear Resistance

Factors Affecting the Mechanical Properties of Ceramics



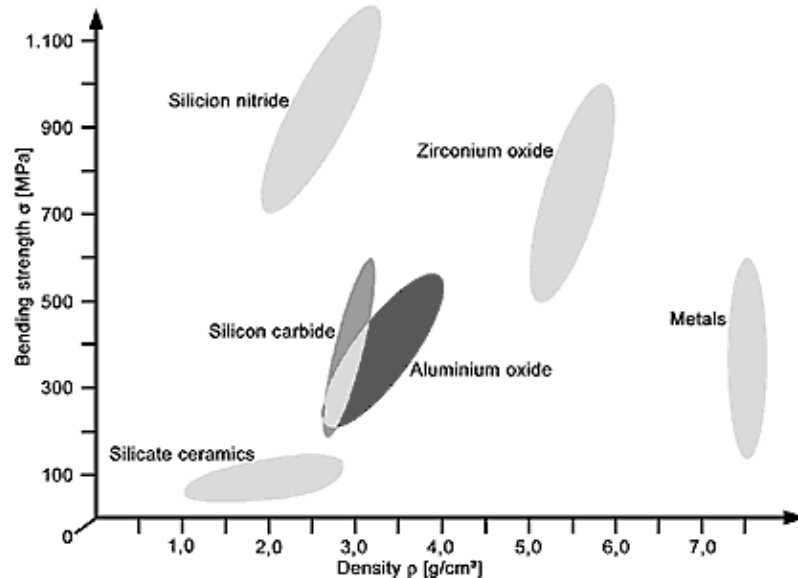
(A) Effect of Porosity on the strength of polycrystalline Al_2O_3 .
(B) Effect of grain size on the strength of BeO .

Density and Porosity

The **density** [g/cm^3] of technical ceramics lies between 20 and 70% of the density of steel. This presents the possibility of weight reductions, which is an advantage for moving parts in machine construction.

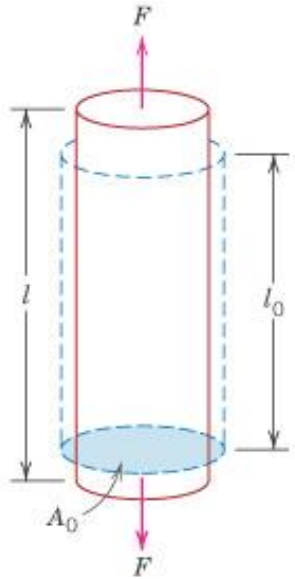
The relative density, d [%], has a significant effect on the properties of the ceramic. For this reason it can be important, for instance, for mechanical applications to achieve a density close to the theoretical figure without generating density gradients, in order to obtain the best possible mechanical properties.

On the other hand, relatively low densities resulting from uniform porosity can, for instance, have a positive effect on the ability to withstand thermal shock.

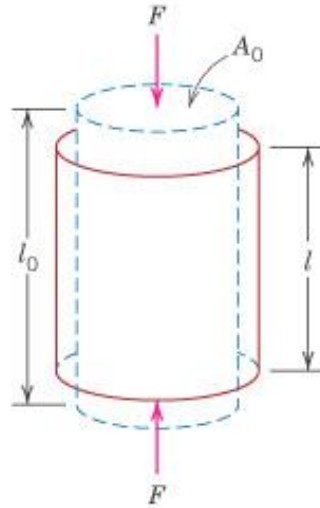


Porosity: As a rule, technical ceramic materials have no open porosity, and are thus gas tight. For some materials, however, porosity is desirable. This may be because thermal stresses are more easily managed by porous materials, or because larger surface areas are required. Porosity can be deliberately generated through the appropriate selection of raw materials, the manufacturing process, and in some cases through the use of additives. This allows closed and open pores to be created with sizes from a few nm up to a few μm .

Stress for Tension and Compression



Tensile load

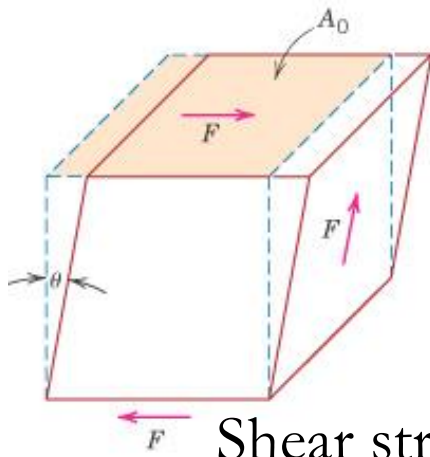


Compressive load

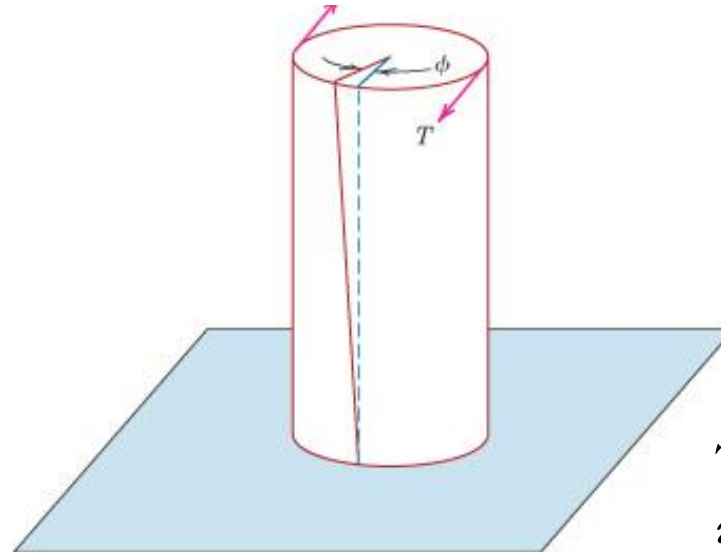
$$\sigma = \frac{F}{A_0}$$

Strain for tension and compression

$$\varepsilon = \frac{\delta}{L_0}$$



Shear strain

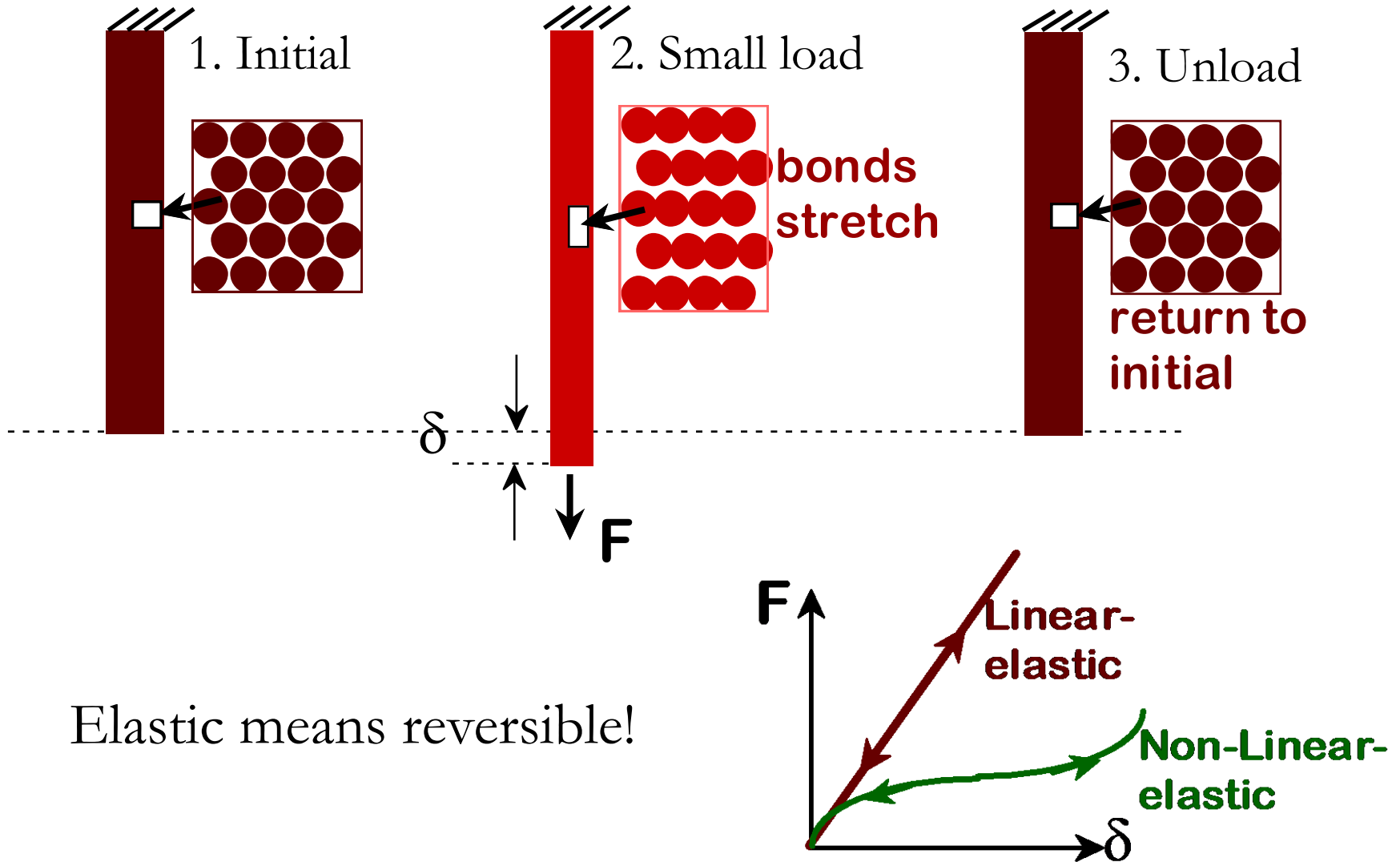


Shear stress

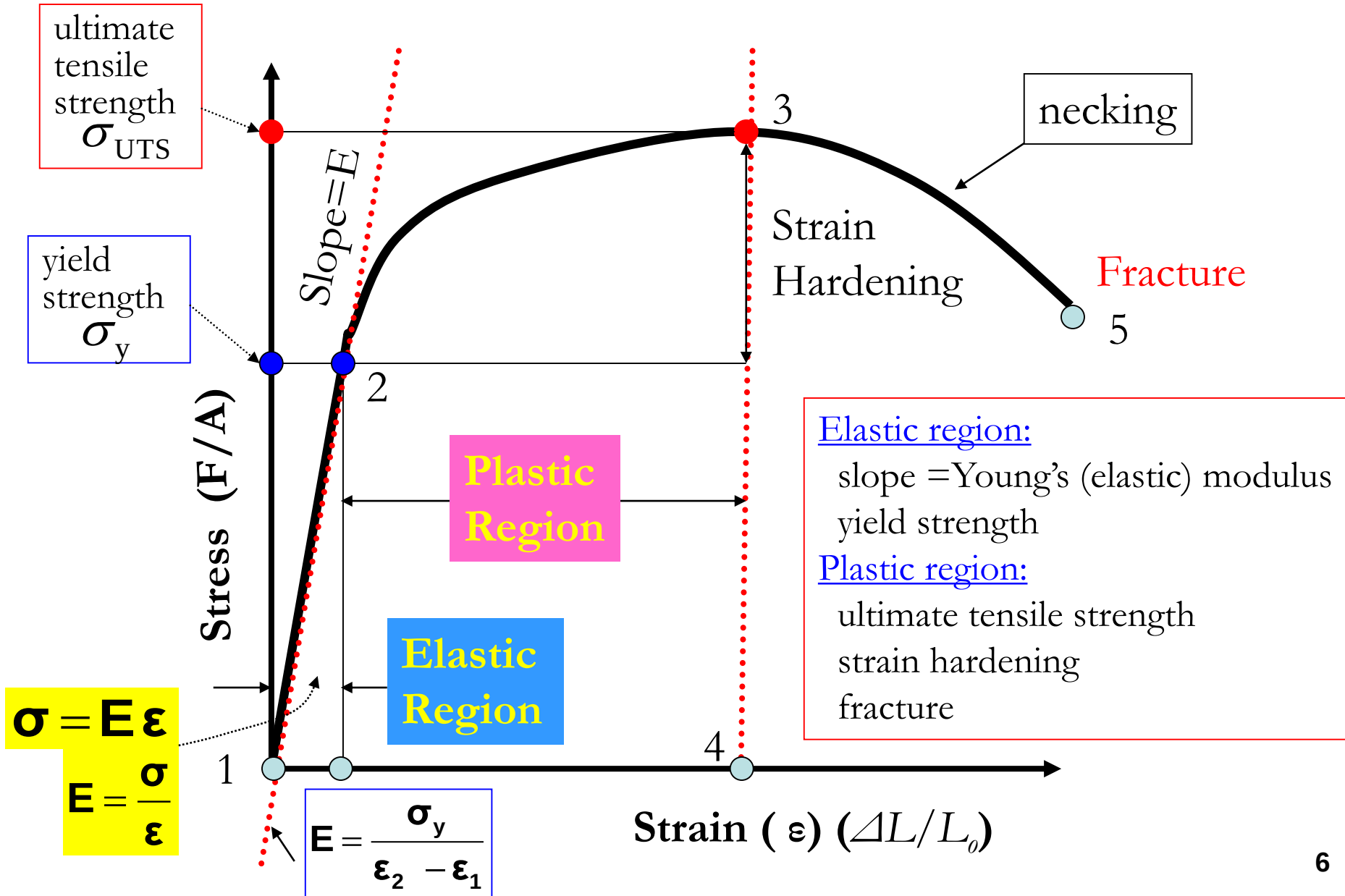
$$\tau = \frac{F_s}{A_0}$$

Torsional deformation
angle of twist, ψ

Elastic Deformation



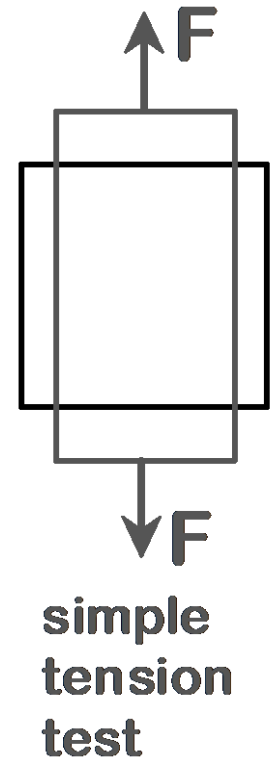
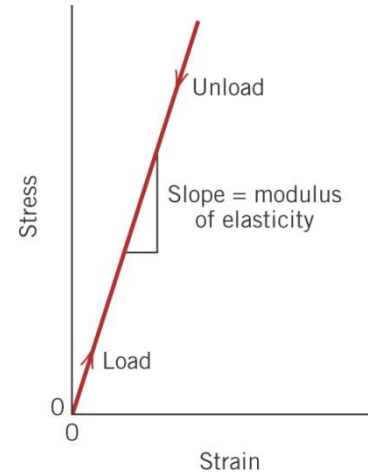
Stress-Strain Curve



Elastic Constants

- Elastic Modulus, E :
(Young's modulus)

$$\sigma = E \epsilon$$



- Poisson's ratio, ν :

Negative ratio of transverse strain to longitudinal strain

$$\nu = -\epsilon_T / \epsilon_L$$

Metals: ~ 0.33

Ceramics: 0.18-0.30

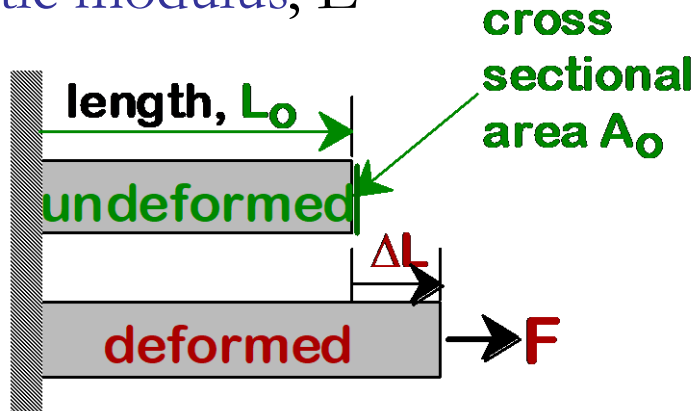
Polymers: ~ 0.40

Units:

E : GPa; ν : dimensionless

Properties for bonding: E

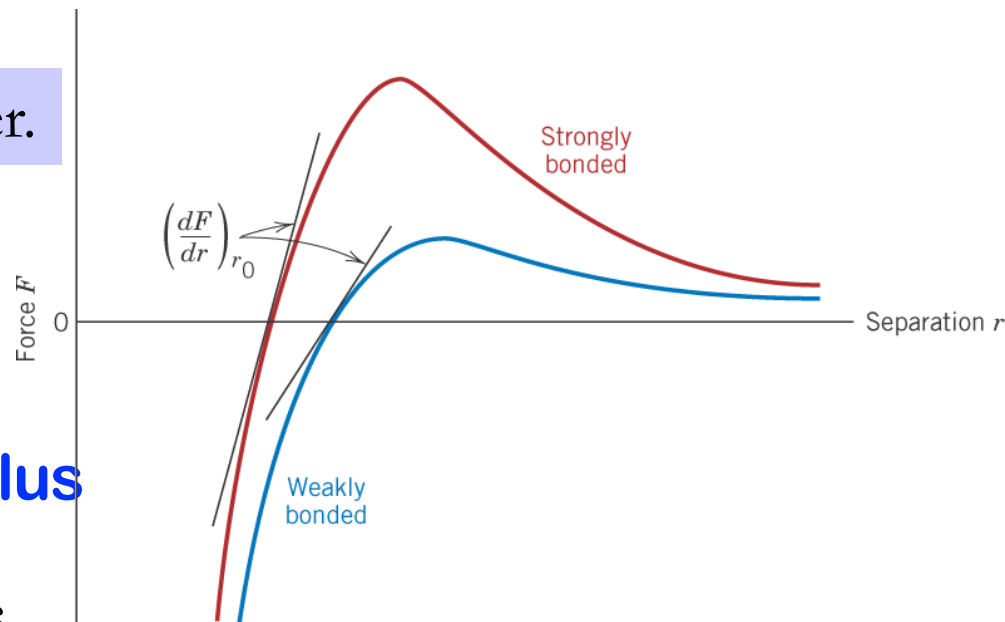
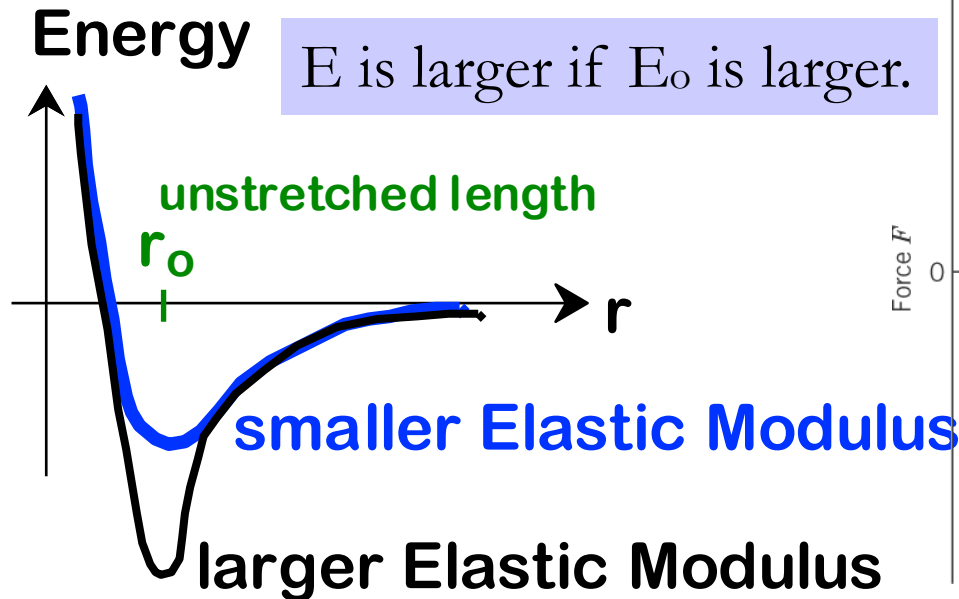
- Elastic modulus, E



Elastic modulus

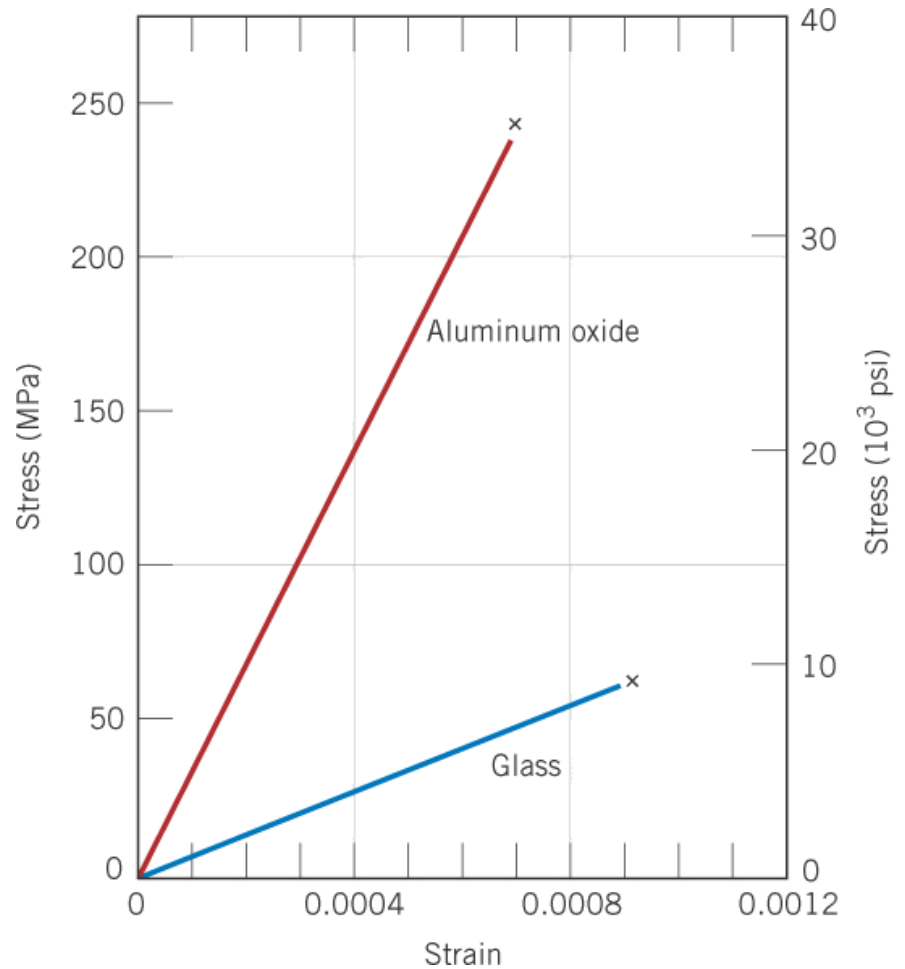
$$\frac{F}{A_0} = E \frac{\Delta L}{L_0}$$

Energy $\sim$ curvature at r_0



Elastic Behavior for Ceramics

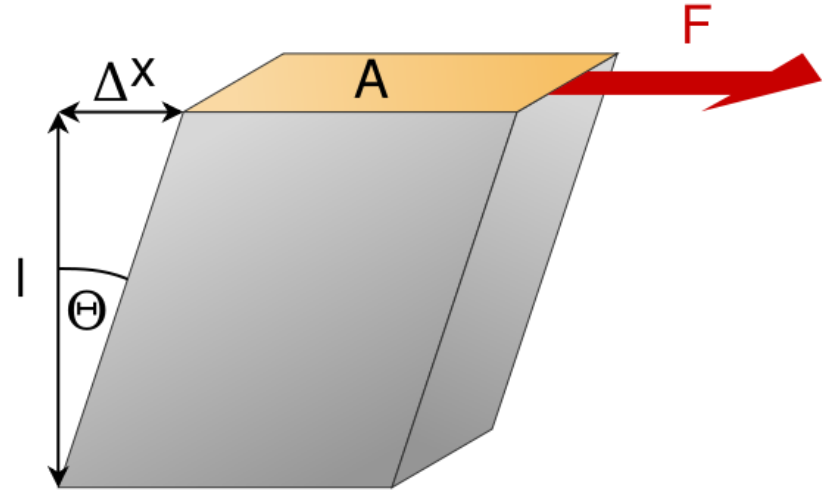
- Similar to tensile test for metals;
- Linear stress-strain;
- Moduli of elasticity for ceramics are slightly higher than for metals;
- No plastic deformation prior to fracture.



Shear Modulus and Bulk Modulus

Shear modulus, G or μ , is defined as the ratio of shear stress to the shear strain:

$$G \stackrel{\text{def}}{=} \frac{\tau_{xy}}{\gamma_{xy}} = \frac{F/A}{\Delta x/l} = \frac{Fl}{A\Delta x}$$

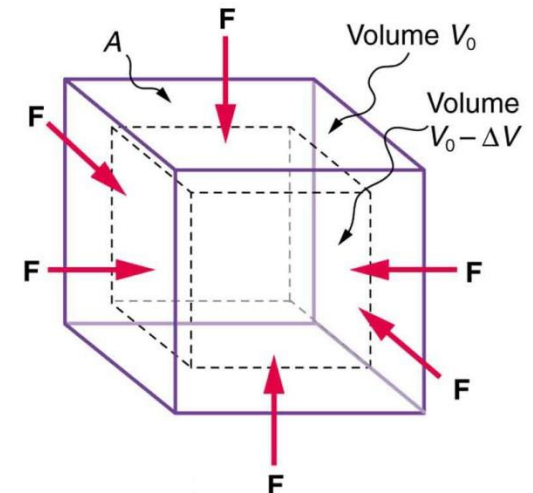


The **bulk modulus** measures the Material's resistance to uniform/hydrostatic compression.

$$K = -P/(\Delta V/V_0)$$

In isotropic materials, the bulk modulus is related to elastic modulus by

$$G = E/3(1-2\nu)$$



Effect of Microstructure on Elastic Modulus

In ceramics, usually more than one phase present, the overall modulus ?

Voigt Model

Assumption: strain in each constituent is the same, representing upper bound of Young's modulus:

$$E = f_1 E_1 + (1 - f_1) E_2, \text{ where } f_1 = V_1 / (V_1 + V_2)$$

Reuss Model

Assumption: Stress in each phase is the same, representing lower bound of Young's modulus.

$$E = (f_1 / E_1 + (1 - f_1) / E_2)^{-1}$$

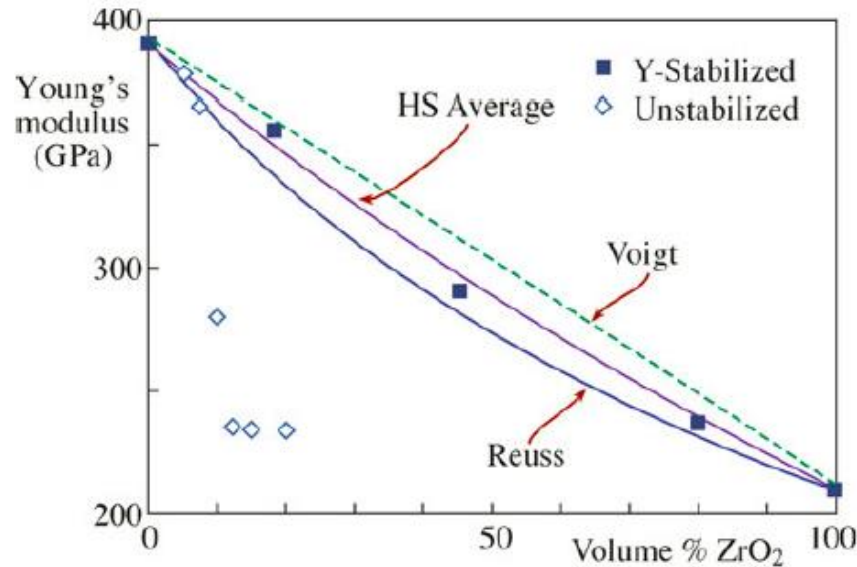
Hashin-Shtrikman bounds

The tightest bounds possible from range of composite moduli for a two-phase material

$$K_{\text{HS}}^{\pm} = K_2 + \frac{\phi}{(K_1 - K_2)^{-1} + (1 - \phi)(K_2 + \frac{4}{3}\mu_2)^{-1}}$$

$$\mu_{\text{HS}}^{\pm} = \mu_2 + \frac{\phi}{(\mu_1 - \mu_2)^{-1} + \frac{2(1 - \phi)(K_2 + 2\mu_2)}{5\mu_2(K_2 + \frac{4}{3}\mu_2)}}$$

Comparison of predicted values and experimental data



Comparison of predicted values of Young's modulus for an Al₂O₃-ZrO₂ composite with experimental data.

If Second phase is porosity ?

$$E = E_0(1 - aP)$$

Linear decrease in Young's modulus with porosity when P is small $a \sim 4$

$$E = E_0(1 - aP + bP^2)$$

For a low concentration of spherical pores ($a \sim 1.9$, $b \sim 0.9$)

$$E = E_0(1 - aP)^b$$

For solid foams with very high porosity $P > 0.7$ ($a = 1$, $b = 2$)

$$E = E_0 \left[(1 - P)^2 / 1 + (a - 1)P \right]$$

a is a shape factor with values depending on porosity:

$a = 2.5$ for interconnected porosity ($1/a = 0.4$)

$a = 3.3 - 1.4$ for porosity that resembles ribbons ($1/a = 0.3 - 0.7$)

$a = 0.6 - 1.0$ for isolated pores ($1/a = 0.6 - 1.0$)

($1/a$ is known as the Nielson shape factor)

Empirical for oxides with porosity in the range 0–40% ($a \sim 4$)

$$E = E_0 \exp(-aP)$$

Room temperature Elastic and Shear Moduli, Poisson's Ratio for Various Materials

Material	Modulus of Elasticity		Shear Modulus		Poisson's Ratio
	GPa	10 ⁶ psi	GPa	10 ⁶ psi	
Metal Alloys					
Tungsten	407	59	160	23.2	0.28
Steel	207	30	83	12.0	0.30
Nickel	207	30	76	11.0	0.31
Titanium	107	15.5	45	6.5	0.34
Copper	110	16	46	6.7	0.34
Brass	97	14	37	5.4	0.34
Aluminum	69	10	25	3.6	0.33
Magnesium	45	6.5	17	2.5	0.35
Ceramic Materials					
Aluminum oxide (Al ₂ O ₃)	393	57	—	—	0.22
Silicon carbide (SiC)	345	50	—	—	0.17
Silicon nitride (Si ₃ N ₄)	304	44	—	—	0.30
Spinel (MgAl ₂ O ₄)	260	38	—	—	—
Magnesium oxide (MgO)	225	33	—	—	0.18
Zirconia (ZrO ₂) ^a	205	30	—	—	0.31
Mullite (3Al ₂ O ₃ ·2SiO ₂)	145	21	—	—	0.24
Glass-ceramic (Pyroceram)	120	17	—	—	0.25
Fused silica (SiO ₂)	73	11	—	—	0.17
Soda-lime glass	69	10	—	—	0.23
Polymers ^b					
Phenol-formaldehyde	2.76–4.83	0.40–0.70	—	—	—
Poly(vinyl chloride) (PVC)	2.41–4.14	0.35–0.60	—	—	0.38
Poly(ethylene terephthalate) (PET)	2.76–4.14	0.40–0.60	—	—	0.33
Polystyrene (PS)	2.28–3.28	0.33–0.48	—	—	0.33
Poly(methyl methacrylate)(PMMA)	2.24–3.24	0.33–0.47	—	—	0.37–0.44
Polycarbonate (PC)	2.38	0.35	—	—	0.36
Nylon 6,6	1.59–3.79	0.23–0.55	—	—	0.39
Polypropylene (PP)	1.14–1.55	0.17–0.23	—	—	0.40
Polyethylene—high density (HDPE)	1.08	0.16	—	—	0.46
Polytetrafluoroethylene (PTFE)	0.40–0.55	0.058–0.080	—	—	0.46
Polyethylene—low density (LDPE)	0.17–0.28	0.025–0.041	—	—	0.33–0.40

^a Partially stabilized with 3 mol% Y₂O₃.

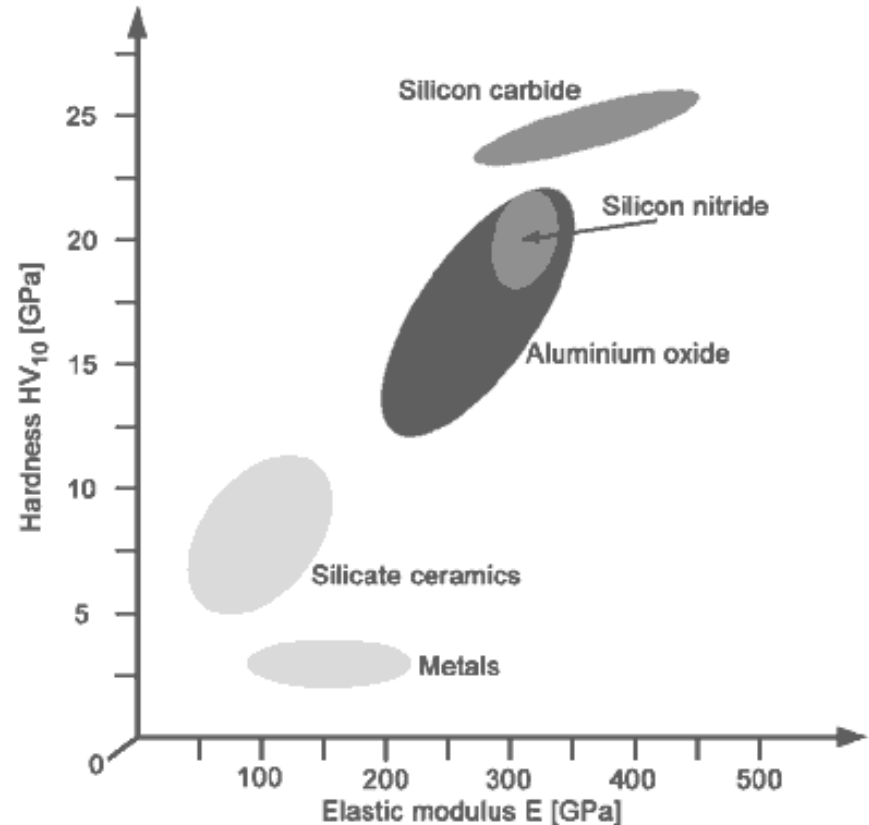
^b **Source:** *Modern Plastics Encyclopedia* '96. Copyright 1995. The McGraw-Hill Companies. Reprinted with permission.

Elastic Modulus of Ceramics

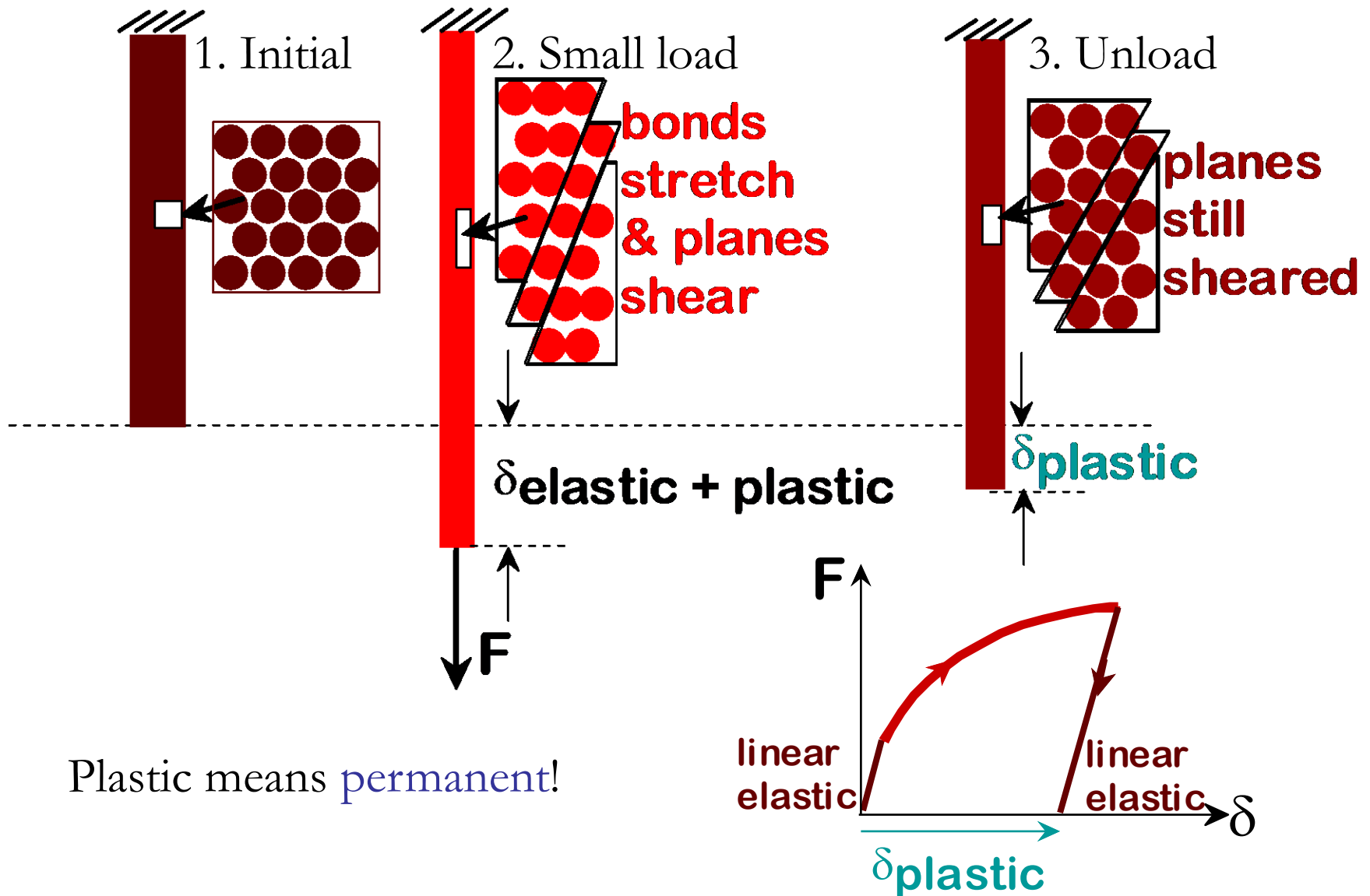
The **elastic modulus E** [GPa] of almost all oxide and non-oxide ceramics is consistently higher than that of steel.

This results in an elastic deformation of only about 50 to 70 % of what is found in steel components.

The high stiffness implies, however, that forces experienced by bonded ceramic/metal constructions must primarily be taken up by the ceramic material.



Plastic Deformation



Hardness

Measure of material resistance to localized plastic deformation, also refers to resistance of material to scratching and abrasion.

Tests:

- Mohs Hardness

- Rockwell Hardness

- Brinell Hardness

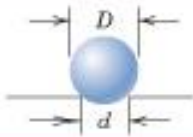
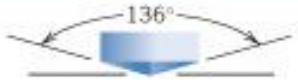
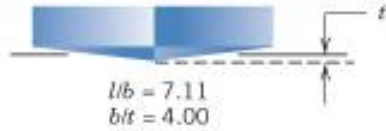
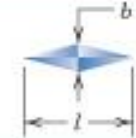
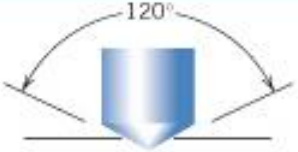
- Knoop & Vickers Microindentation Hardness

- Nanoindentation

Principle: forcing an indenter into the sample surface followed by measuring dimensions of the indentation (depth or actual surface area of the indentation).

Hardness of ceramics is determined by **their chemical compositions** and **microstructure** characteristics: porosity, grain size, grain-boundary.

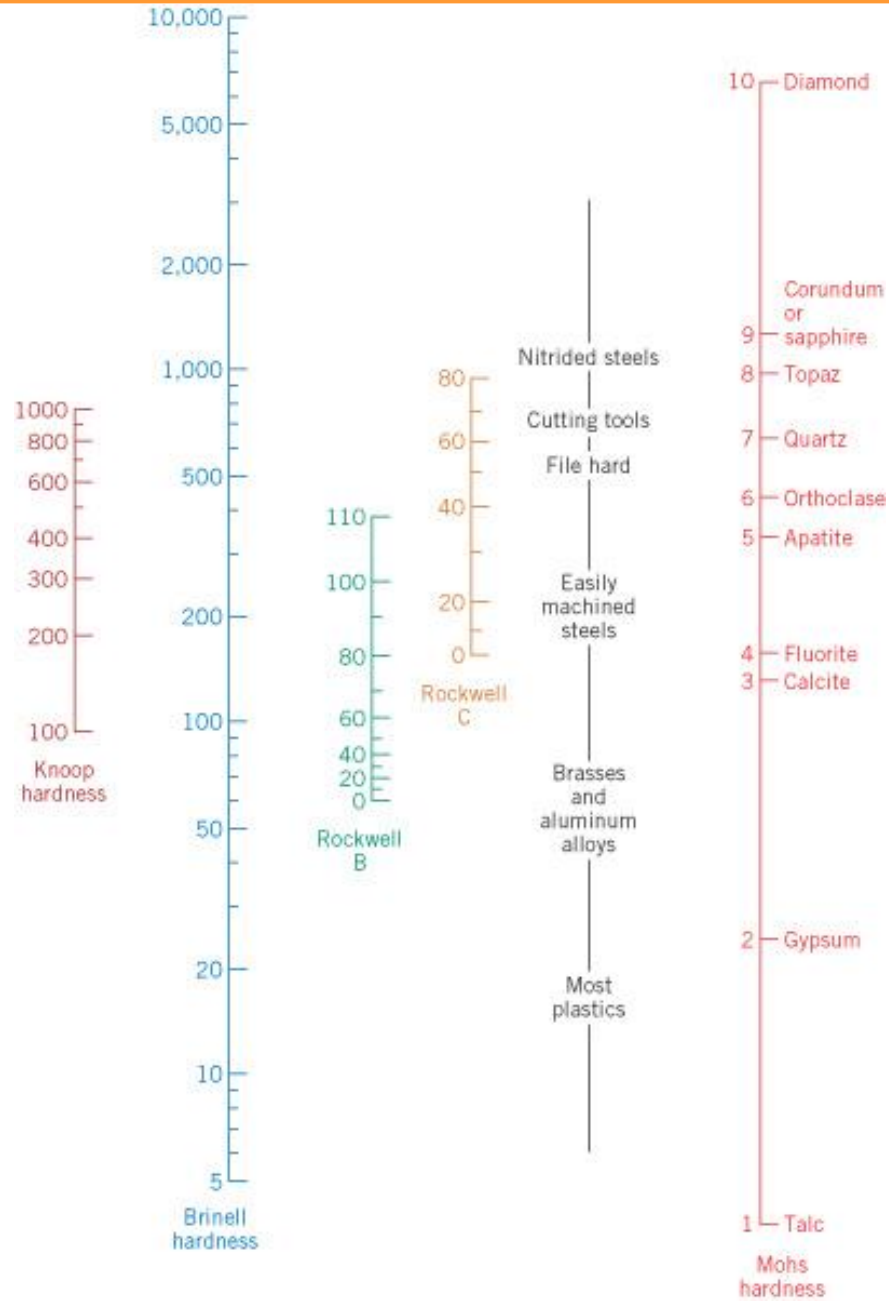
Hardness-Testing Techniques

Test	Indenter	Shape of Indentation		Load	Formula for Hardness Number ^a
		Side View	Top View		
Brinell	10-mm sphere of steel or tungsten carbide			P	$HB = \frac{2P}{\pi D[D - \sqrt{D^2 - d^2}]}$
Vickers microindentation	Diamond pyramid			P	$HV = 1.854P/d_1^2$
Knoop microindentation	Diamond pyramid			P	$HK = 14.2 P/l^2$
Rockwell and Superficial Rockwell	{ Diamond cone; $\frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}$, in. diameter steel spheres }	 	 	$\left. \begin{array}{l} 60 \text{ kg} \\ 100 \text{ kg} \\ 150 \text{ kg} \end{array} \right\} \text{Rockwell}$ $\left. \begin{array}{l} 15 \text{ kg} \\ 30 \text{ kg} \\ 45 \text{ kg} \end{array} \right\} \text{Superficial Rockwell}$	

^a For the hardness formulas given, P (the applied load) is in kg, while D , d , d_1 , and l are all in mm.

Source: Adapted from H. W. Hayden, W. G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behavior*. Copyright © 1965 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.

Hardness Conversion



Hardness of Ceramic Materials

Approximate Knoop Hardness (100 g load) for Seven Ceramic Materials

<i>Material</i>	<i>Approximate Knoop Hardness</i>
Diamond (carbon)	7000
Boron carbide (B_4C)	2800
Silicon carbide (SiC)	2500
Tungsten carbide (WC)	2100
Aluminum oxide (Al_2O_3)	2100
Quartz (SiO_2)	800
Glass	550

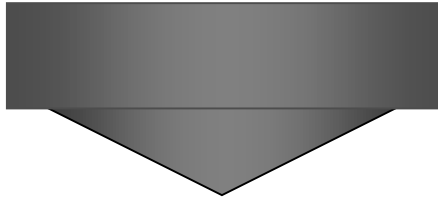
Why Nanoindentation?

Definition: Mechanical probing of a material surface to nm-scale depths, while simultaneously monitoring LOAD and DEPTH.

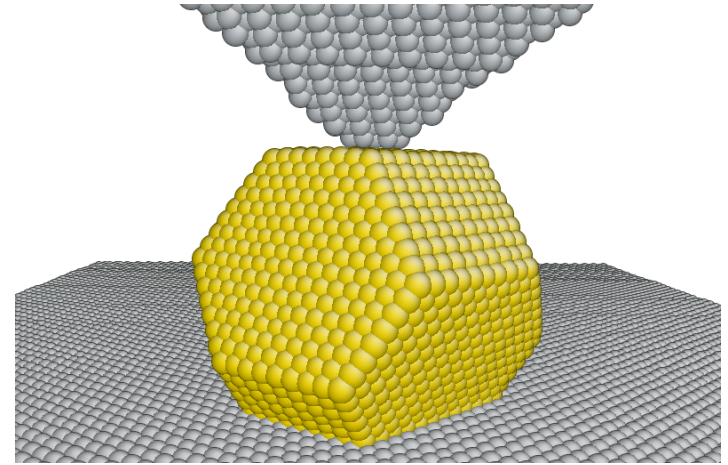
1. Materials engineering: Optimizing material composition, structure, processing for particular applications (bulk and thin films)
2. Mechanics of small volumes: Understanding whether thin films, lines, dots have different mechanical properties than bulk counterparts
3. Material physics: Measuring deformation processes such as dislocation nucleation, crack growth, etc.

Why Nanoindentation?

1. Materials engineering



10 nm – 1 μm t_f



MD simulations of nanoindentation of gold nanoparticles (D. Mordehai)

Microelectronic thin films (early 1980s)

Mechanical properties important: electromigration

Questions:

Is E a function of t_f ?

Is σ_y a function of t_f ?

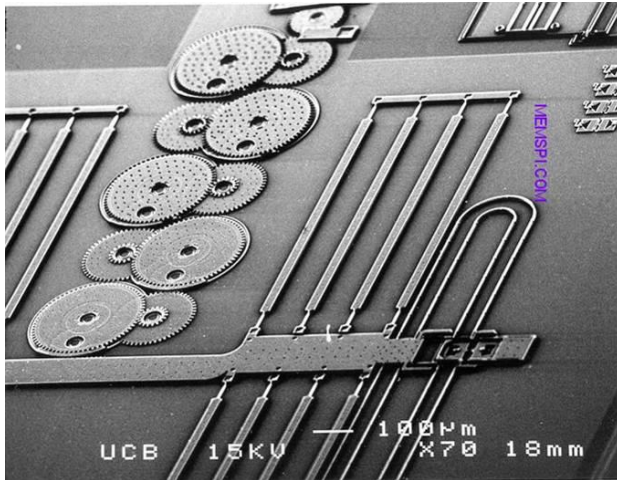
Is n a function of t_f ?

Use nanoindentation to measure thin film props

Consider sharp indenters to mitigate substrate effects

Why Nanoindentation ?

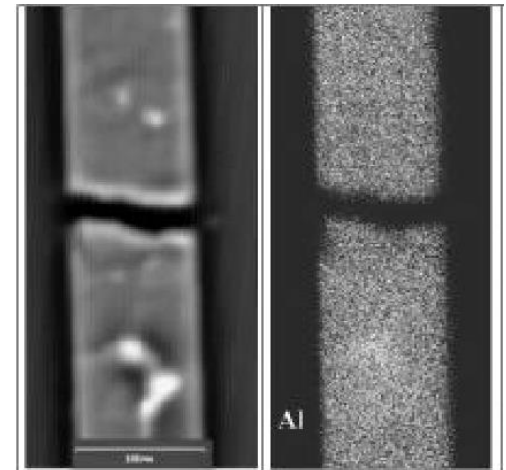
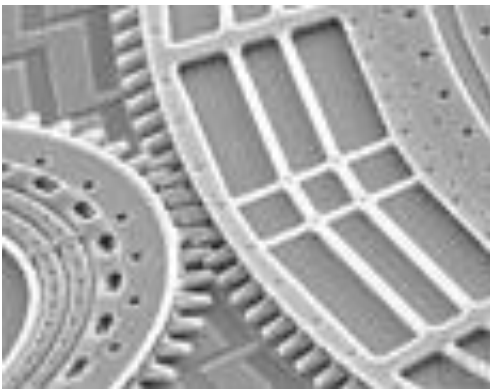
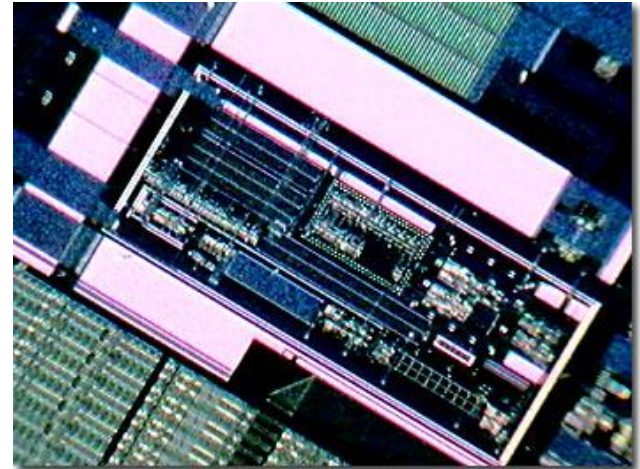
2. Unique mechanics of small volumes



Are mechanical properties
affected by constraint?

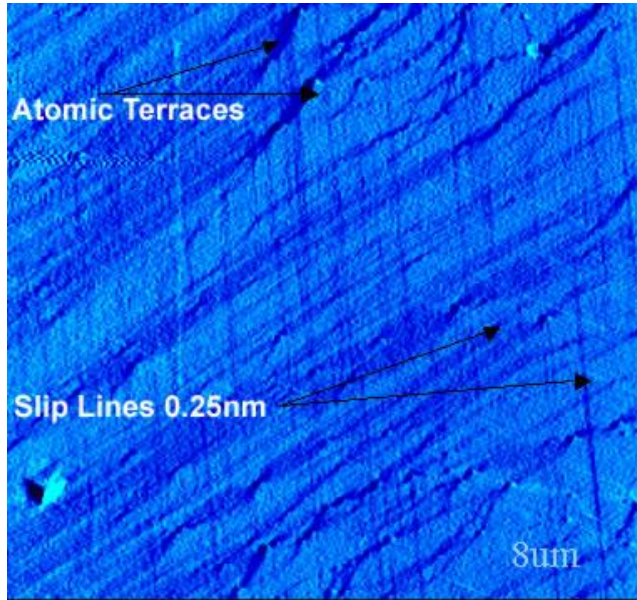
Are E , σ_y , n $f(\text{size})$?

When structure dimensions
approach atomic
dimensions, is mechanical
behavior quantized?

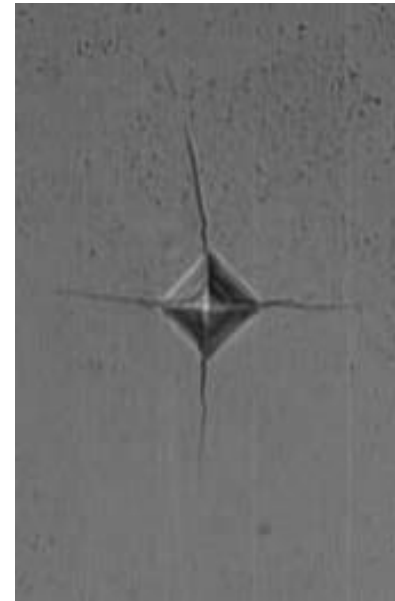


Why Nanoindentation?

3. Material physics



Metals: Dislocation nucleation, material flow



Ceramics: Fracture initiation

By probing materials near atomic and molecular dimensions, can we learn about fundamental deformation mechanisms?

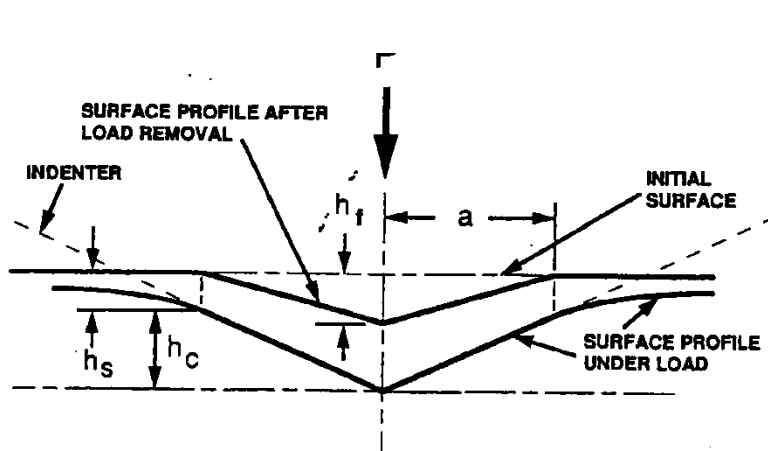
Polymers: Entropic elasticity

Biologicals: Ligand/receptor binding

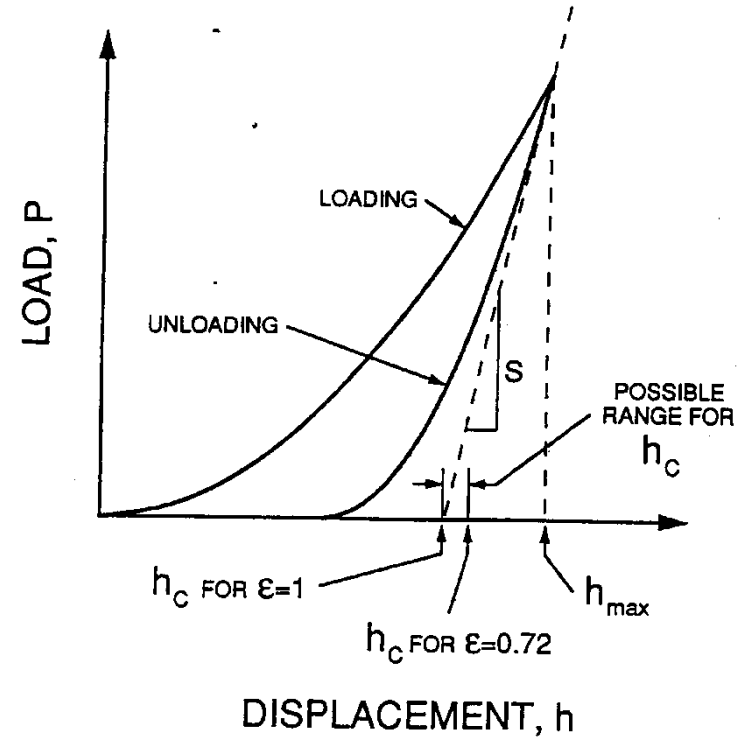
Progress due to rapid increases in microelectronics (data acquisition, signal processing)

Utility due to rapid progress in computational modeling (same time/length scales)

What is Measured?



For calculation of nanohardness the knowledge of h_c (contact depth at maximal load) is necessary.

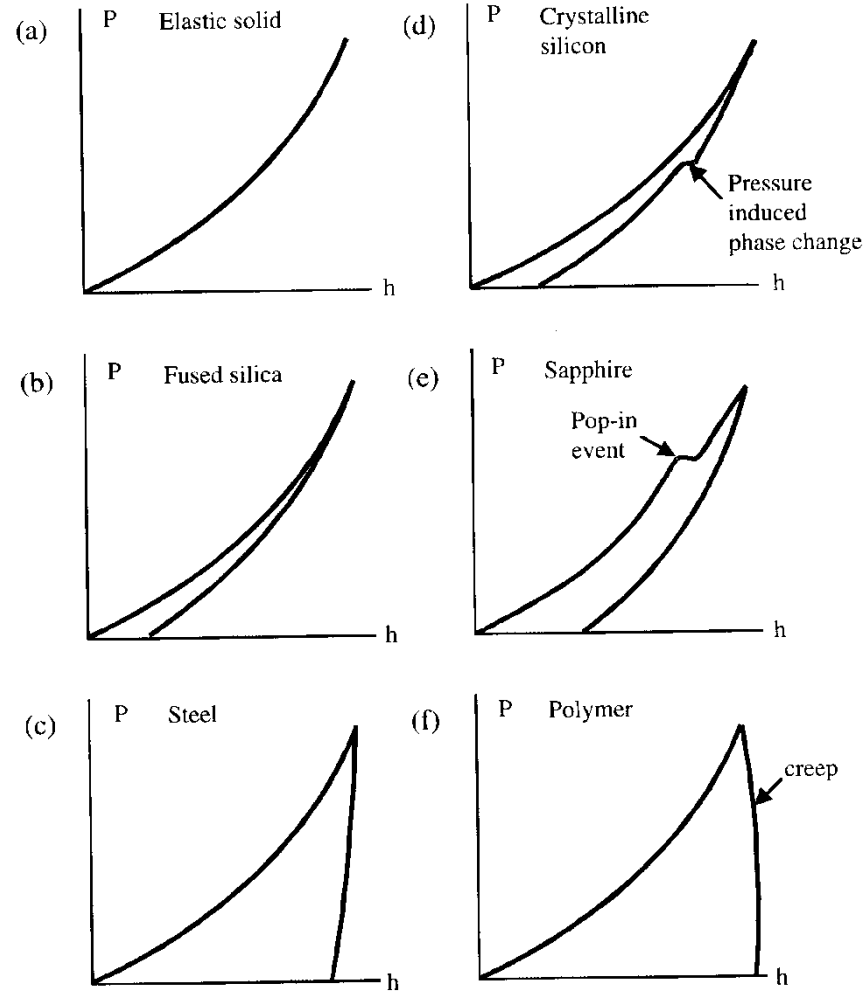


According to Oliver and Pharr (1992)

$$h_c = h_{max} - \epsilon \frac{P_{max}}{S}$$

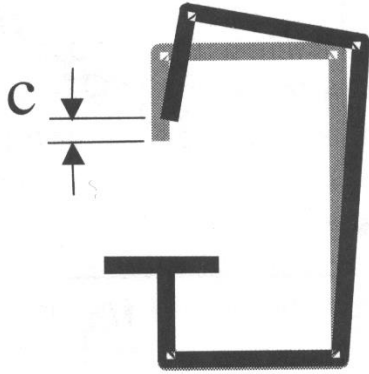
where $\epsilon=0.72$ for conical indenter and $\epsilon=0.75$ for paraboloid of revolution. S is a contact stiffness.

Loading curves for various materials



Oliver-Pharr method

Determining machine compliance:



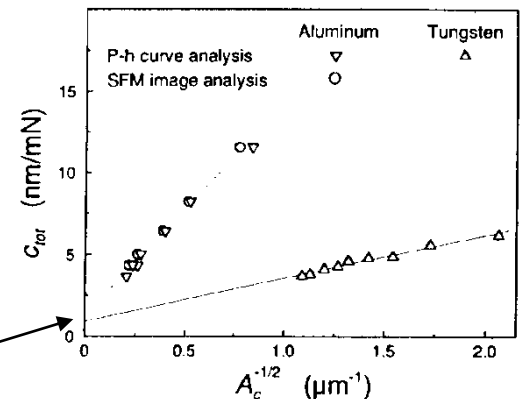
$$c_{tot} = c_f + \frac{\sqrt{\pi}}{2} \frac{1}{E_r \sqrt{A_c}}$$

A series of LARGE indents is made in Al single crystal. The contact area for Berkovich indenter is determined according to $A_c = 24.5 h_c^2$.

Canonical Sneddon stiffness equation:

$$S = \frac{2}{\sqrt{\pi}} E_r \sqrt{A_c}$$

1. A fraction of unloading curve is fitted with $P = A(h - h_f)^m$, where A and m are fitting parameters.
2. The unloading curve fit is differentiated analytically to find the stiffness at maximal load: $S = (dP/dh)_{P=P_{max}}$



machine compliance

Determining tip shape area function

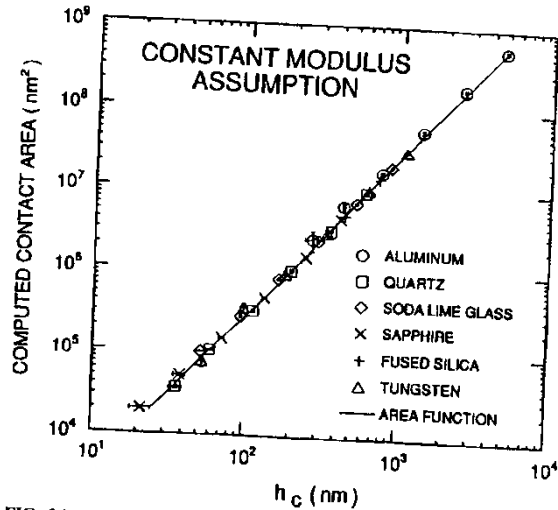


FIG. 24. The computed contact areas versus contact depths for all six materials. The error bars are two standard deviations in length.

After machine compliance was found, the empirical tip shape area function is determined Using the set of indents in fused silica:
 $E_r = 69.6$ GPa for diamond indenters.

$$A_c = \frac{\pi}{4} \frac{1}{E_r^2 (c_{tot} - c_f)^2}$$

The tip shape area function is determined by fitting the obtained (A_c , h_c) data to a polynomial of the form:

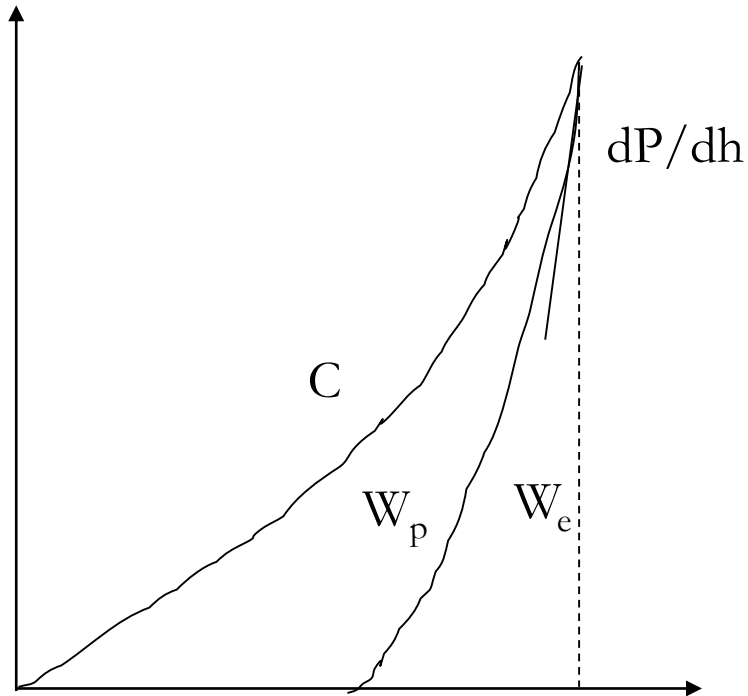
$$A_c = a_0 h_c^2 + a_1 h_c + a_2 h_c^{1/2} + a_3 h_c^{1/4} + a_4 h_c^{1/8} + a_5 h_c^{1/16}$$

After tip shape area function is determined, the nanohardness is determined according to

$$H = \frac{P_{\max}}{A_c(h_c)} \quad \text{where} \quad h_c = h_{\max} - \varepsilon \frac{P_{\max}}{S}$$

The elastic modulus is determined by Sneddon stiffness equation.

What is Nanoindentation?



Show variables in response:

C = loading curvature

dP/dh = unloading stiffness

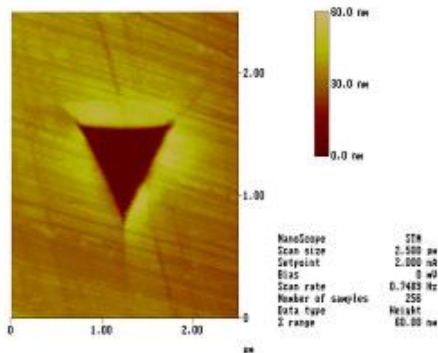
$P_{\max}, h_{\max}$

W_p = plastic work of indentation

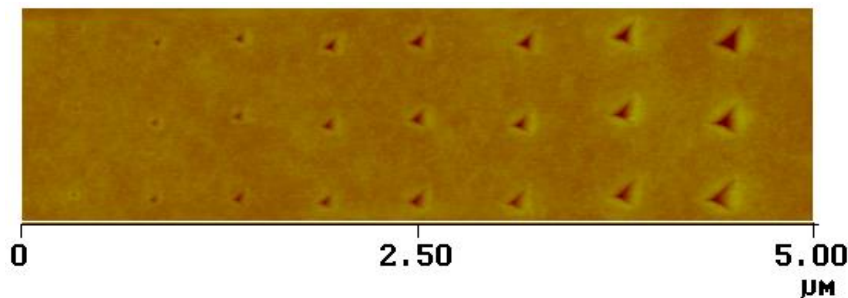
W_t = total work of indentation

Materials engineering goal: Interpret P-h data in terms of continuum scale properties.
That is, test a small material volume to obtain parameters for a large material volume
Translate a complicated, multiaxial stress state into uniaxial properties.

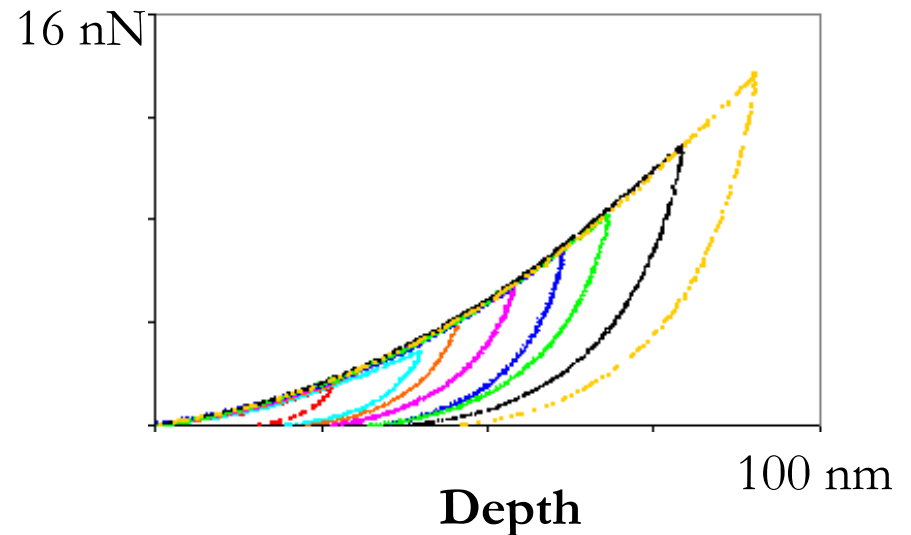
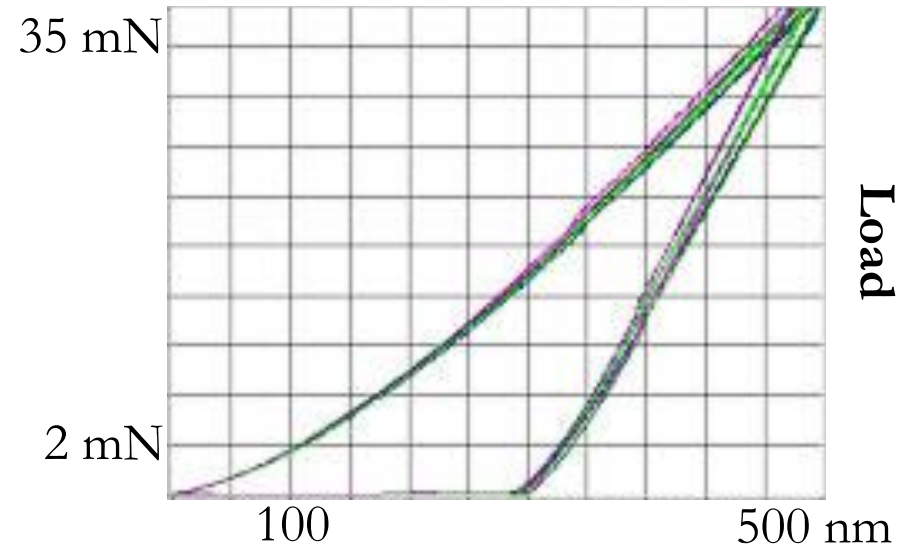
What is Nanoindentation ?



SiC (ceramic)



PE (polymer)



How does Nanoindentation work?

A. Material assumptions

Continuum: No structural length scales e.g. grain size, film thickness

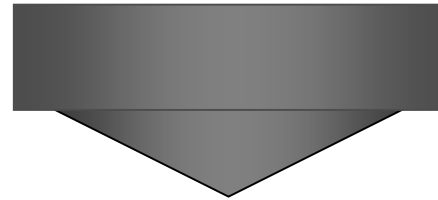
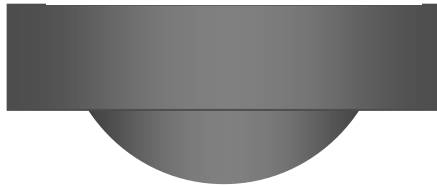
Homogeneous: One phase

Elastoplastic: Deforms via plastic yielding, rather than fracture, phase transformation

B. Indenter considerations

$E_i \gg E_s$: Ideally, deformation occurs only in sample material

Primarily diamond (hardest natural material)

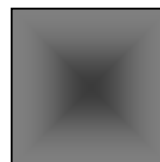


Spherical:

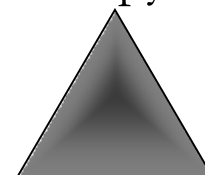
- * Elastic, THEN plastic
- * Difficult to machine diamond

Sharp:

- * Immediately elastic AND plastic
- * Can facet diamond into pyramids, not cones



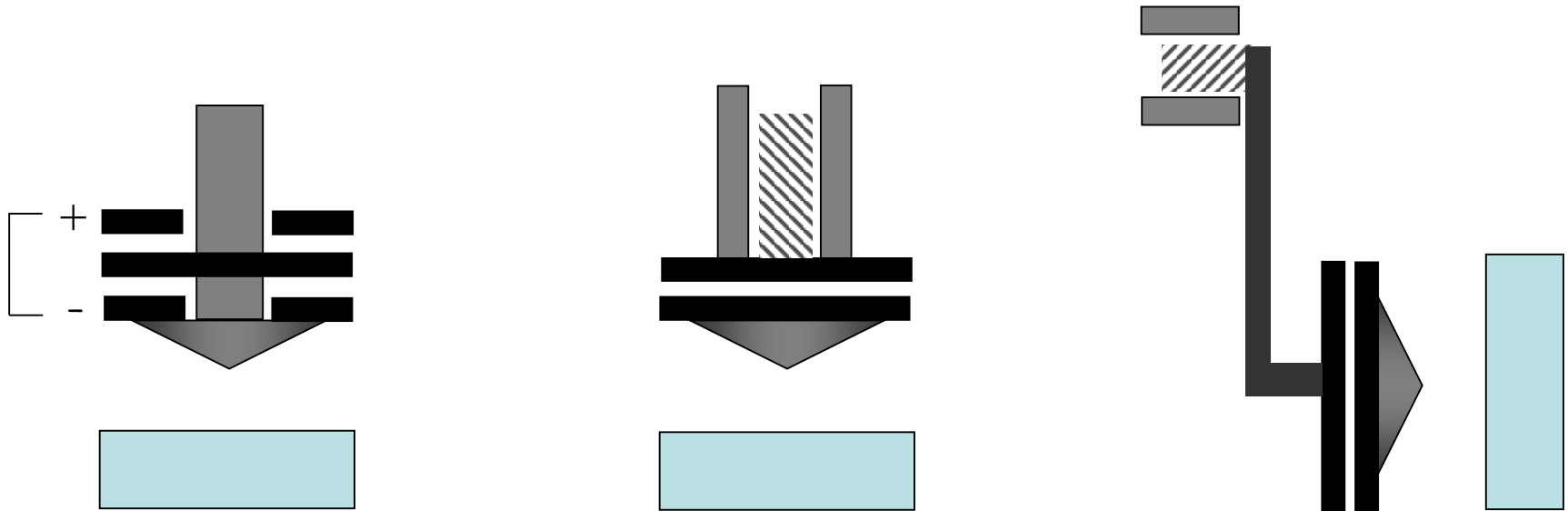
Vickers



Berkovich

How does Nanoindentation work?

C. Instrumentation

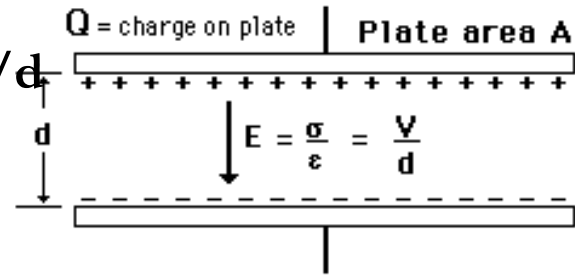


3-plate capacitor
 P , h via same device

How does nanoindentation work?

Capacitance of Parallel Plates:

Electric field between two parallel plates: $E = V/d$



Voltage difference between two plates as $+q$ moves from positive to negative plate:

$$V = \frac{\text{work done}}{\text{charge}} = \frac{Fd}{q} = Ed$$

Capacitance is the amount of charge Q stored per unit voltage V applied to plates:

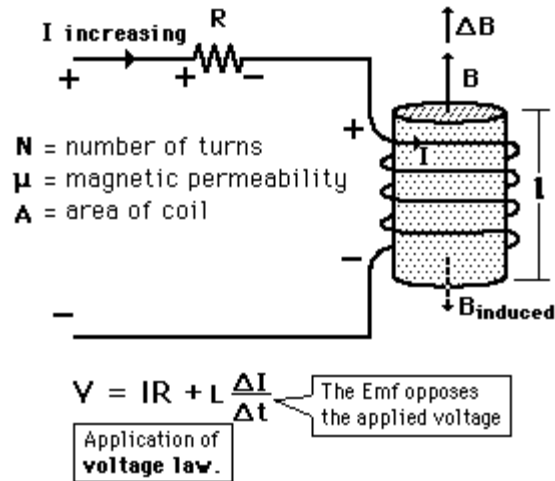
$$C = \frac{Q}{V} = \frac{Q}{Ed} = \frac{Q\epsilon}{\sigma d} = \frac{QA\epsilon}{Qd} = \frac{A\epsilon}{d}$$

Relevance to nanoindentation:

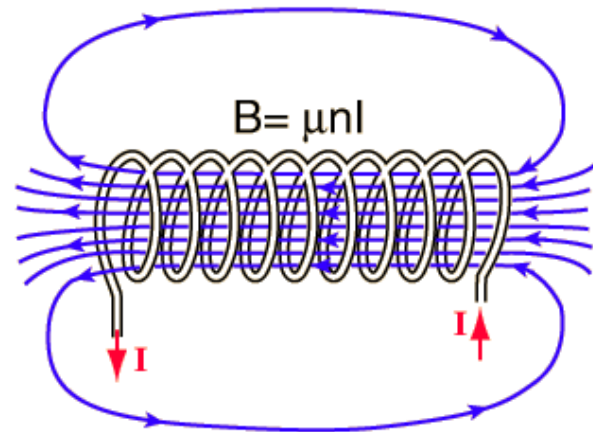
- (1) As plates move closer, d decreases and V , C increase
- (2) Temperature/noise insensitive displacement meter

How does nanoindentation work?

Inductance of a solenoid



Inductance is the change in $B(V)$ created by a change in I



For fixed A , a change in current I leads to:

Magnetic field of a solenoid:

Thus

$$\text{Emf} = -N \frac{\Delta \Phi}{\Delta t} = -N A \frac{\Delta B}{\Delta t}$$

$$B = \mu \frac{N}{l} I$$

$$\text{Emf} = - \frac{\mu N^2 A}{l} \frac{\Delta I}{\Delta t}$$

$$\text{Emf} = -L \frac{\Delta I}{\Delta t}$$

$$L = \frac{\mu N^2 A}{l}$$

Relevance to nanoindentation:

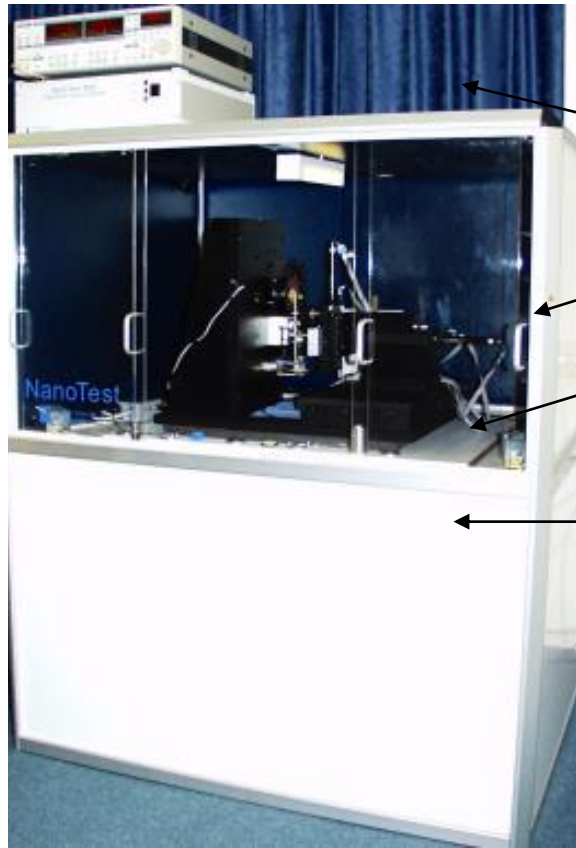
- (1) Can be used to exact motion (change in I)
- (2) Can be used to measure load (change in I or V)

How does Nanoindentation work?

E. Environmental control



Hysitron



MicroMaterials

Humidity control ($< 55\%$)

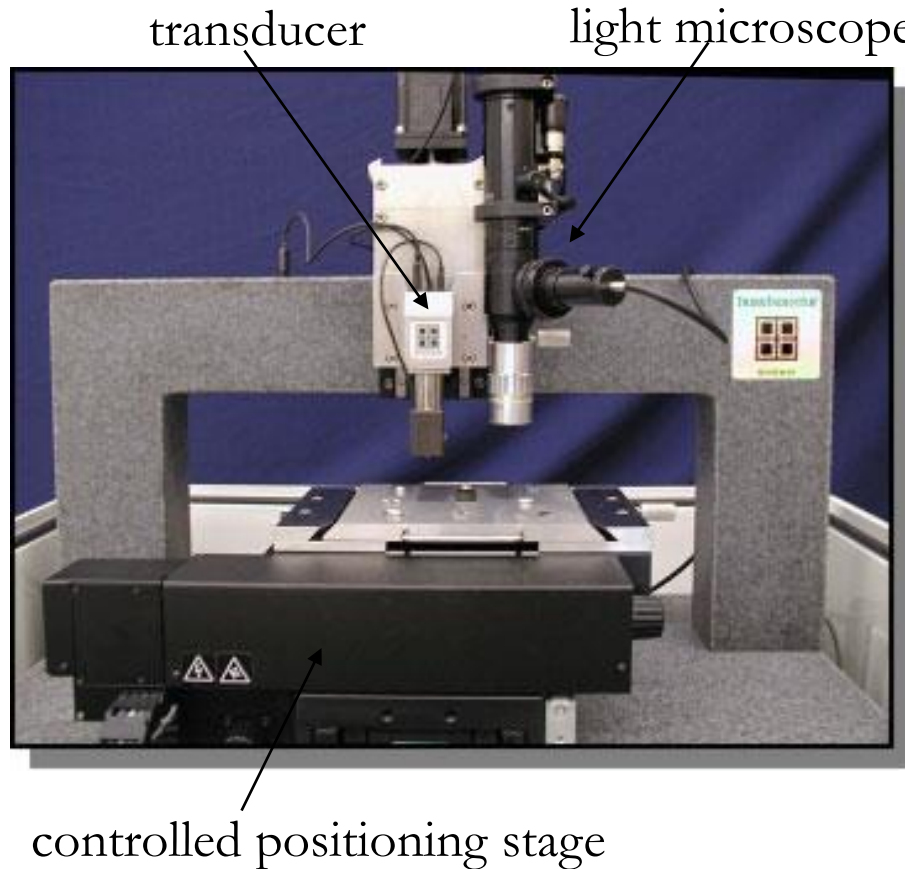
Acoustic isolation (< 1 Hz)

Thermal control (< 0.5 C)

Vibration isolation (< 1 Hz)

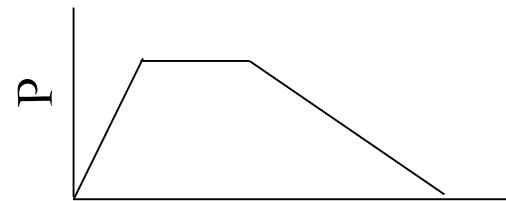


How does Nanoindentation work?



Procedure:

1. Prepare sample surface:
Flat, Parallel faces, Smooth
2. Find region of interest on surface
3. Program load profile



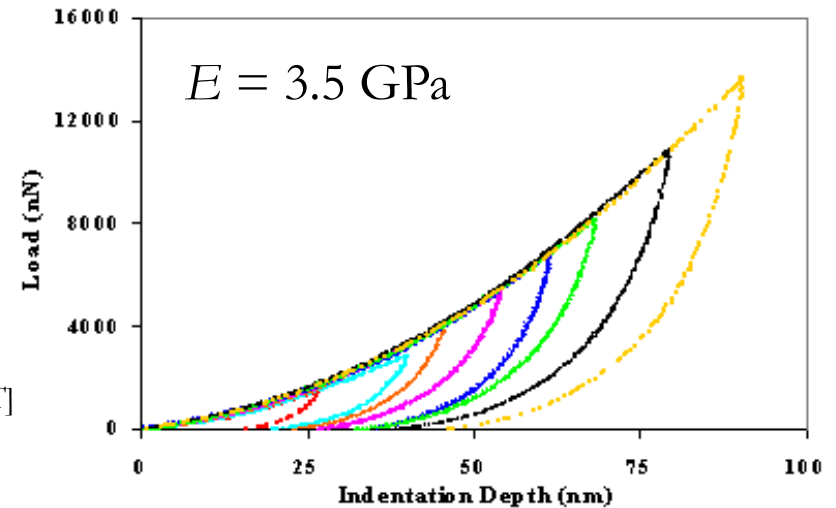
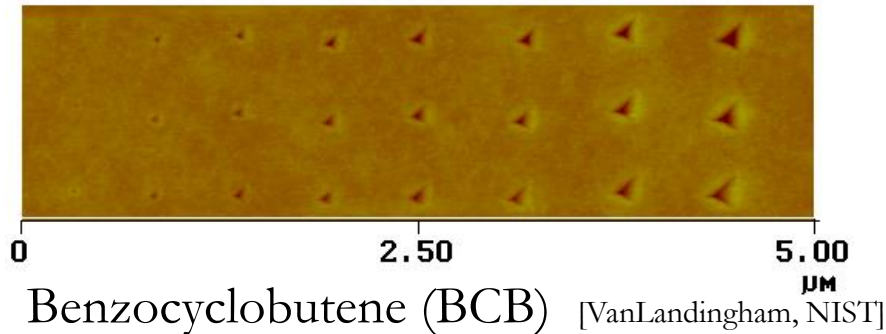
4. Program spatial matrix



5. Execute experiment
6. Analyze data to obtain properties

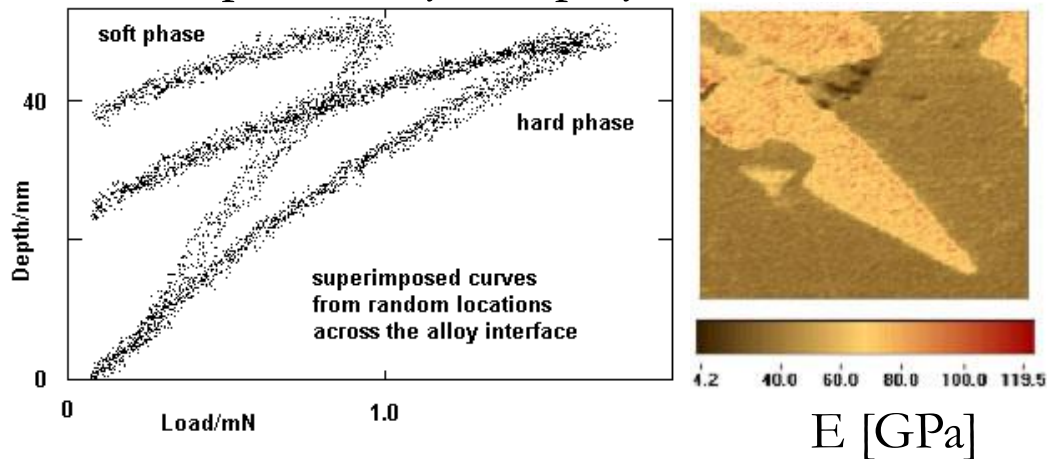
How does Nanoindentation work?

Results in bulk materials



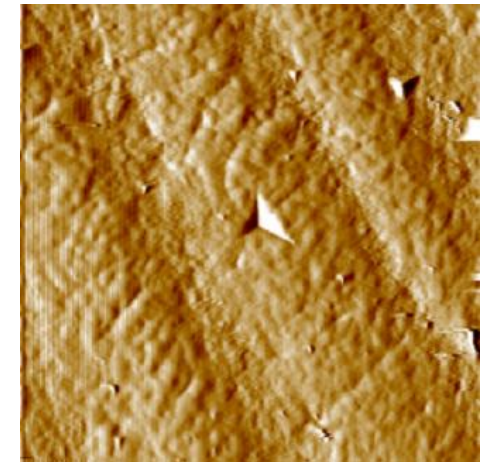
Results in composites

1. Multiphase alloys/Copolymers



[Smith, MicroMaterials]

2. Trabecular bone

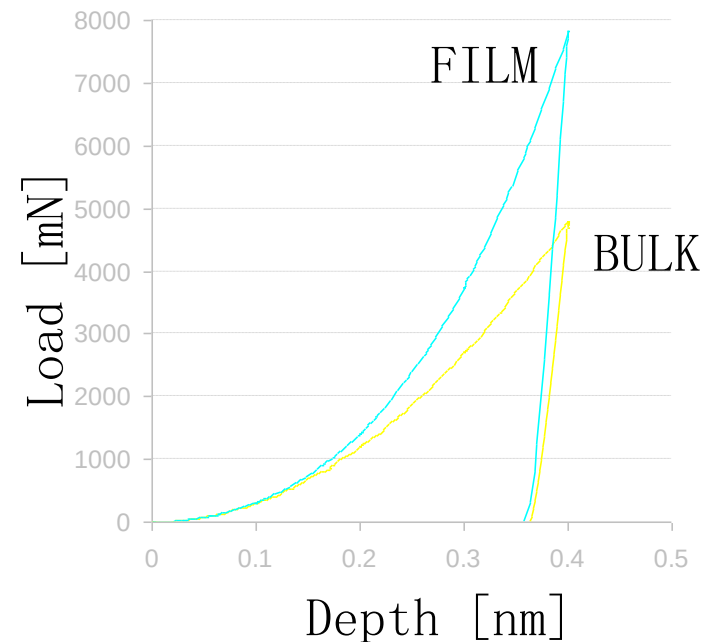
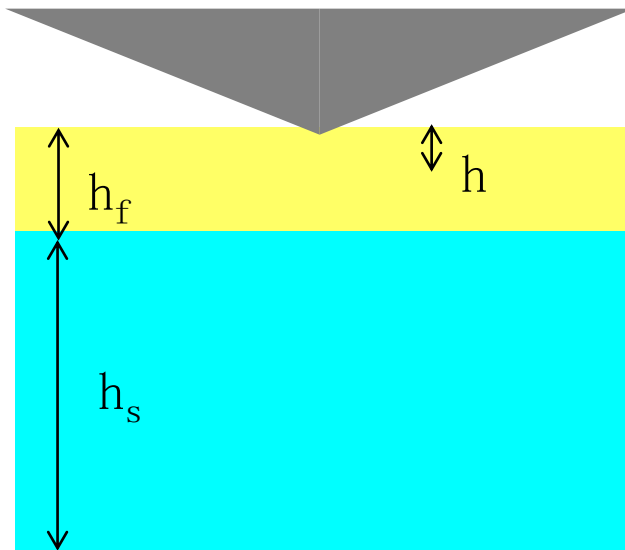


[Wyrobek, Hysitron]

How does Nanoindentation Work?

Results in thin films

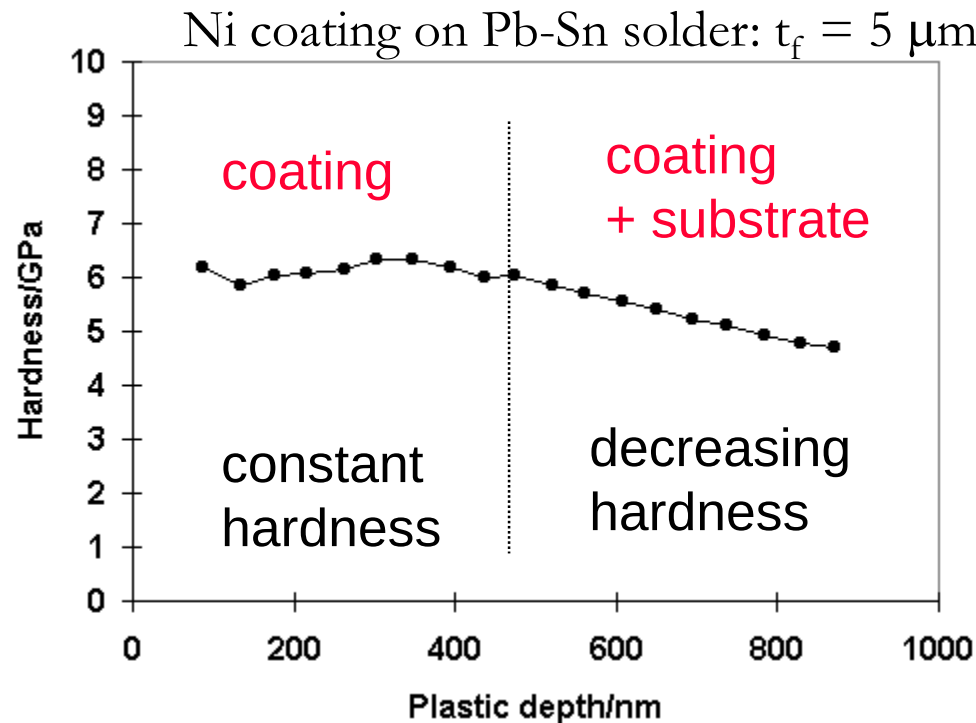
- Substrate affects indentation response
- Complicated relationship due to n
- Empirical rule: $h_{\max} < 10\% t_f$ for continuum analysis to hold



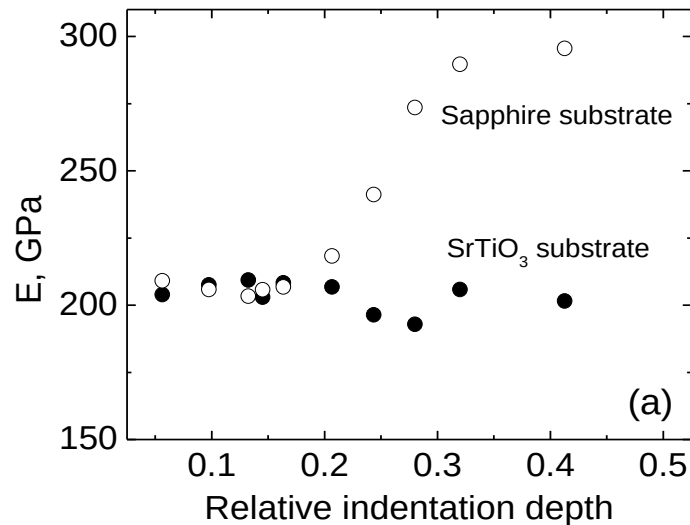
How does Nanoindentation work?

Results in thin films

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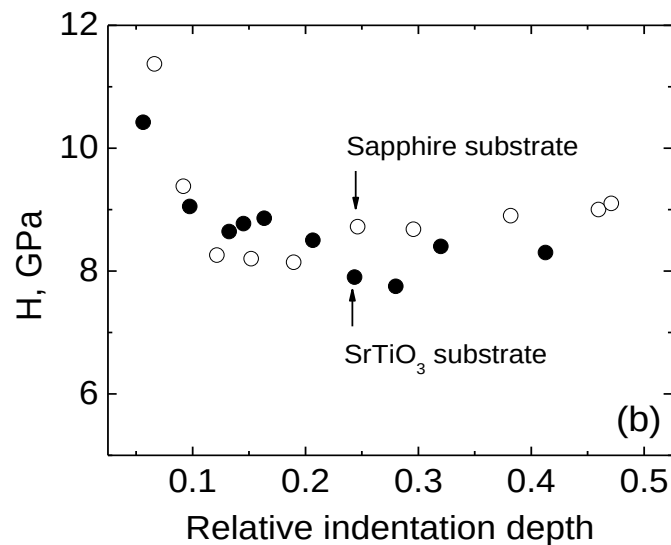


Measuring Mechanical Properties of Ceramic Thin Films



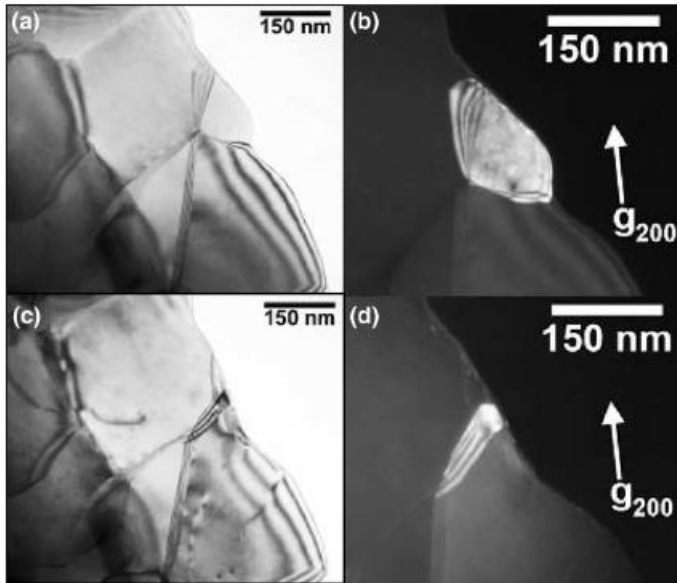
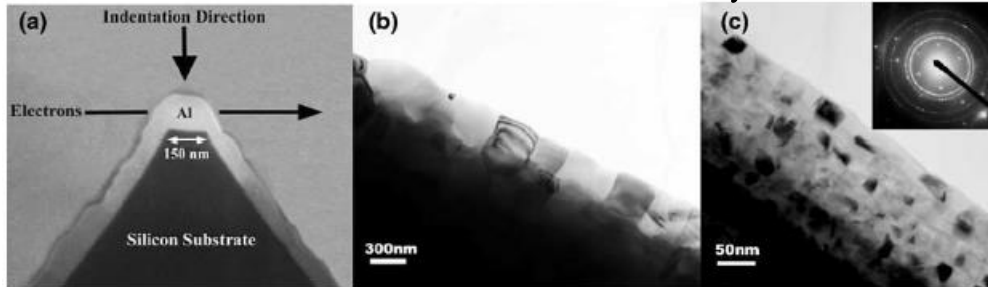
The indentation depth should be less than $1/3$ of film thickness. Otherwise, a combination of film and substrate mechanical properties is measured.

For “soft” films on hard substrates the Young’s modulus is more sensitive to substrate than the hardness.



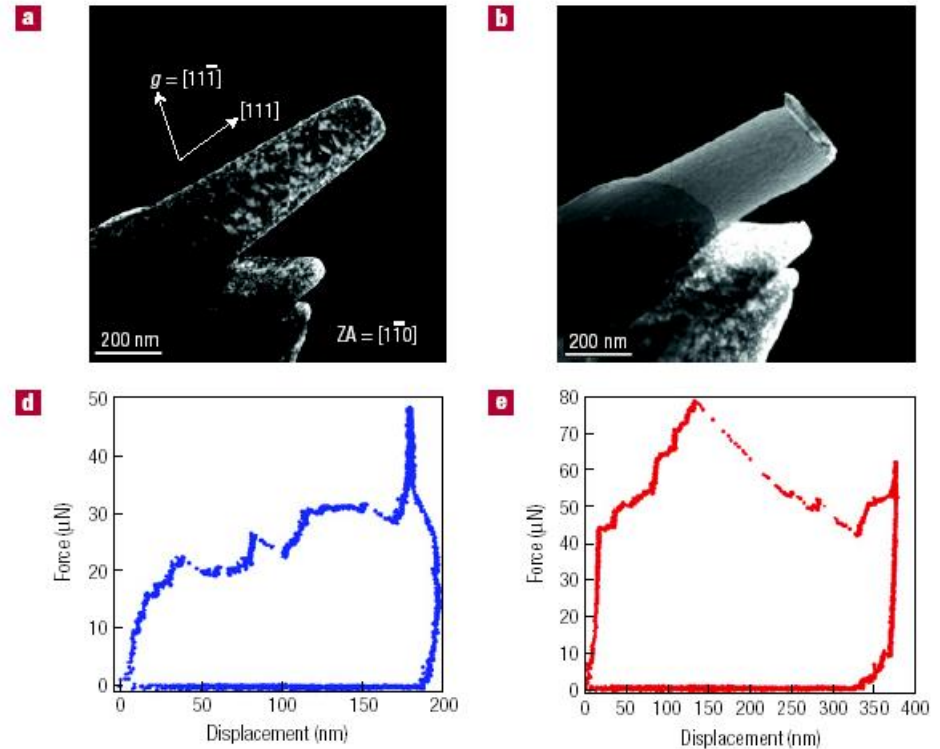
In-situ Nanoindentation in TEM

“Local” recrystallization

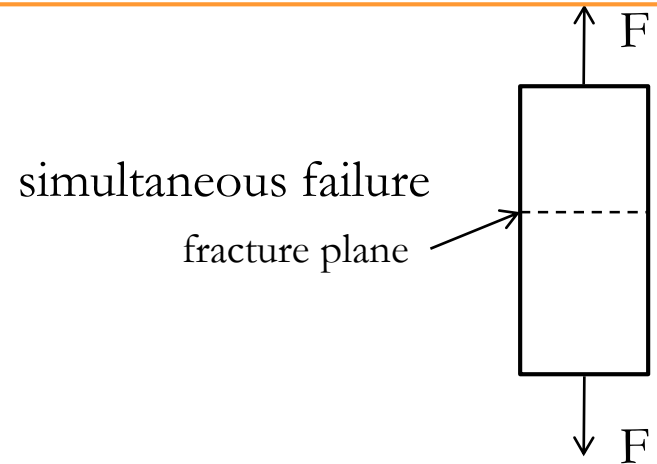
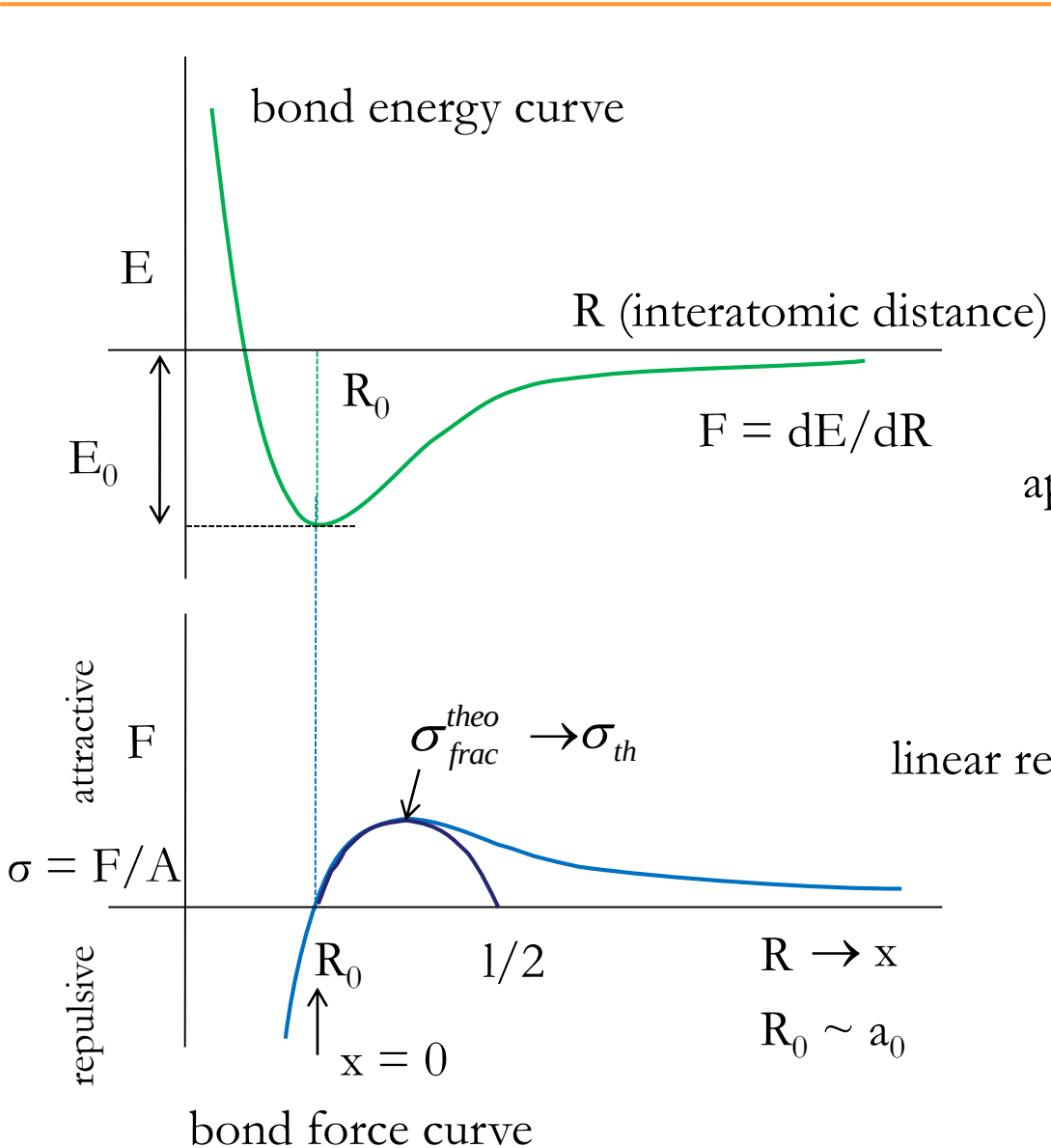


M. Jin et al., Acta mater. 2004

Shan et al., Nature mater. 2008



Theoretical Strength



approximate s as sinusoidal

$$\sigma = \sigma_{th} \sin\left(\frac{2\pi x}{\lambda}\right)$$

linear region: $\sigma = E\varepsilon = E\left(\frac{\Delta R}{R_0}\right) = E\left(\frac{x}{a_0}\right)$

$$\sigma \sim \sigma_{th} \frac{2\pi x}{\lambda} = E\left(\frac{x}{a_0}\right)$$

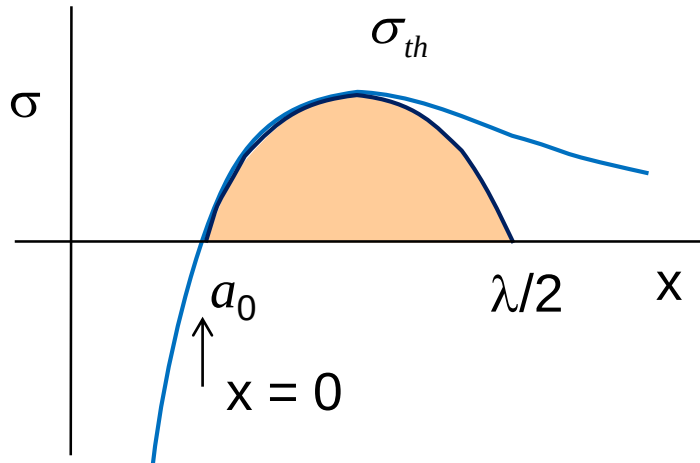
$$\Rightarrow \sigma_{th} = \frac{E \lambda}{2\pi a_0} \quad ???$$

Evaluate σ_{frac} (theoretical)

Obtain 1 by equating mechanical energy (work) of creating two surfaces to their surface energy

$$E = \int F dx \quad E / \text{area} = \int \sigma dx$$

$$\begin{aligned} \text{work} / \text{area} &= \int_0^{\lambda/2} \sigma_{th} \sin\left(\frac{2\pi x}{\lambda}\right) dx \\ &= \frac{\lambda \sigma_{th}}{\pi} \end{aligned}$$

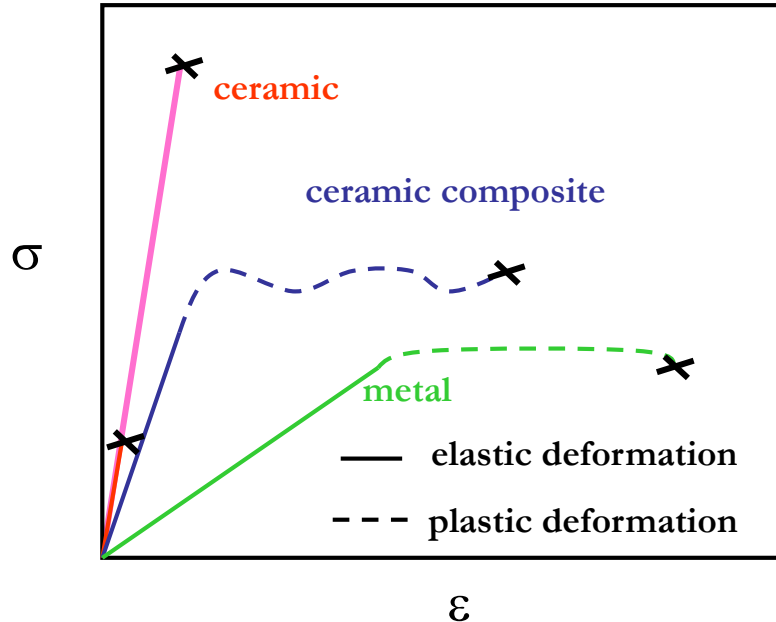


surface energy / area of fracture = 2γ

$$\Rightarrow \lambda = \frac{2\gamma\pi}{\sigma_{th}} \Rightarrow \sigma_{th} = \frac{E}{2\pi a_o} \frac{2\gamma\pi}{\sigma_{th}}$$

$$\Rightarrow \sigma_{th} = \left(\frac{\gamma E}{a_o} \right)^{1/2}$$

Material Strength



Stress-strain relationships for different materials at ambient temperature

The theoretical strength of a material (for a flawless single crystal) is related to the elastic modulus:

$$\sigma_{\max} = \left(\frac{\gamma E}{a_0} \right)^{\frac{1}{2}} \approx \frac{E}{10}$$

γ : surface energy
 a_0 : av. atomic distance

Ceramics have much larger elastic moduli than metals, e.g. they are much less elastically deformed than metal.

Material	Young's modulus (GPa)
diamond	1210, 970
Al ₂ O ₃ (s,p)	460, 390
MgO(s)	250
SiC (p)	560
Glasses	70 - 80
Aluminum	60 -75
Steel	190

The theoretical strength of alumina should therefore be between 40 and 50GPa, the measured values however are only 0.27GPa! Why?

Defects in Ceramics

✚ Point defects ✓

- ☞ vacancies, interstitials, etc.

- ☞ control mass diffusion

✚ Linear defects

- ☞ dislocations (Mechanical properties – yield, metals)

- ☞ control deformation processes

✚ Planar defects ✓

- ☞ grain boundaries, surfaces, interfaces,

- ☞ heterophase boundaries

✚ Volume defects ✓

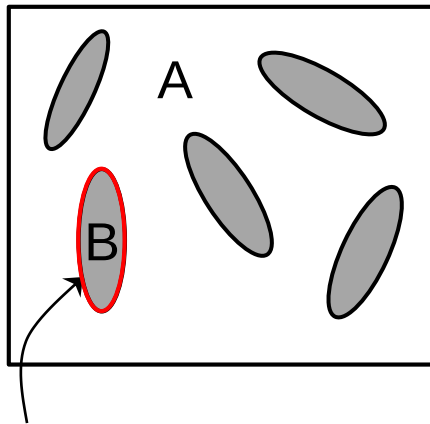
- ☞ voids, secondary components (phases)

(Mechanical properties – fracture, ceramics)

Volume Defects / Heterophase Boundaries

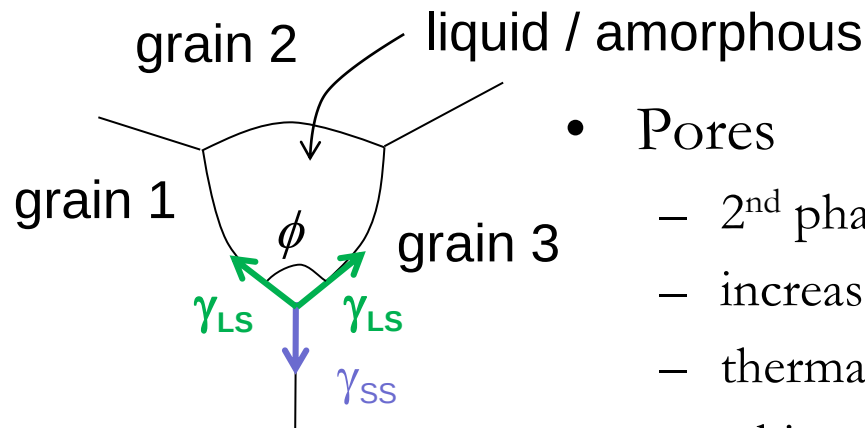
Composites

- Two or more distinct types of materials, “phases”
- Boundary between them is a heterophase interface



interphase boundary

- At grain boundaries
 - Second phase concentrated at triple contacts of host grain boundaries
 - Typical when liquid phase forms at high temperature



Balance of forces

$$\gamma_{SS} = 2\gamma_{SL} \cos(\phi/2)$$

Pores

- 2nd phase is a void
- increases scattering
- thermal insulation
- white, not transparent

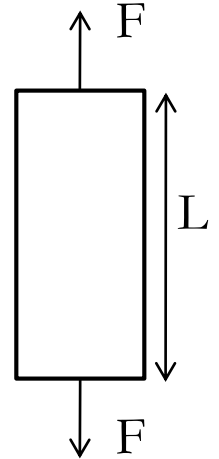
Volume Defects and Mechanics

✦ A secondary (different) material: “phase”

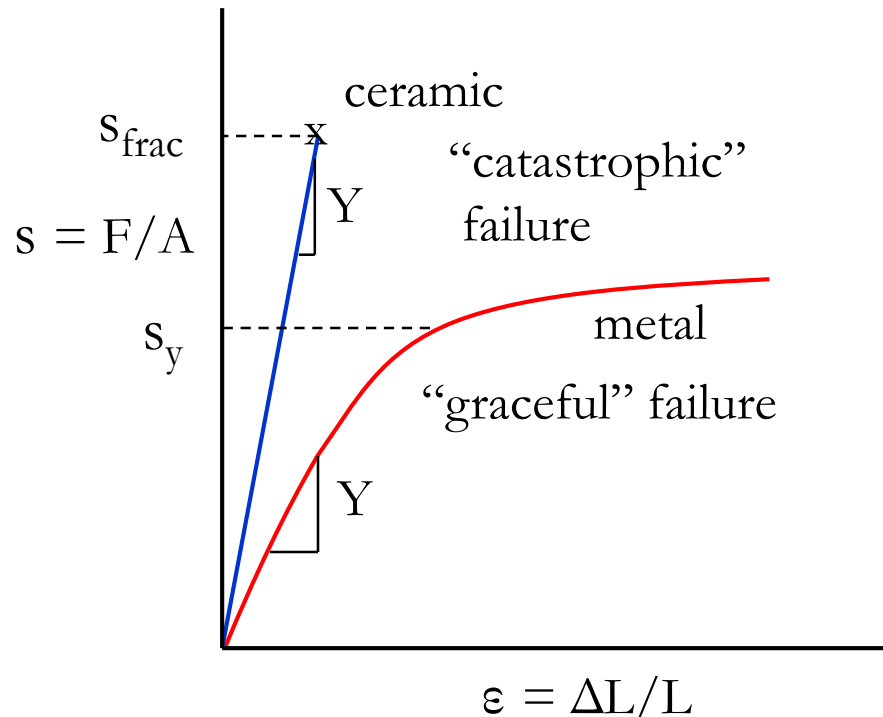
📄 in metals: secondary phases tend to pin dislocations

✦ Pores

📄 in ceramics: tend to be source of failure



Mechanical Behavior



Y – from chemical bonds

s_y – due to dislocation glide

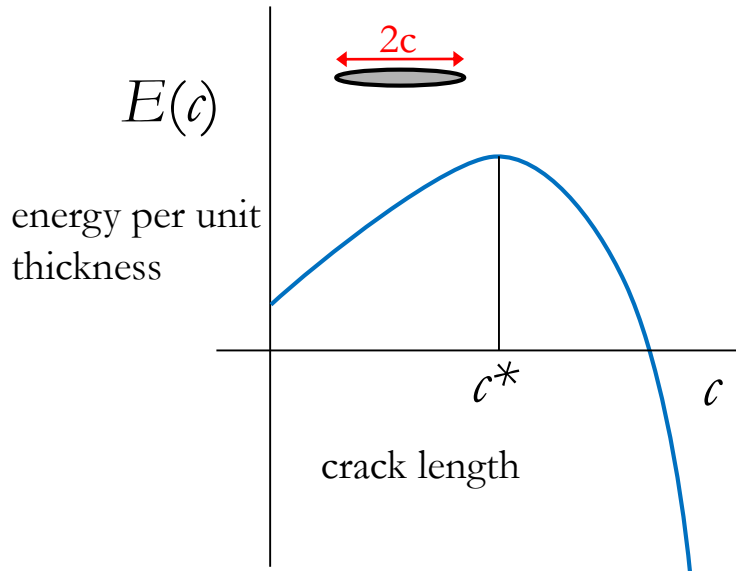
$s_y (\text{obs}) \ll s_y (\text{theo})$

s_{frac} – due to volume defects

$s_{frac} (\text{obs}) \ll s_{frac} (\text{theo})$

Observed Strength

Alternative derivation: again, consider energy balance



crack energy = initial energy + surface energy - released strain energy

$$E(c) = E_0 + 4c\gamma l - \frac{\pi\sigma^2 c^2 l}{E}$$

take fracture to occur when: $c > c^*$

$$\frac{\partial E}{\partial c} = 0 = 4\gamma - \frac{2\pi\sigma^2 c}{E} \quad \sigma_{frac}^{obs}$$

$$\Rightarrow \sigma_{frac}^{obs} = \left(\frac{2\gamma E}{\pi c} \right)^{1/2}$$

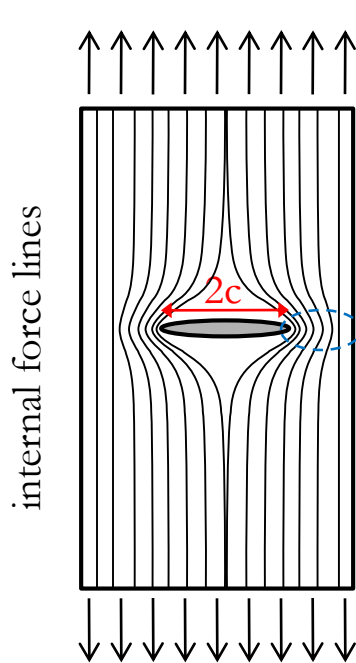
as before: $\sigma_{frac}^{obs} \propto c^{-1/2}$

For a plate which is thick compared with crack size (i.e. plane strain condition), the stress is given as

$$\sigma = \left(\frac{2E\gamma}{(1-\nu^2)c\pi} \right)^{1/2}$$

Evaluate $s_{\text{frac}}(\text{observed})$

$$\sigma_{\text{frac}}^{\text{obs}} \ll \sigma_{\text{frac}}^{\text{th}} \sim \left(\frac{\gamma E}{a_0} \right)^{1/2} \quad \text{Why ??} \quad \text{Stress concentration at crack tips}$$



internal force lines

$\sigma = F/A$

only this region of the material supports the load

can show: $\sigma_{\text{tip}} \approx 2\sigma_{\text{app}} \left(\frac{c}{\rho} \right)^{1/2}$ radius of curvature

take fracture to occur when: $\sigma_{\text{tip}} = \sigma_{\text{frac}}^{\text{th}}$

$$\Rightarrow 2\sigma_{\text{app}} \left(\frac{c}{\rho} \right)^{1/2} = \left(\frac{\gamma E}{a_0} \right)^{1/2} \Rightarrow \sigma_{\text{app}} = \left(\frac{\gamma E \rho}{4a_0 c} \right)^{1/2}$$

$\sigma_{\text{frac}}^{\text{obs}}$ (pointing to σ_{app})

a_0 (pointing to a_0)

atomically sharp crack tip

$$\Rightarrow \sigma_{\text{frac}}^{\text{obs}} = \left(\frac{\gamma E}{4c} \right)^{1/2}$$

in general: $\sigma_{\text{frac}}^{\text{obs}} \propto c^{-1/2}$

measured fracture stress is not an “inherent” material property

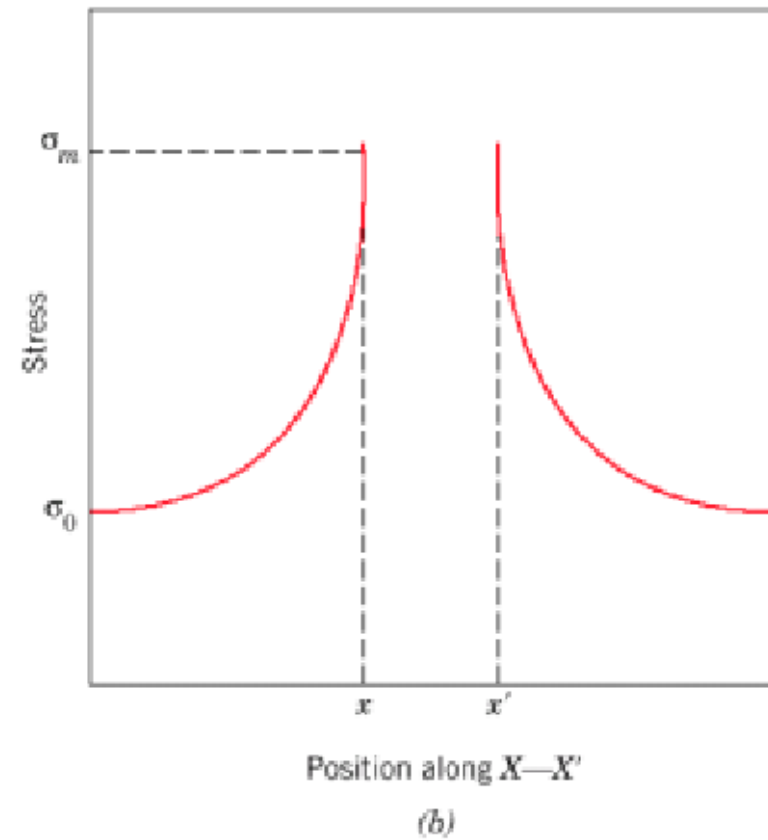
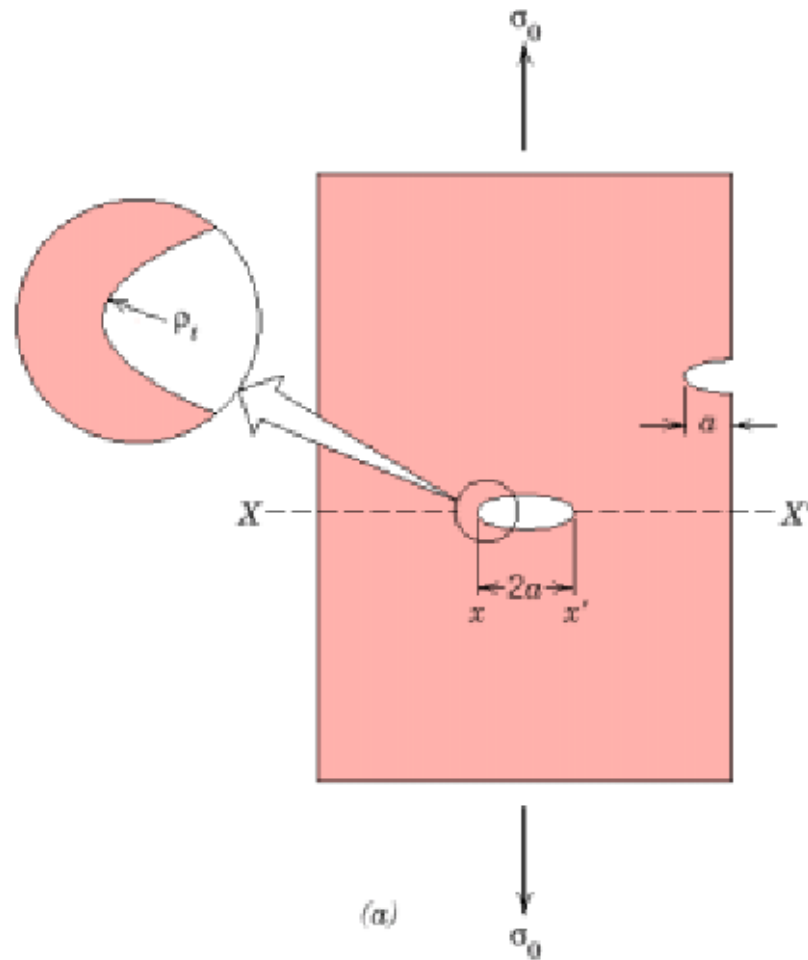
Stress Concentration

For a long crack oriented perpendicular to the applied stress the maximum stress near the crack is:

$$\sigma_m \approx 2\sigma (c/\rho)^{1/2} \quad (1)$$

where σ is the applied external stress, c is the half-length of the crack, and ρ the radius of curvature of the crack tip. (note that c is half-length of the internal flaw, but the full length for a surface flaw).

Stress Concentration



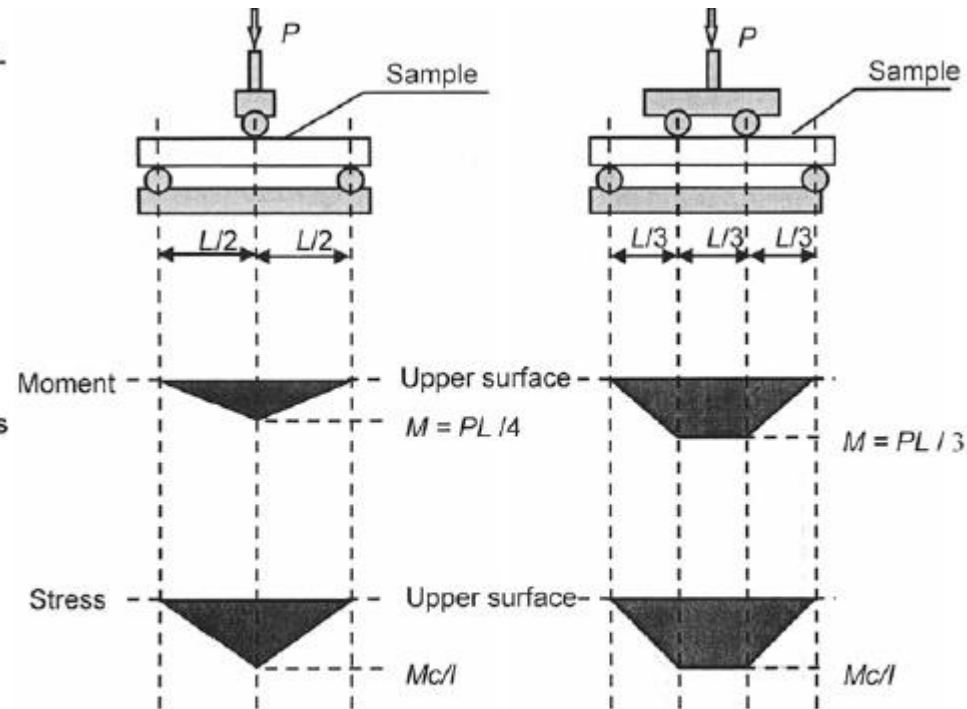
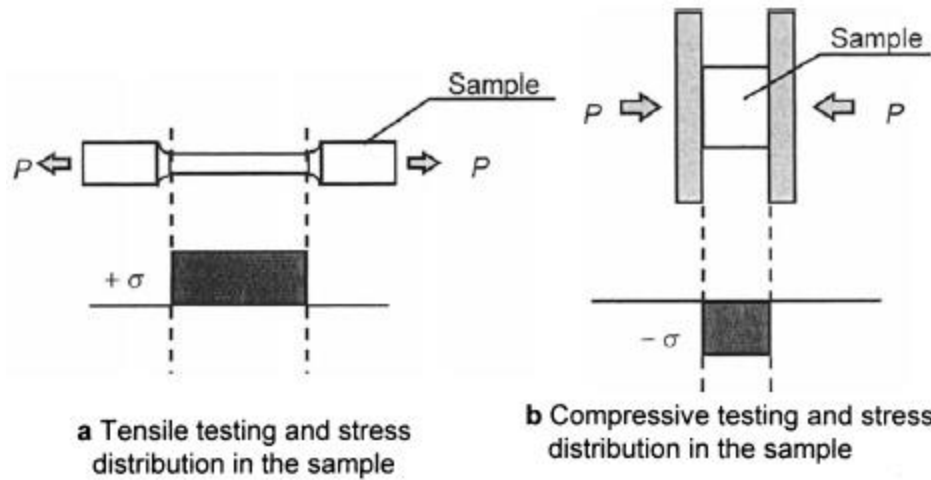
Stress Concentration Factor

The ratio of the maximum stress and the nominal applied tensile stress is denoted as the stress concentration factor, K_t .

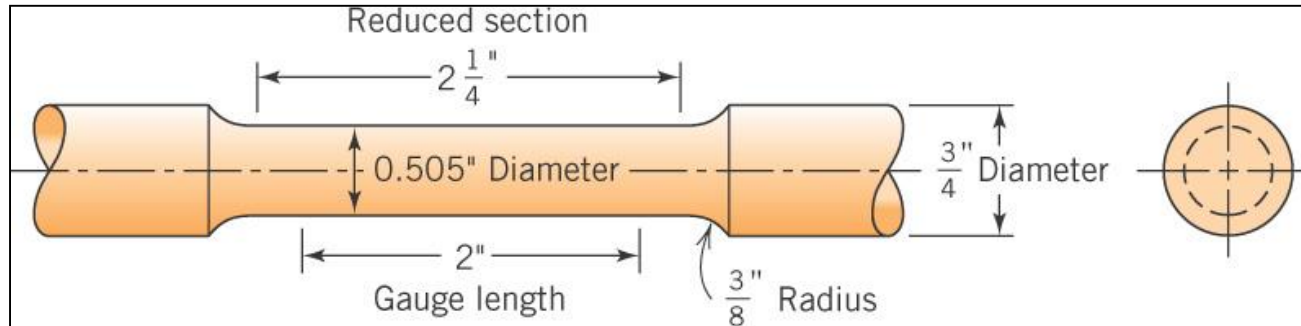
The *stress concentration factor* is a simple measure of the degree to which an external stress is amplified at the tip of a small crack.

$$K_t = \frac{\sigma_m}{\sigma} = 2\left(\frac{c}{\rho}\right)^{1/2}$$

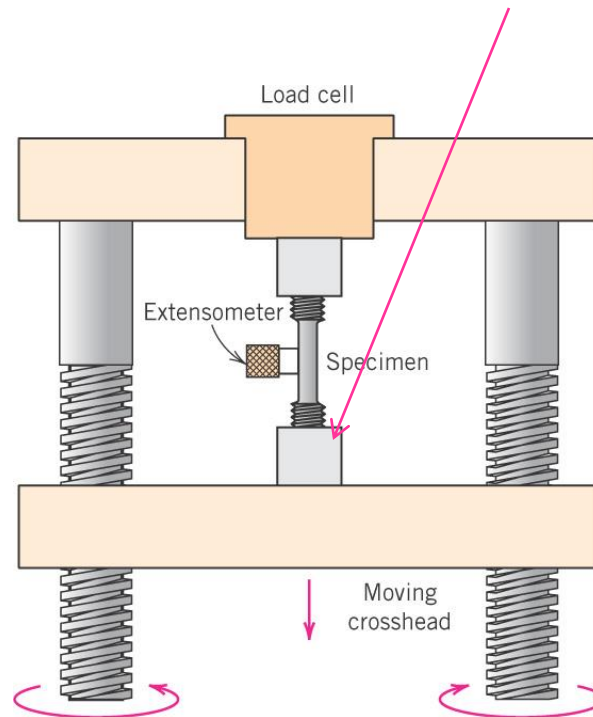
Strength Testing



How to Test the Strength for Ceramics ?



specimen



Machine

Does tensile test work on ceramics ? If not, why ?

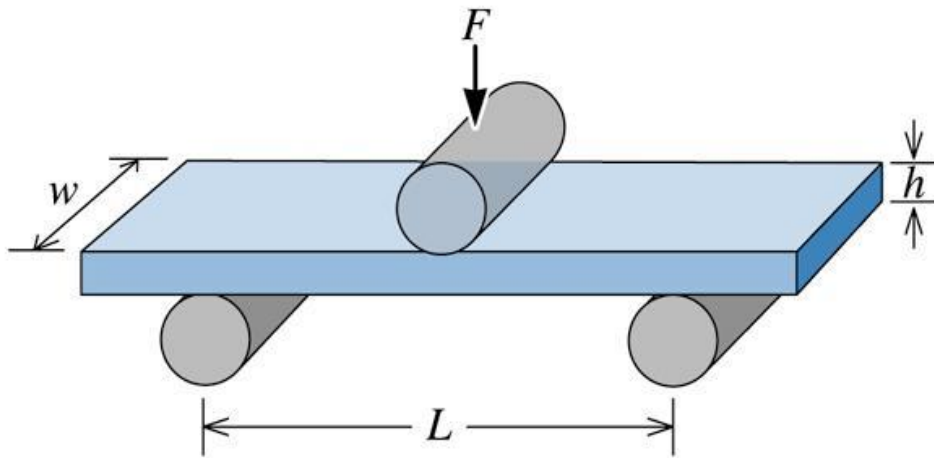
The Bend Test for Ceramic Materials

Bend test - Application of a force to the center of a bar that is supported on each end to determine the resistance of the material to a static or slowly applied load.

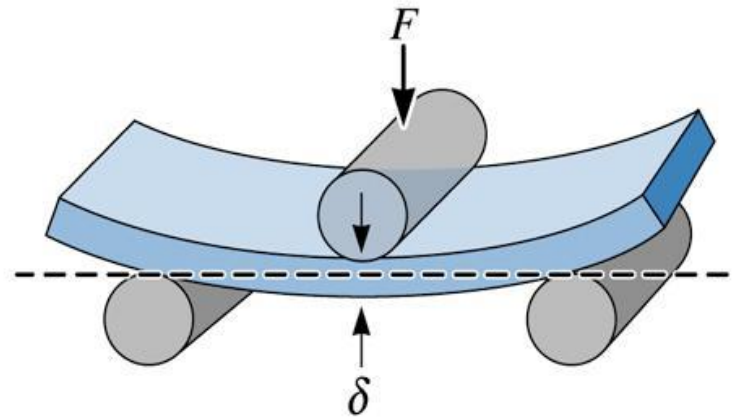
Flexural strength or modulus of rupture -The stress required to fracture a specimen in a bend test.

Flexural modulus - The modulus of elasticity calculated from the results of a bend test, giving the slope of the stress-deflection curve.

The Bend Test



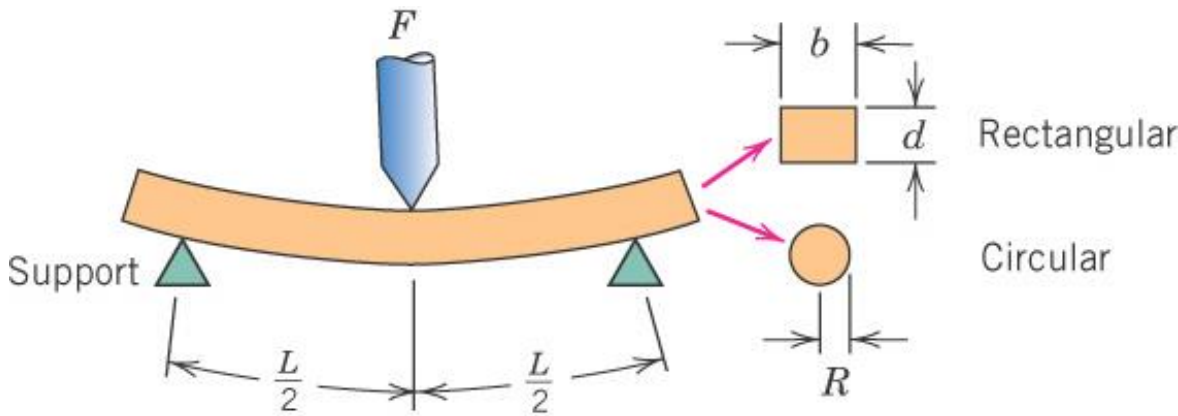
(a)



(b)

(a) The bend test often used for measuring the strength of brittle materials, and (b) the deflection δ obtained by bending

Flexural Strength-3 point bend test



$$\sigma = \text{stress} = \frac{Mc}{I}$$

where M = maximum bending moment

c = distance from center of specimen to outer surface

I = moment of inertia of cross section

F = applied load

	$\frac{M}{}$	$\frac{c}{}$	$\frac{I}{}$	$\frac{\sigma}{}$
Rectangular	$\frac{FL}{4}$	$\frac{d}{2}$	$\frac{bd^3}{12}$	$\frac{3FL}{2bd^2}$
Circular	$\frac{FL}{4}$	R	$\frac{\pi R^4}{4}$	$\frac{FL}{\pi R^3}$

✦ Able to measure the stress-strain behavior and flexural strength of brittle ceramics.

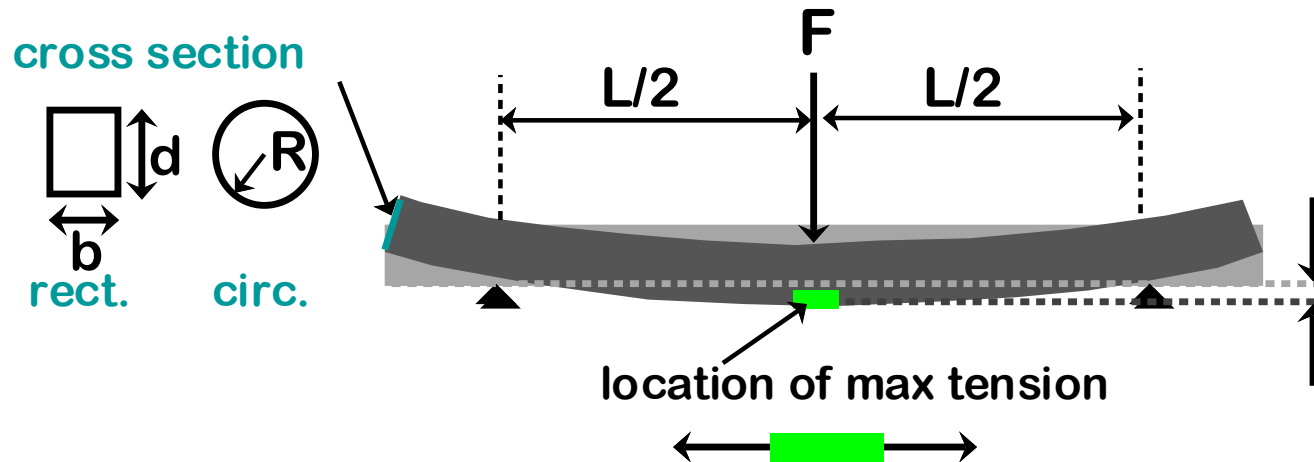
✦ Flexural strength (modulus of rupture or bend strength) is the stress at fracture.

$$\sigma_{fs} = \frac{3F_f L}{2bd^2}$$

$$\sigma_{fs} = \frac{F_f L}{\pi R^3}$$

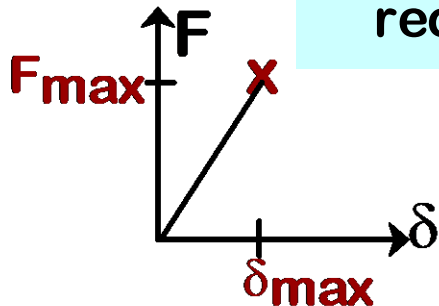
Measuring Flexural Strength

- 3-point bend test to measure room T strength.



- Flexural strength:

$$\sigma_{fs} = \sigma_m^{fail} = \frac{1.5F_{max}L}{bd^2_{rect.}} = \frac{F_{max}L}{\pi R^3}$$



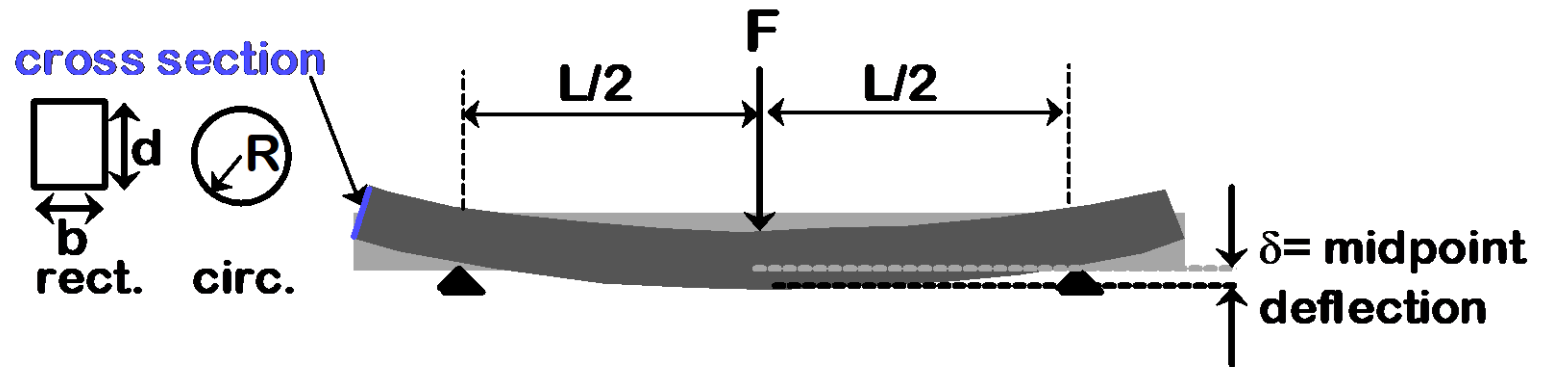
- Typ. values:

Material	σ_{fs} (MPa)	E(GPa)
Si nitride	700-1000	300
Si carbide	550-860	430
Al oxide	275-550	390
glass (soda)	69	69

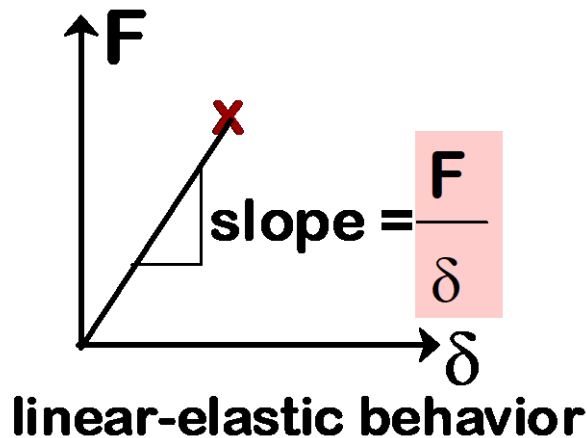
Data from Table 12.5, *Callister 6e*.

Measuring Flexural Modulus

- Room T behavior is usually elastic, with brittle failure.
- 3-Point Bend Testing often used.
--tensile tests are difficult for brittle materials.



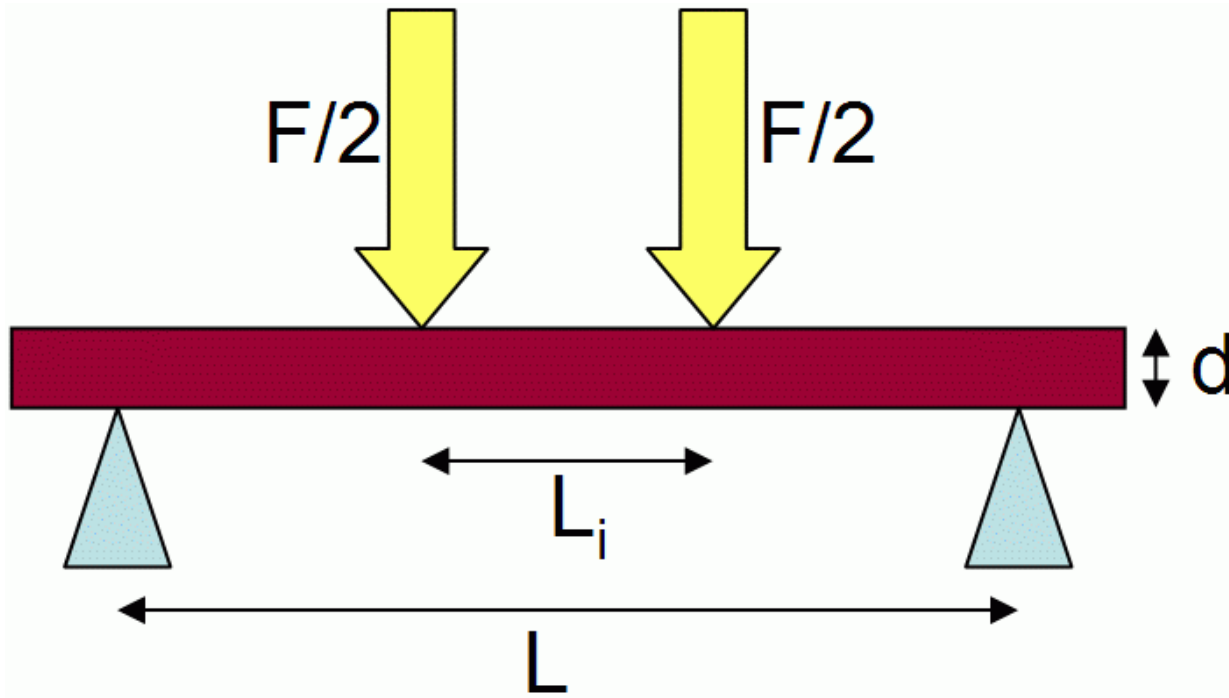
- Determine elastic modulus according to:



$$E = \frac{F}{\delta} \frac{L^3}{4bd^3} = \frac{F}{\delta} \frac{L^3}{12\pi R^4}$$

rect. cross section circ. cross section

4-point Flexure Test



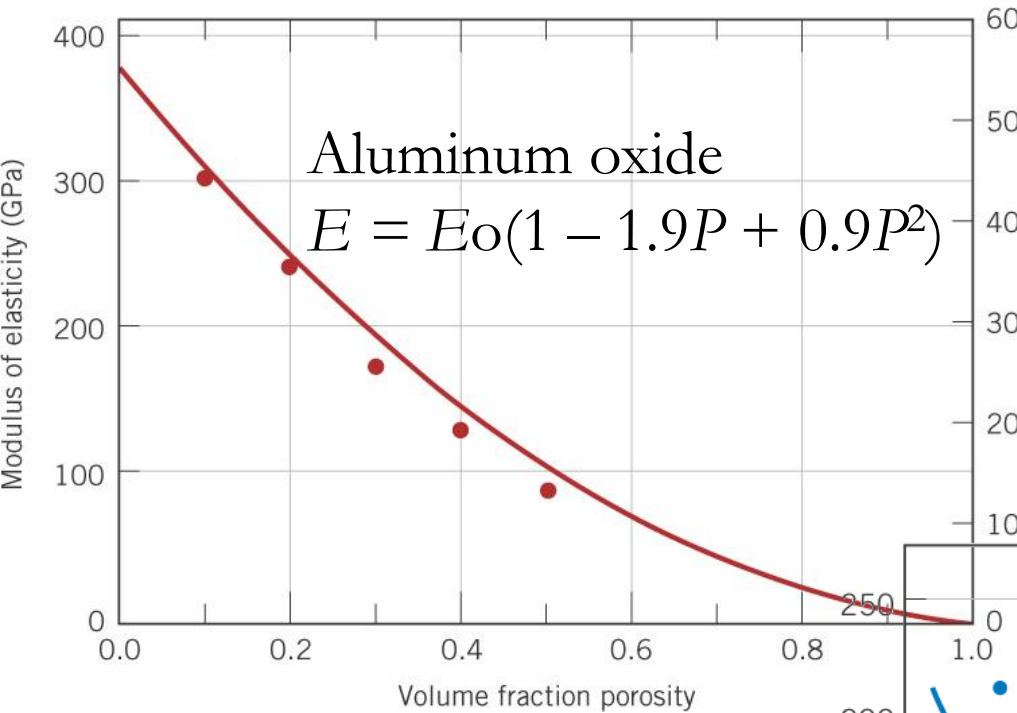
For a rectangular sample where the loading span is $1/3$ of the support span: $\sigma = \frac{FL}{bd^2}$

if the loading span is $1/2$ of the support span (i.e. $L_i = 1/2 L$); $\sigma = \frac{3FL}{4bd^2}$

If neither $1/3$ nor $1/2$ the support span: $\sigma = \frac{3F(L - L_i)}{2bd^2}$

- L_i is the length of the loading (inner) span, F is the load (force) at the fracture point
- L is the length of the support (outer) span, b is width, d is thickness

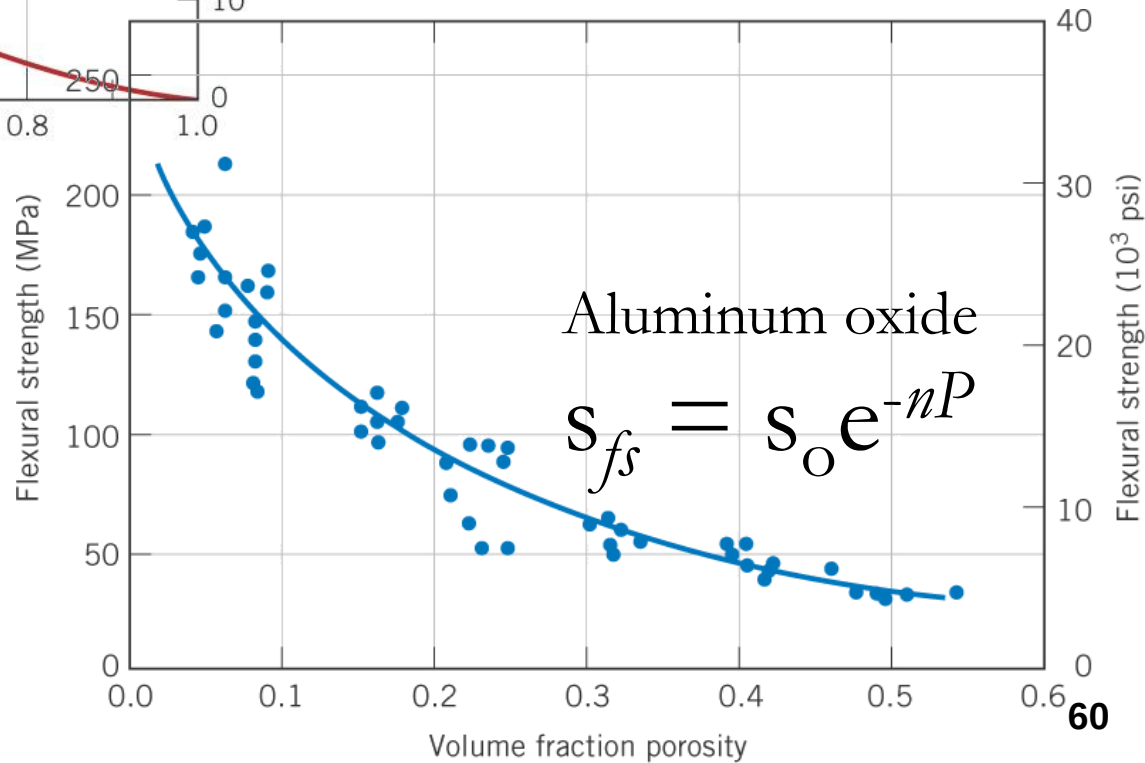
Influence of Porosity on the Mechanical Properties of Ceramics



- Powder as precursor
- Compaction to desire shape
- Pores or voids elimination incomplete
- Residual porosity remains
- Deleterious influence on elasticity and strength
- Volume fraction porosity P

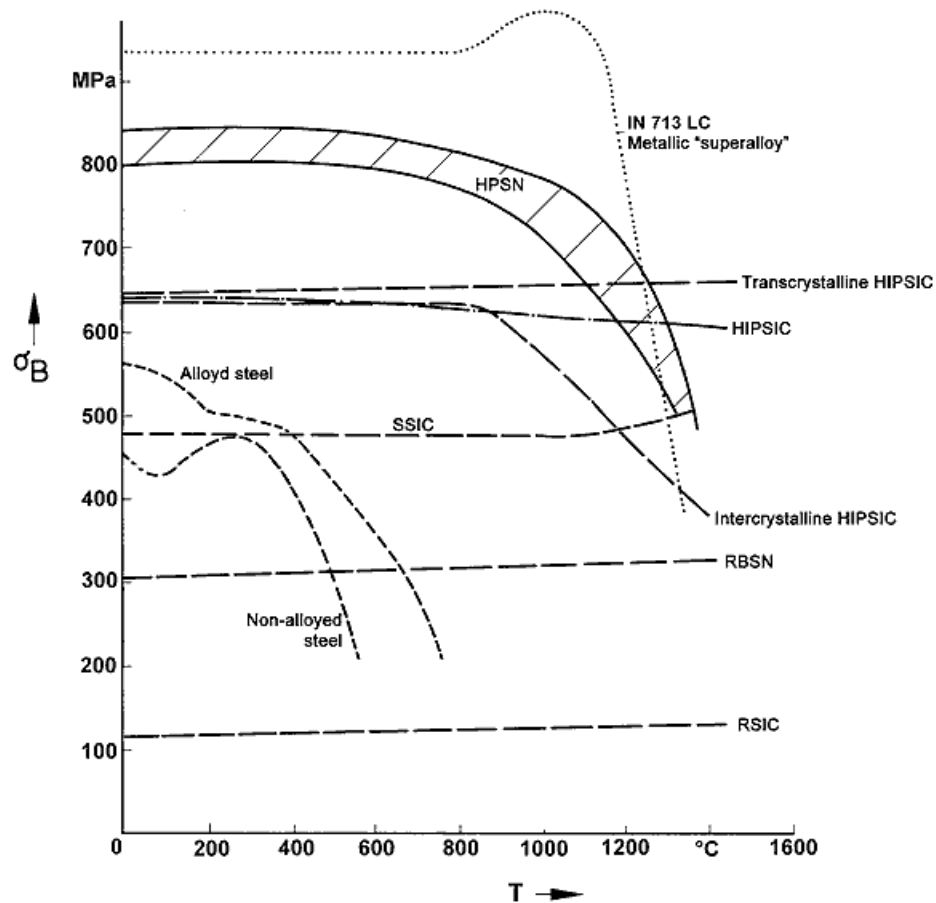
E_0 = modulus of elasticity of the non porous material

- Pores reduce the area
- Pores are stress concentrators
- tensile stress doubles in an isolated spherical pore



Effect of temperature on bending strength

Ceramics are distinguished by their very good high-temperature strength, with maximum working temperatures that even exceed those of metallic superalloys.



Ceramic materials are thus ideal for applications in high-temperature processes.

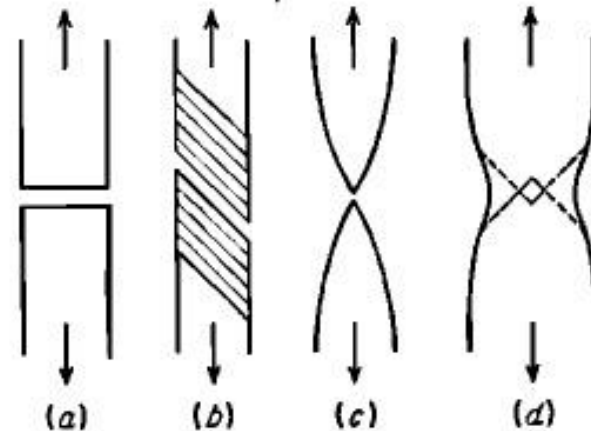
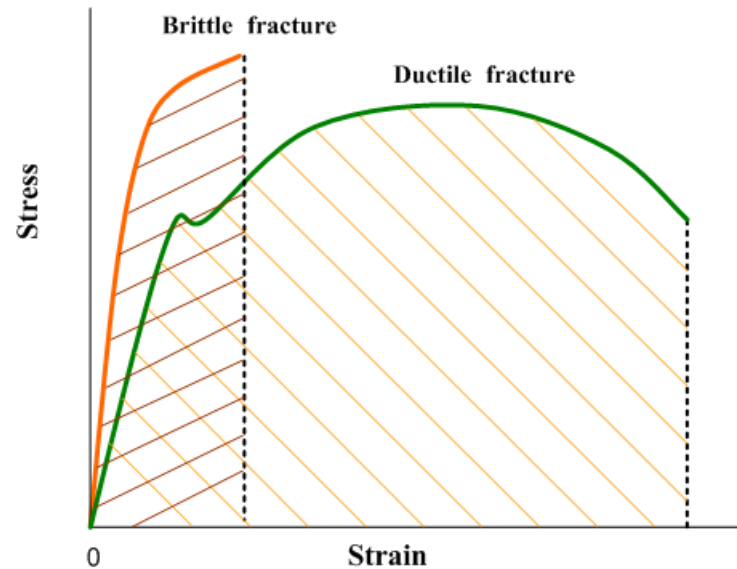
Fracture Behavior

Fracture: Separation of a body into pieces due to stress, at temperatures below the melting point.

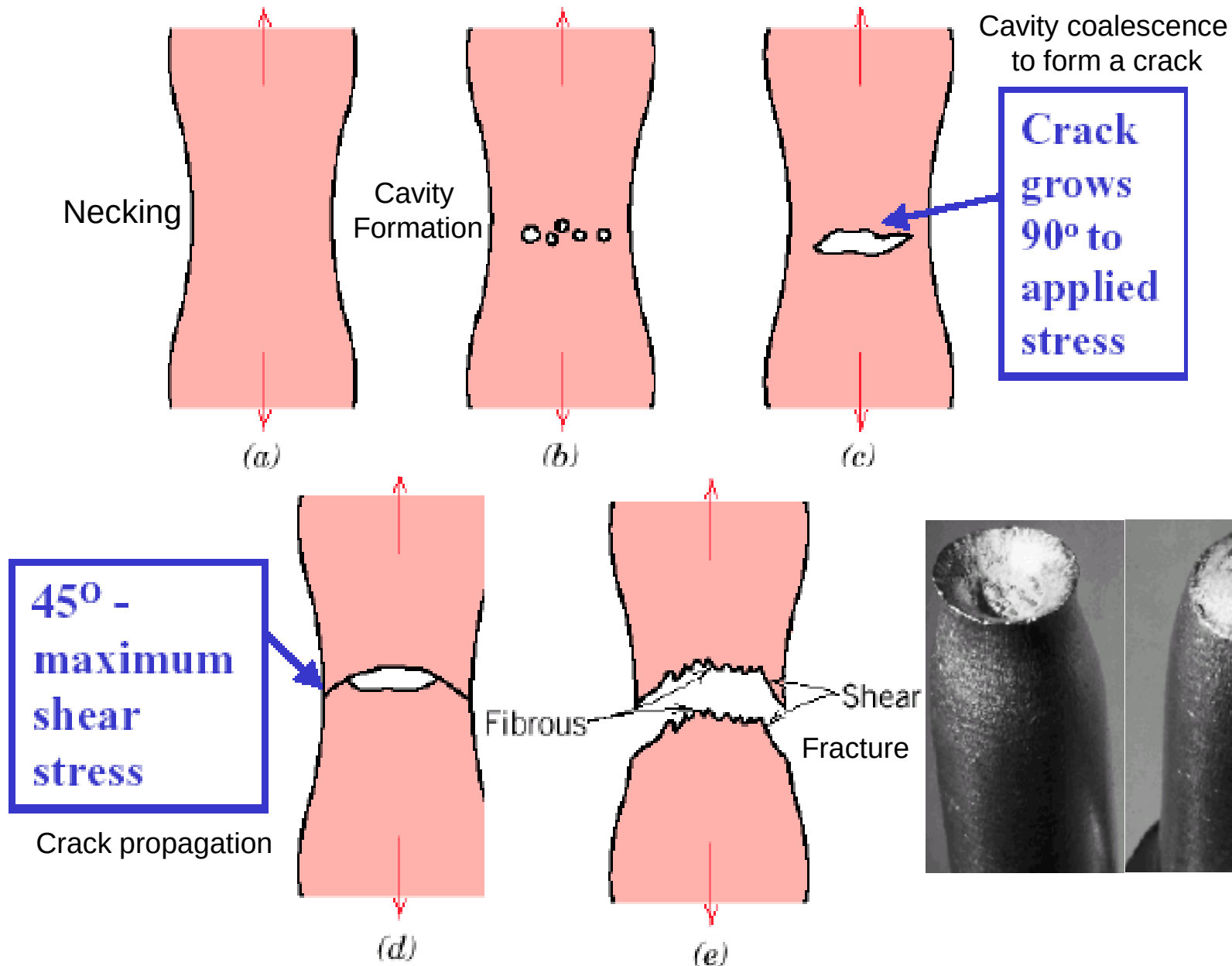
Steps in fracture:

- Crack formation
- Crack propagation

Depending on the ability of material to undergo plastic deformation before the fracture two fracture modes can be defined - ductile or brittle

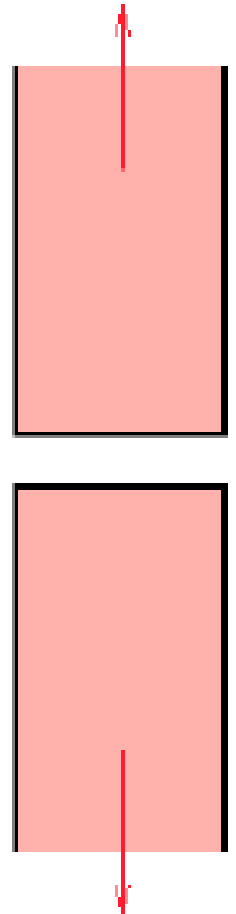
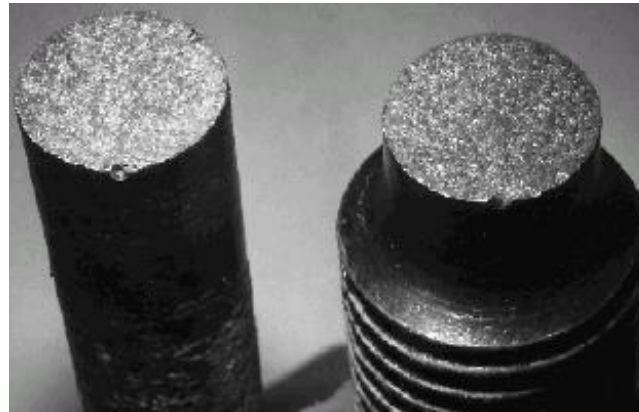


Ductile Fracture



Brittle Fracture

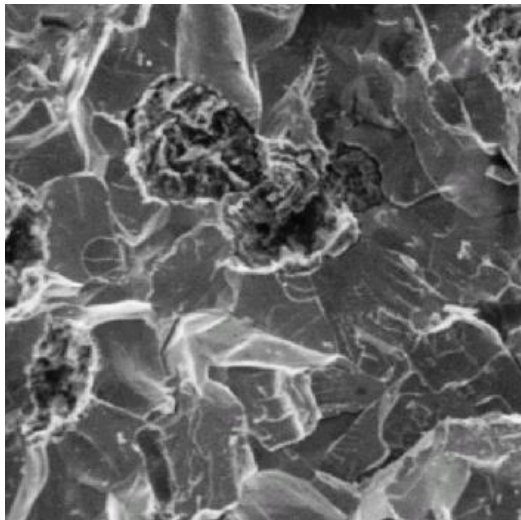
- No appreciable plastic deformation;
- Crack propagation is very fast;
- Crack propagates nearly perpendicular to the direction of the applied stress;
- Crack often propagates by cleavage – breaking of atomic bonds along specific crystallographic planes (cleavage planes);



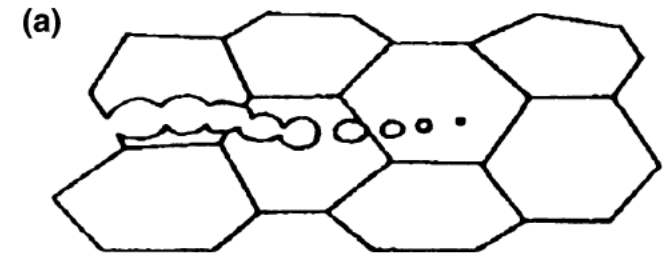
Brittle Fracture

A. **Transgranular fracture:** Fracture cracks pass through grains. Fracture surface have faceted texture because of different orientation of cleavage planes in grains.

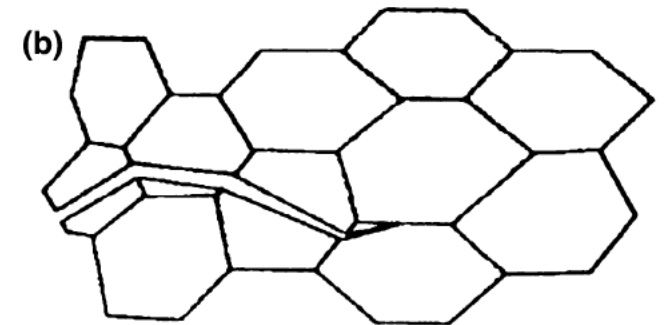
B. **Intergranular fracture:** Fracture crack propagation is along grain boundaries (grain boundaries are weakened or embrittled by impurities segregation etc.)



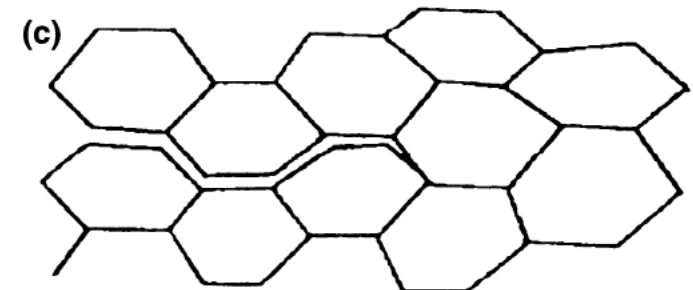
Brittle fracture and crack propagation in the polycrystalline ceramic materials



Coalescence of microcavities

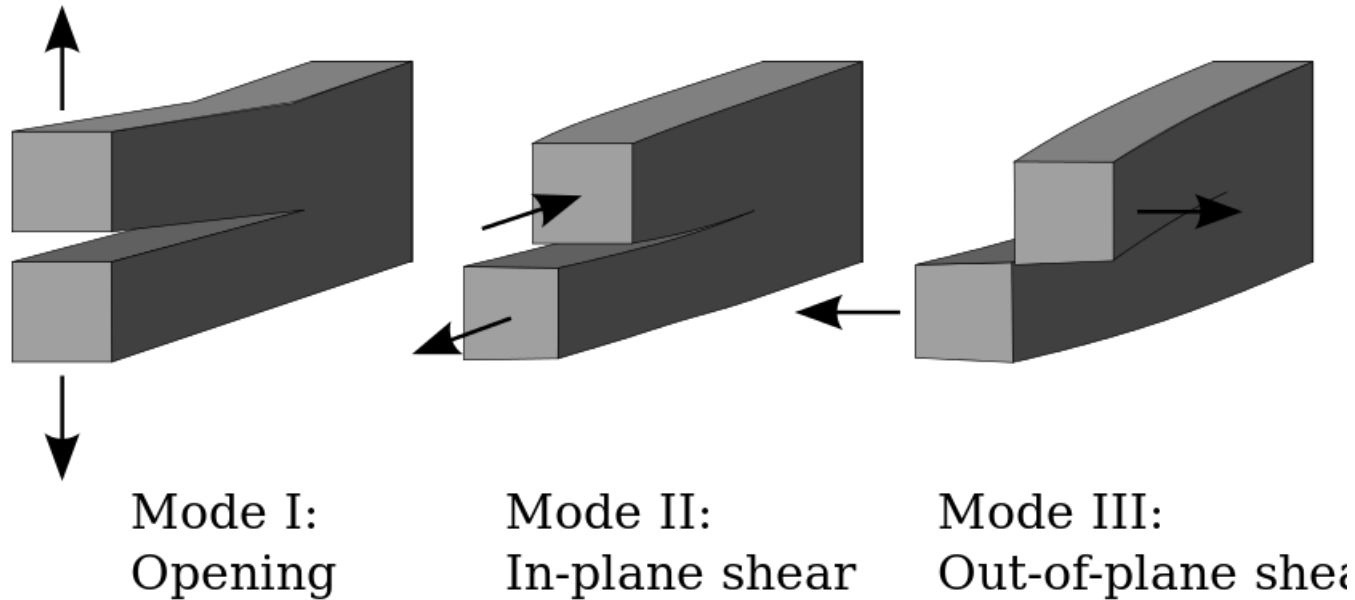


Transgranular crack propagation



Intergranular crack propagation

Fracture Mode



Mode I fracture – Opening mode (a tensile stress normal to the plane of the crack);

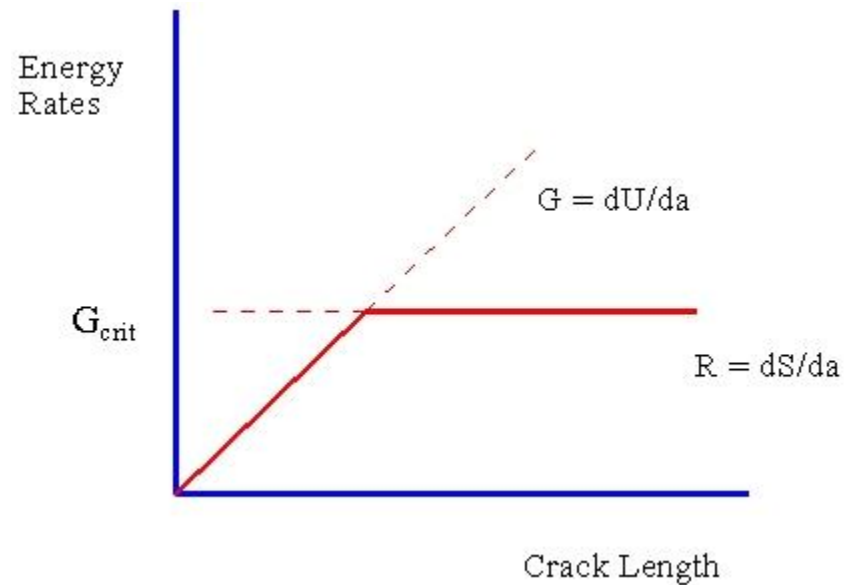
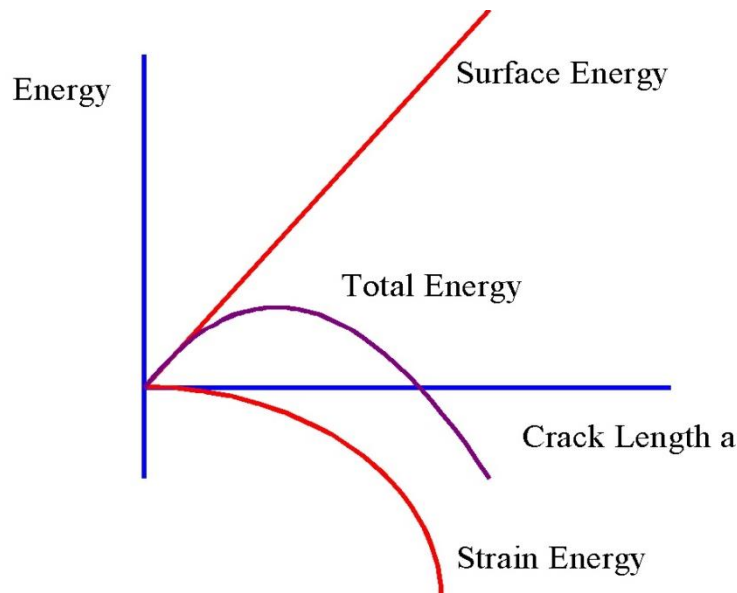
Mode II fracture – Sliding mode (a shear stress acting parallel to the plane of the crack and perpendicular to the crack front);

Mode III fracture – Tearing mode (a shear stress acting parallel to the plane of the crack and parallel to the crack front).

Griffith Theory of Brittle Fracture

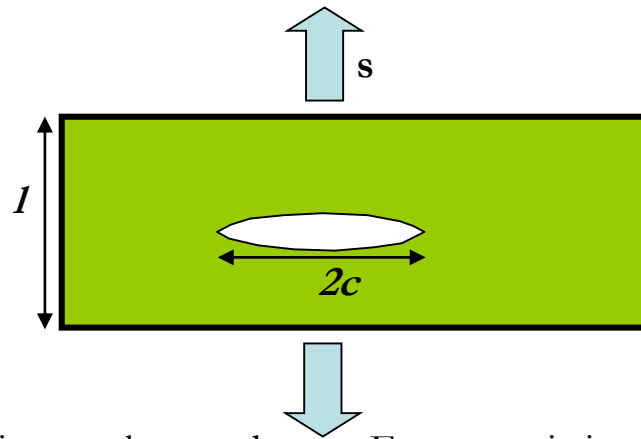
- The stress needed to fracture bulk glass is around 100 MPa ;
- The theoretical stress needed for breaking atomic bonds is approximately 10,000 MPa. Why ???

The Griffith theory states that a crack will propagate when the reduction in potential energy that occurs due to crack growth is greater than or equal to the increase in surface energy due to the creation of new free surfaces.



Fracture Strength I

The maximum strength is based on the assumption that a body fails by simultaneous separation of all bonds, actual fracture in brittle material however occur by enlargement of preexisting flaws (cracks).



Energy promoting crack growth
= elastic energy release stored at
the crack tip

$$-\frac{\pi c^2 \sigma^2 l}{E}$$

Energy resisting crack growth
= surface energy

$$4\gamma cl$$

The energy U is the sum of the strain energy and surface energy terms,

$$U = -\frac{\pi c^2 \sigma^2 l}{E} + 4\gamma cl$$

Fracture Strength II

Above a certain crack size, it is energetically downhill for the crack to propagate ($dU/dc < 0$), and catastrophic failure occurs. The critical stress for a certain crack size c already present in the material is given by:

$$\frac{d}{dc} \left(-\frac{\pi c^2 \sigma_c^2 l}{E} + 4\gamma c l \right) = 0 \longrightarrow \frac{\pi c \sigma_c^2}{E} = 2\gamma \longrightarrow \sigma_c = \left(\frac{2E\gamma}{\pi c} \right)^{\frac{1}{2}} = \frac{K_{IC}}{\sqrt{\pi c}}$$

Energy Release Rate G R Crack Resistance

The much lower than theoretically predicted strength of ceramic materials is due to the fact, that it is impossible to manufacture perfect ceramic parts which contain no cracks. A second problem is, that the number and the size of cracks present in a ceramic part are usually not known.

$$\frac{\pi c \sigma^2}{E} \geq \frac{\pi c \sigma_c^2}{E} = G_c = R$$

K_{IC} is called the fracture toughness for opening mode loading, e.g. tensile stresses perpendicular to the crack axis. Fracture propagation prevention = toughening through microstructural adjustments

Material	Fracture toughness (MPa /m ²)
Al ₂ O ₃ (s,p)	4.5, 3.5 - 4
MgO(s)	1
SiC(p)	4 - 6
Glasses	0.7 - 2
Aluminum	35 -45
Steel	40 - 60

Stress Intensity Factor and Fracture Toughness

From linear elastic theory, Irwin showed that the resulting stresses vary with distance r and angle θ (in polar coordinates) away from the crack tip, and take on the form:

$$\sigma_{ij} = \frac{K}{\sqrt{2\pi r}} f_{ij}(\theta) + \dots$$

K is the *stress intensity factor*, more generally we can rewrite Griffith's result in the terms of Stress intensity factor as

$$K = \sigma \sqrt{2\pi c} f\left(\frac{c}{w}\right)$$

The critical stress intensity factor, causing fracture propagation is highly material dependent, and is termed as *fracture toughness*, can be determined experimentally by measuring the stress necessary for crack extension for a particular specimen/crack symmetry.

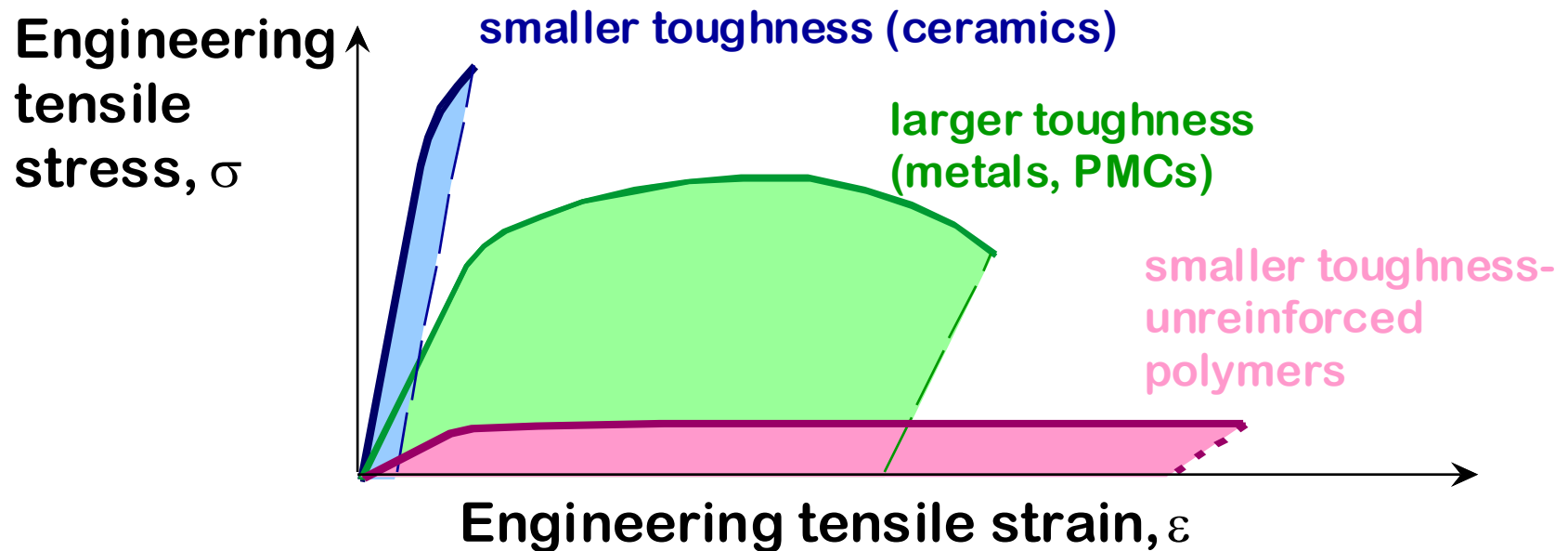
$$K_c = \sigma_c \sqrt{\pi c}$$

Critical Energy Release Rate

$$G_c = \frac{K_c^2}{E}$$

Fracture Toughness

- Ability to absorb energy up to fracture

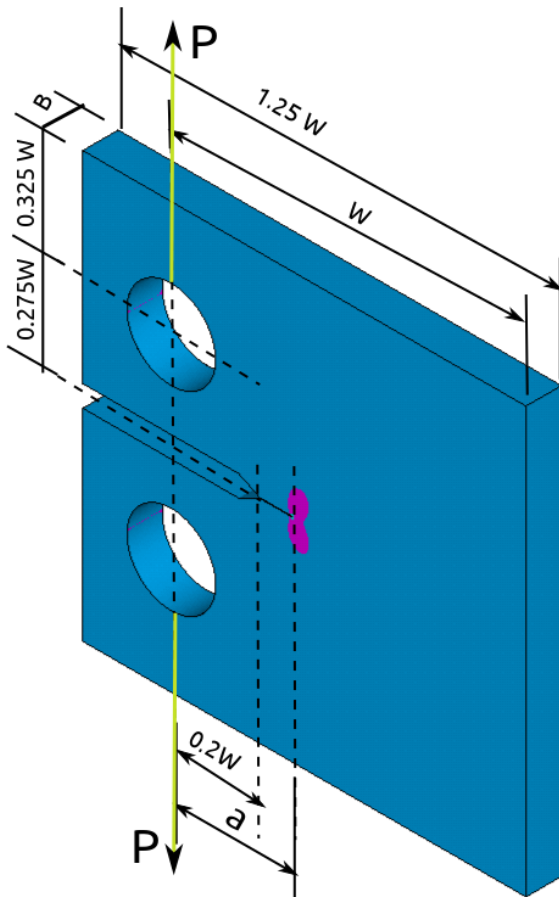


Usually ductile materials are tougher than brittle ones
Areas below the curves

Fracture Toughness Testing

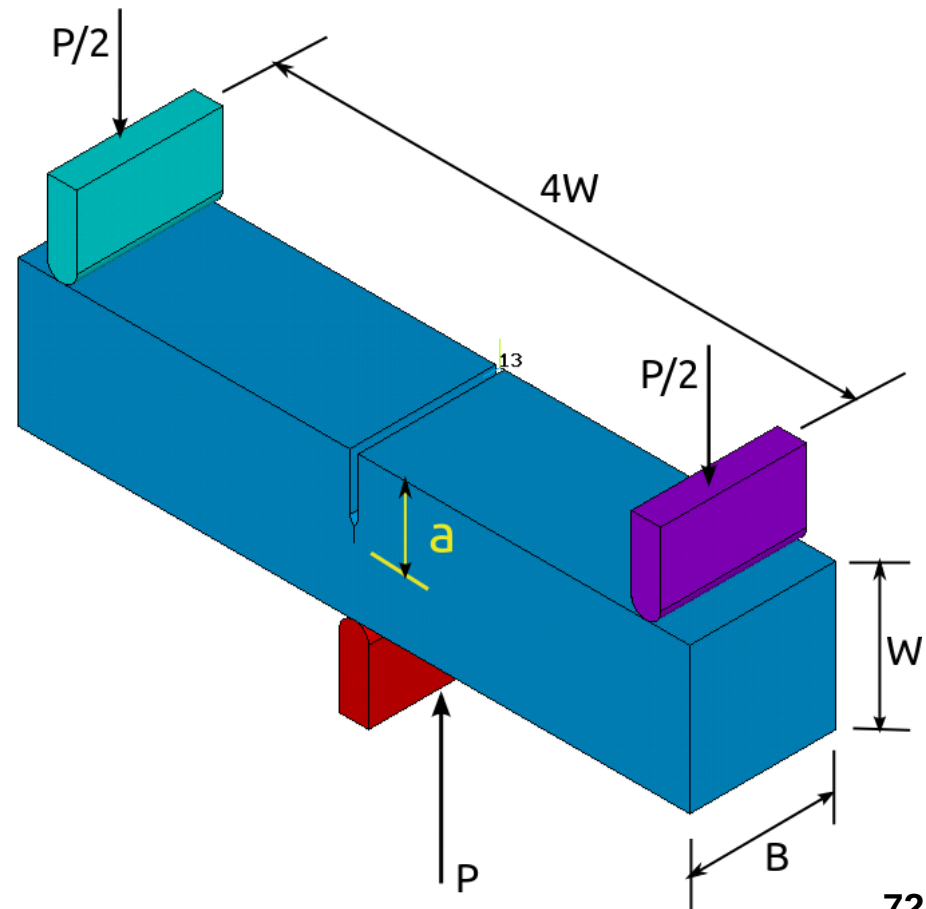
Compact tension specimen

$$K_I = \frac{P}{B} \sqrt{\frac{\pi}{W}} \left[16.7 \left(\frac{a}{W} \right)^{1/2} - 104.7 \left(\frac{a}{W} \right)^{3/2} + 369.9 \left(\frac{a}{W} \right)^{5/2} - 573.8 \left(\frac{a}{W} \right)^{7/2} + 360.5 \left(\frac{a}{W} \right)^{9/2} \right]$$



Single edge notch bending specimen

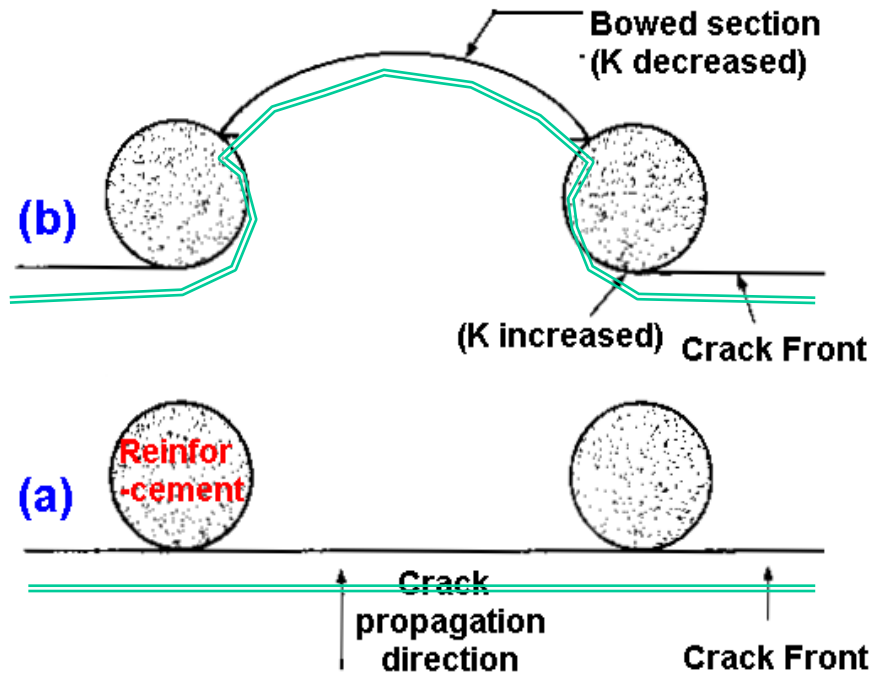
$$K_I = \frac{4P}{B} \sqrt{\frac{\pi}{W}} \left[1.6 \left(\frac{a}{W} \right)^{1/2} - 2.6 \left(\frac{a}{W} \right)^{3/2} + 12.3 \left(\frac{a}{W} \right)^{5/2} - 21.2 \left(\frac{a}{W} \right)^{7/2} + 21.8 \left(\frac{a}{W} \right)^{9/2} \right]$$



Toughening mechanisms- Introduction

- ✦ There are many different toughening mechanisms.
- ✦ One or more toughening mechanisms may operative in a composite.
- ✦ The effectiveness of the toughening mechanisms depends on:
 - 📄 Size, morphology and volume fraction of the reinforcement;
 - 📄 Interfacial bond;
 - 📄 Properties (e. g. mechanical, thermal expansion) of the matrix and the reinforcement;
 - 📄 Phase transformation
 - 📄

Toughening mechanisms- crack bowing



Crack bowing toughening mechanism (a) crack approaches (b) crack bowed at reinforcement

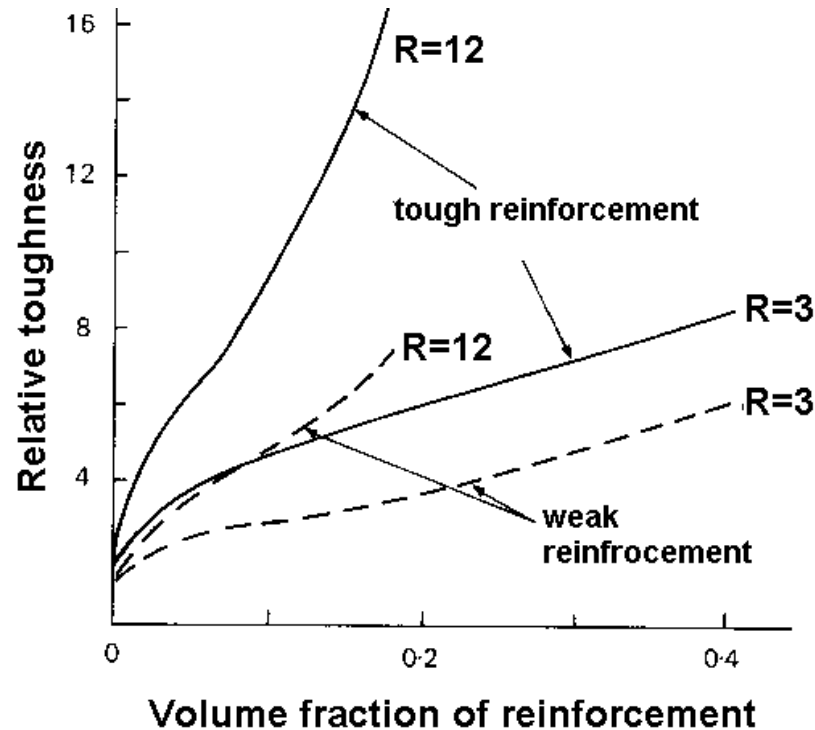
Crack bowing

- ✦ (a) Crack approaches to reinforcements.
- ✦ (b) the crack bowed under stress to form a nonlinear crack front.
 - 📄 Decrease in the **stress intensity K** along the bowed section in the matrix
 - 📄 Increase in the stress intensity K at the reinforcement
 - 📄 K reached to the fracture toughness of the reinforcement → the reinforcement breaks

✦ Bowing needs more energy to increase toughness

Toughening mechanisms

- crack bowing

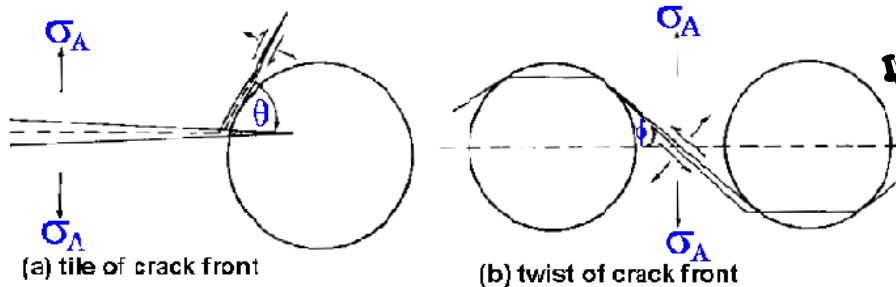


Theoretical plots: effects of vol. fraction, toughness & aspect ratio (R) of a disc-shaped reinforcement on crack bowing toughening

- Crack bowing toughening ↑
- with ↑ the volume fraction of reinforcement (more reinforcements)
 - with ↑ aspect ratio of the reinforcement
 - with ↑ the properties of reinforcement

Toughening mechanisms

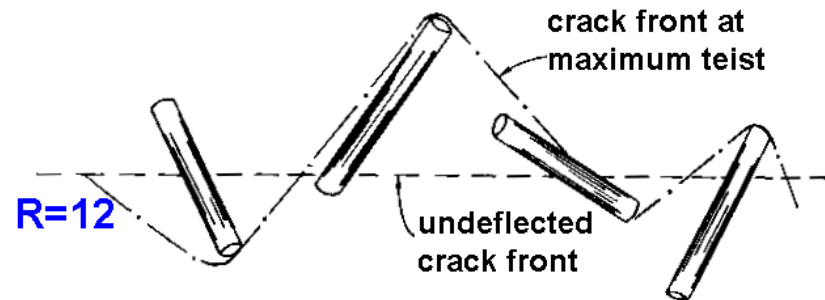
- Crack deflection



Crack deflects and becomes non-planar, due to interaction between the reinforcement and crack front.

📄 (a) Tilt of crack front

📄 (b) Twist of crack front



$R=5$

(c) effect of aspect ratio on twist

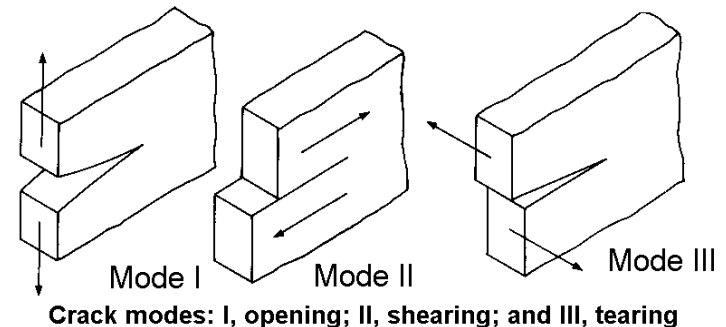
Crack deflection toughening

📌 There are 3 crack modes

📄 Flat crack propagates in mode I.

📄 Tilt crack in modes I and II

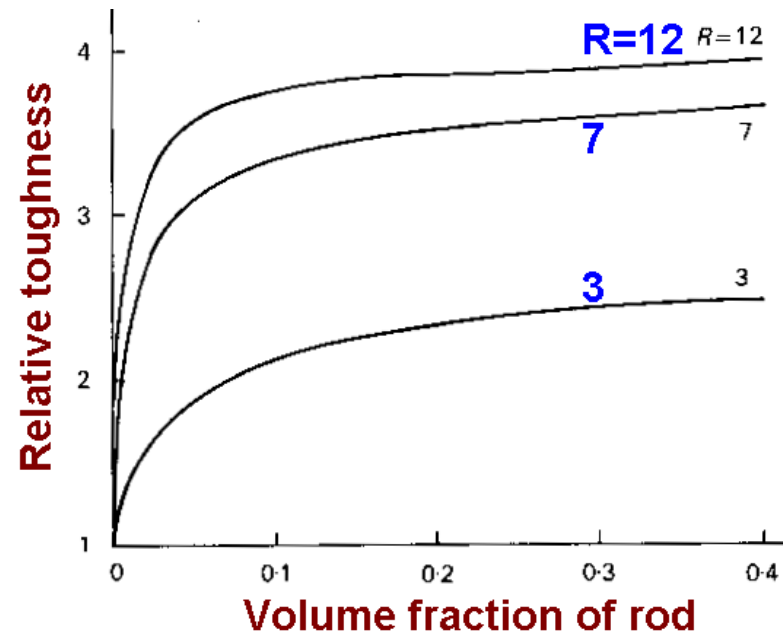
📄 Twist crack in modes I and III



Toughening mechanisms

- Crack deflection

- 1 Deflection occurs when the interaction of the crack with the residual stress fields due to differences in the thermal expansion coefficients or elastic moduli between the matrix and reinforcement.
- 1 Deflection toughening ↑:
 - 📄 with ↑ volume fraction of reinforcement
 - 📄 With ↑ aspect ratio of reinforcement
 - 📄 Dominated by twisting rather than tilting of the crack



Theoretical plot the effects of volume fraction & aspect ratio of a rod-shaped reinforcement on crack deflection toughening

Toughening mechanisms

- Debonding toughening

✚ **Debonding:** Reinforcement fibre separates from matrix.

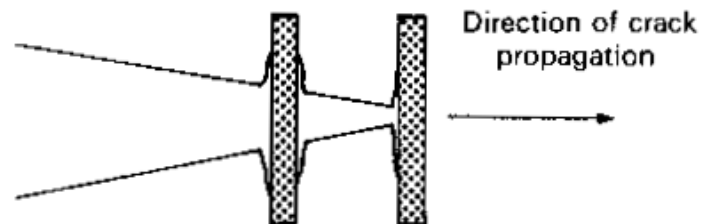
✚ **Debonding toughening:** New surface in the composite require energy in debonding.

✚ **Debonding toughening** ↑

📄 Weak interface of matrix and reinforcement

📄 Strong reinforcement

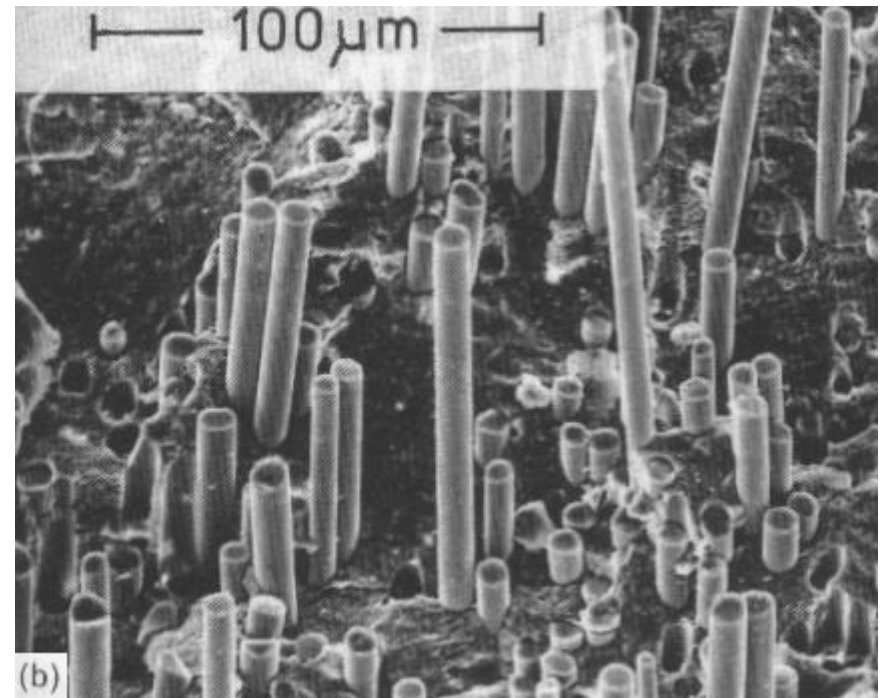
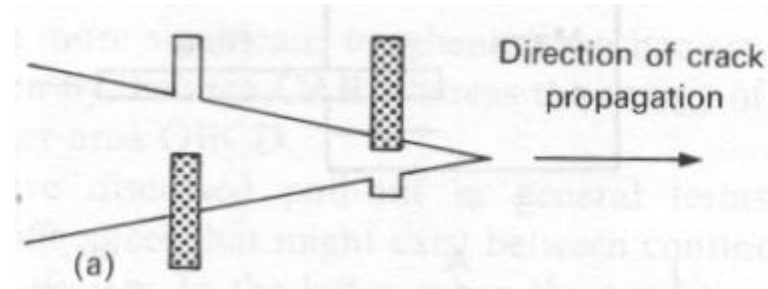
📄 Large volume fraction of reinforcement.



Toughening mechanisms

- Pull-out toughening

- ✦ Pull out a fibre
 - 📄 Pull-out
 - 📄 Debonding
 - 📄 Fibre fracture for long fiber
- ✦ The normal frictional forces have to be overcome during pull-out.
- ✦ The maximum pull-out length of a fibre is $\frac{1}{2}$ the critical length (l_c).
- ✦ If embedded length is greater than l_c fibre will **break**.



Toughening mechanisms

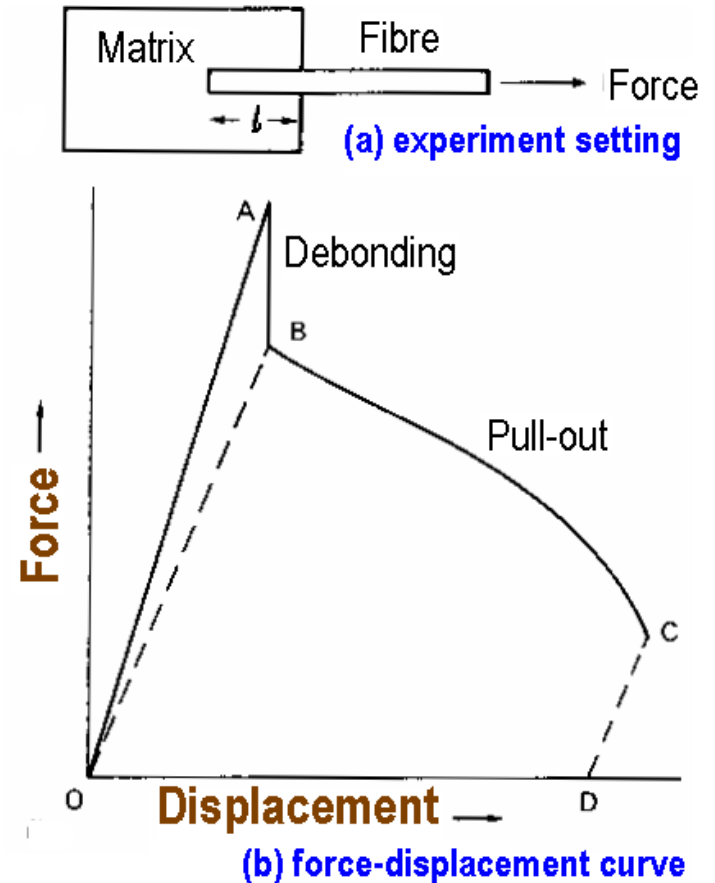
- Pull-out toughening

- Maximum work to pull out a fibre is

$$W_{pull-out}(\max) = \frac{\pi D^2 \sigma_{Tf} l_c}{16} \text{ per fibre}$$

Where D , l_c and σ_{Tf} are diameter, critical length and fracture strength of the fibre, respectively.

- The energy of pull-out is greater than that of debonding.

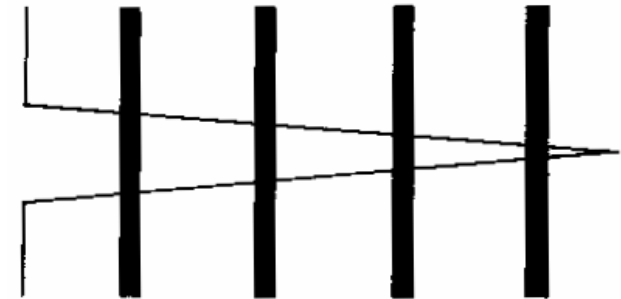


Pulling a fibre out of the matrix

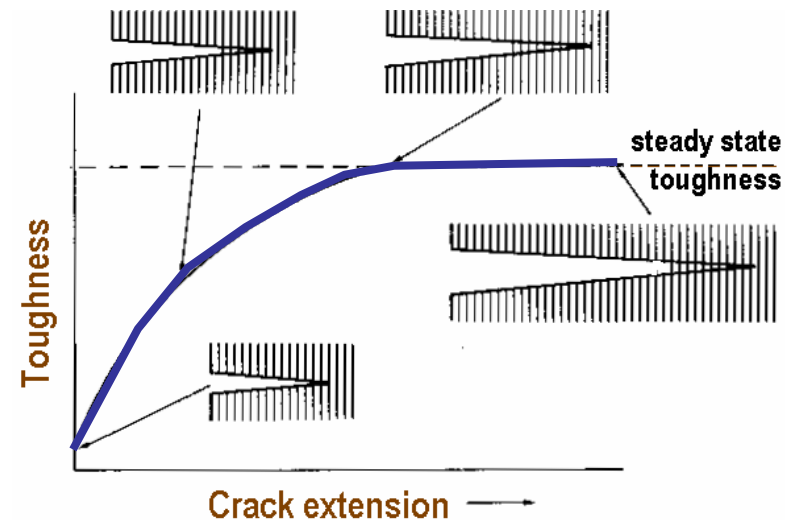
Toughening mechanisms

- Fibre bridging toughening

- 📌 Fibre bridging: some fibres debonds but not break.
- 📌 Fibres carry out stresses under load.
- 📌 Reduce the stresses at crack tip and hinder crack propagation.
- 📌 Toughness-crack extension curve:
 - 📄 Toughness increase with crack extension at initial cracking
 - 📄 Constant toughness maintains when crack reaches to critical value.



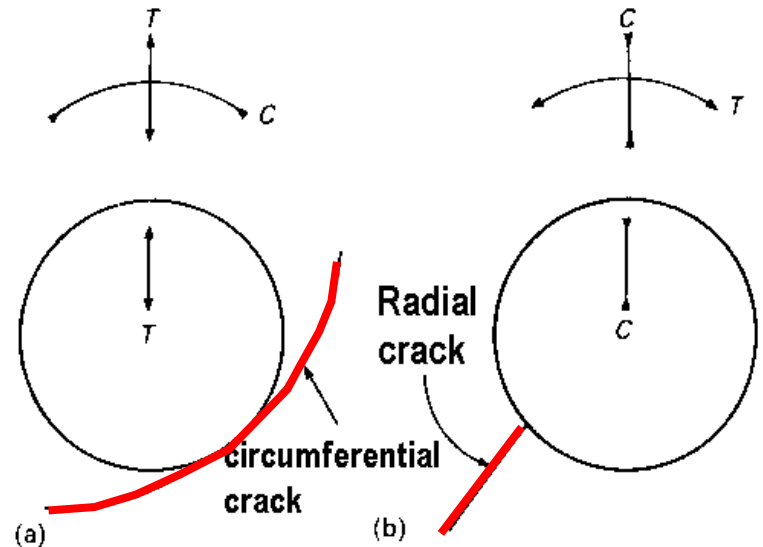
bridging toughening due to fibres



Crack resistance curve (R-curve) due to fibre bridging

Toughening Mechanisms- Microcrack toughening

- ✦ Thermal stress forms between matrix and reinforcement during cooling, due to difference in coefficient of thermal expansion (α).
- ✦ $\alpha_f > \alpha_m$
 - 📄 Tangential compressive and a radial tensile stresses in matrix
 - 📄 Circumferential crack forms under high tensile stress.
- ✦ $\alpha_f < \alpha_m$
 - Tangential tensile stress in matrix cause radial crack under high tensile stress.



Stress distribution and microcrack formation around spherical particles when (a) $\alpha_f > \alpha_m$, (b) $\alpha_f < \alpha_m$, C and T for compressive and tensile stresses

Toughening mechanisms

- microcrack toughening

- ✦ The toughness of a materials can be enhanced by the presence of microcracks, due to crack blunting, branching and deflection.
- ✦ The microcrack toughening is effective on the limited density and size of cracks.
- ✦ Toughness of materials increases and strength decreases in the microcrack toughening.

Toughening mechanisms

- Transformation toughening

✚ Metastable *tetragonal*-ZrO₂ at room temperature in ZTA

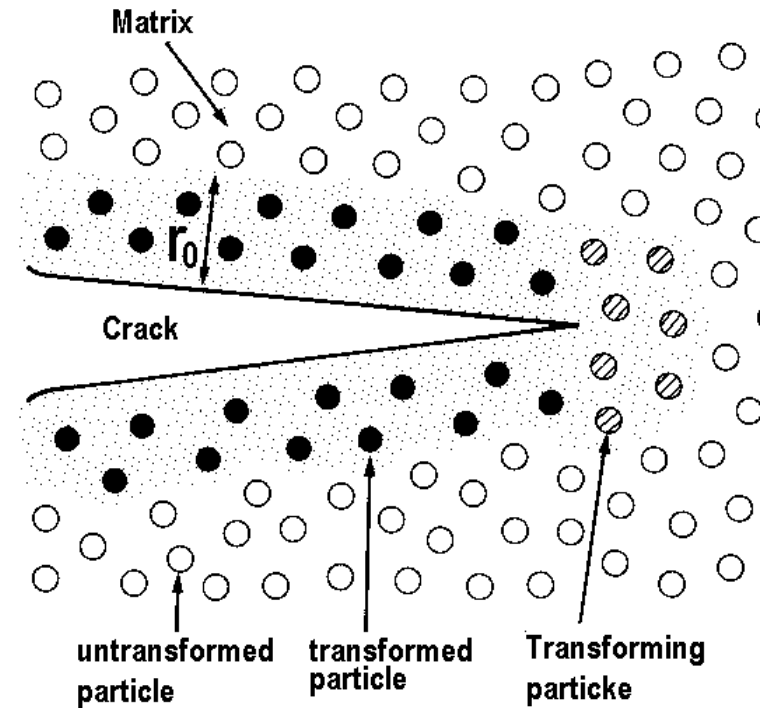
✚ **transformation toughening:** ZrO₂ particles at a crack tip will transfer to *monoclinic*-ZrO₂ under stress. Energy is absorbed ahead of the primary crack owing to the transformation.

✚ Giving an increase in toughness

$$\Delta K_{TT} = 0.3 v_{zirc} \Delta \epsilon E_m r_o^{1/2}$$

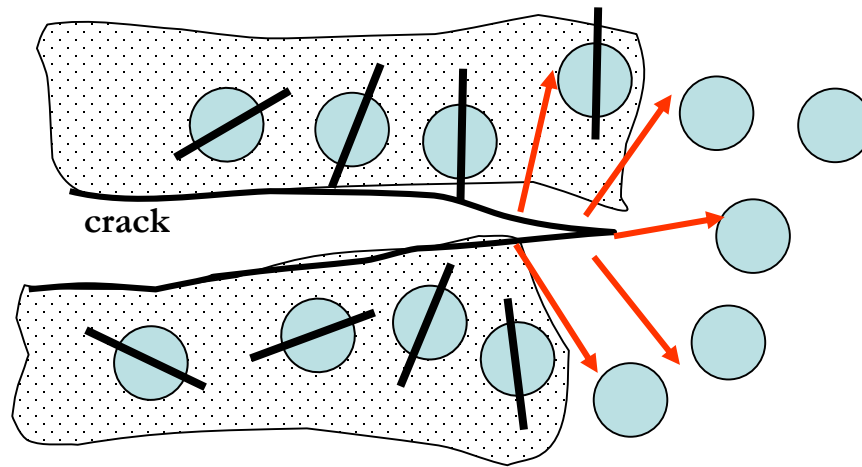
Where v_{zirc} is the volume fraction of metastable particles; $\Delta \epsilon$ is unconstrained strain accompany the transformation; E_m is young's alumina matrix and r_o is the width of zone in the crack.

✚ Strength and toughness of materials increase at same time.

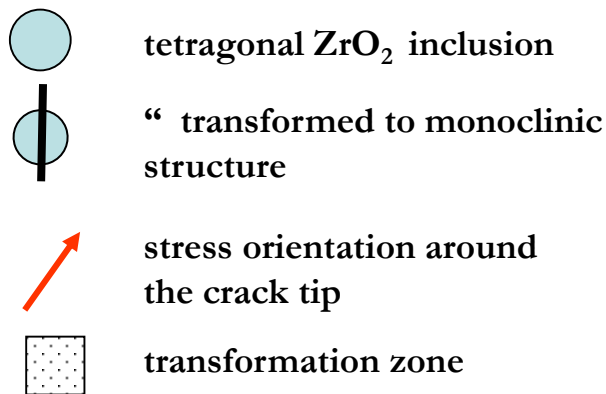


Transformation of metastable particles at the crack tip gives a Zone, of width r_o , of transformed particles

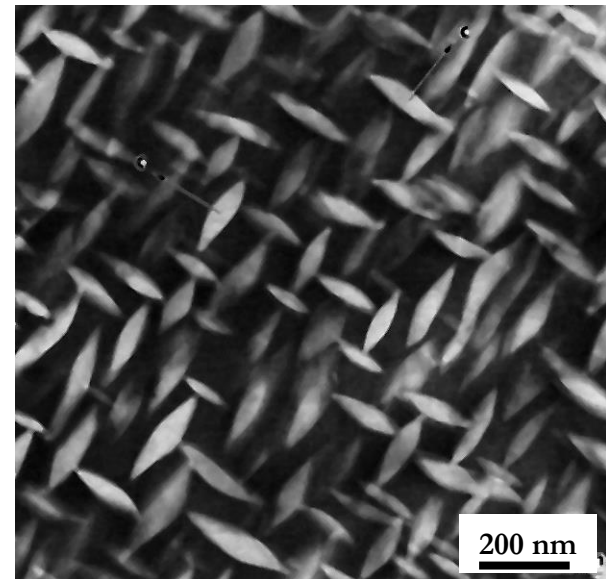
Transformation toughening I



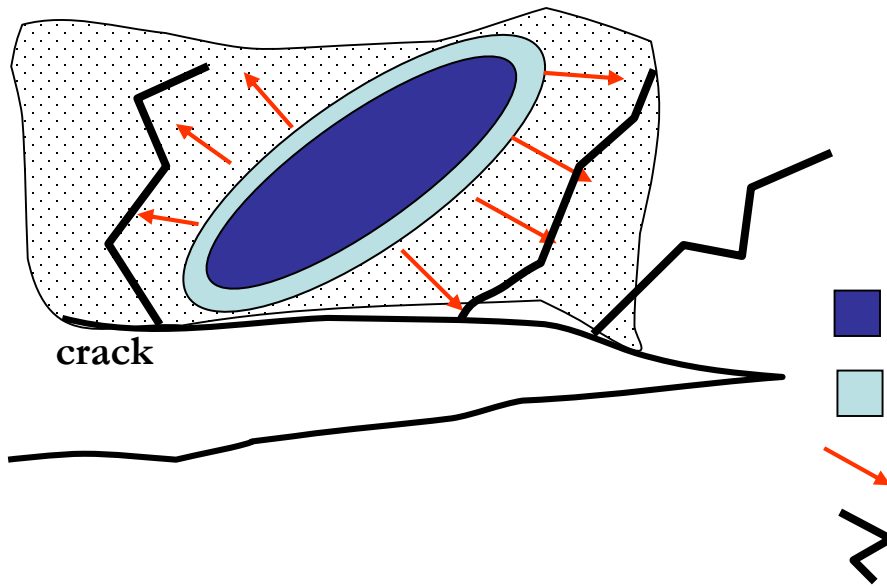
1. The stresses concentrated at the crack tip transform the surrounding tetragonal ZrO_2 inclusions to the monoclinic polymorph. The transformation absorbs fracture energy and slows down crack propagation.



Lense-shaped tetragonal inclusions in a matrix (black) of cubic zirconia (A. Heuer).

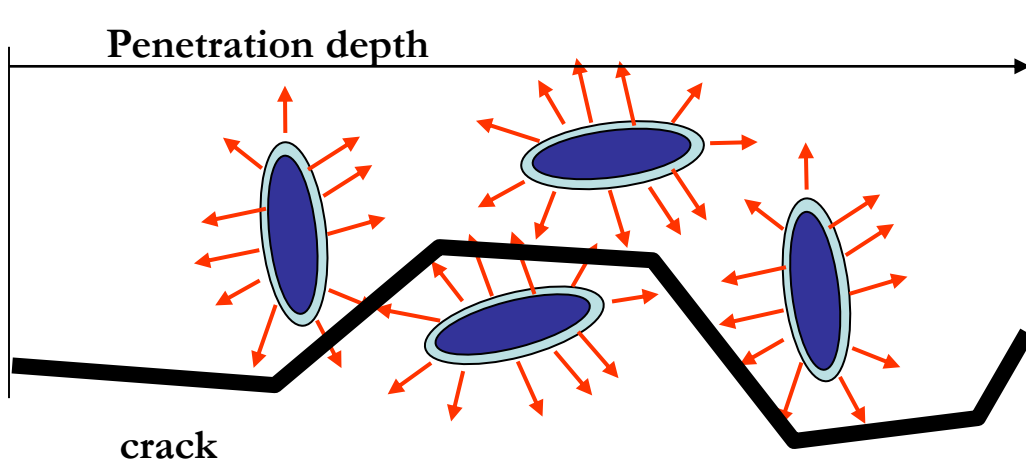


Transformation Toughening II



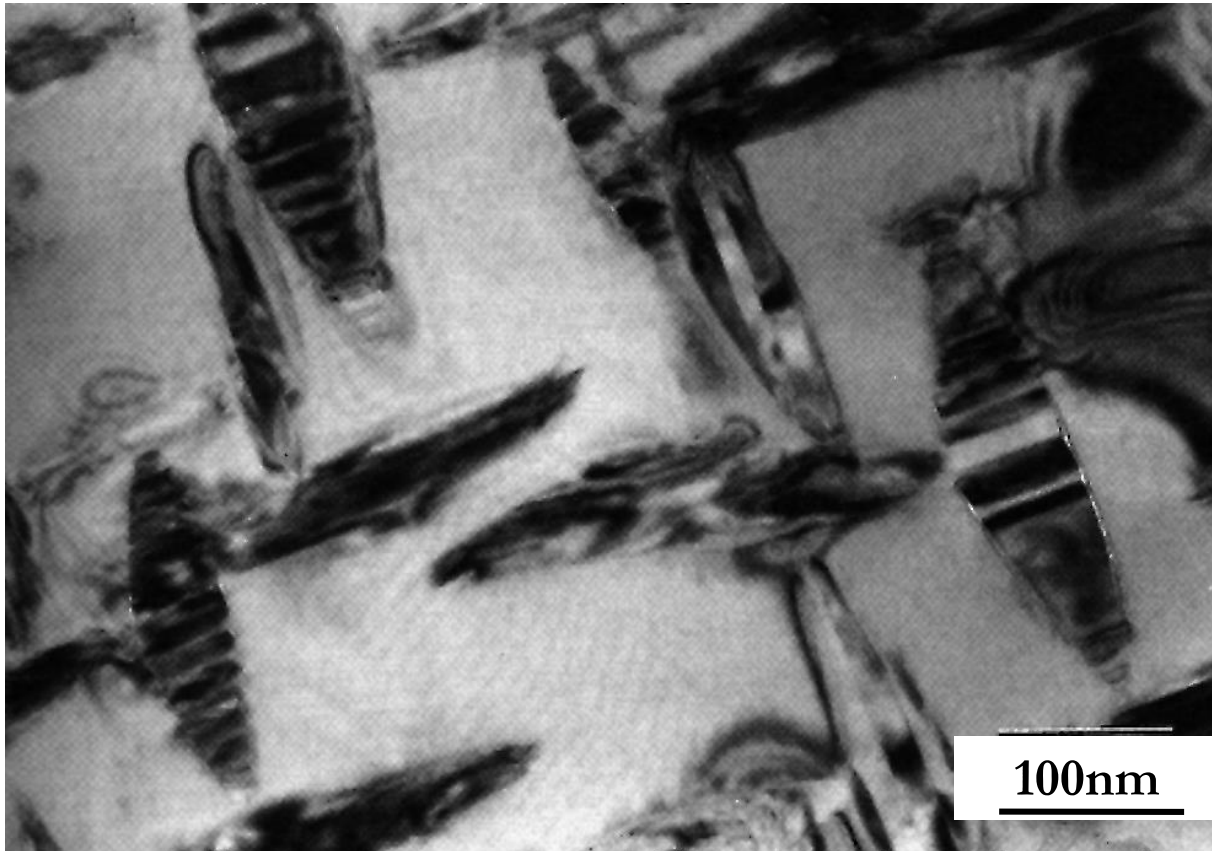
2. Microcracking around the transformed inclusions: The volume stresses resulting from the tetragonal-monoclinic transformation delocalize also the stresses from the crack tip.

volume of the tetragonal zirconia inclusion
volume after transformation to monoclinic
stresses due to the volume increase
microfracture due to the volume stresses



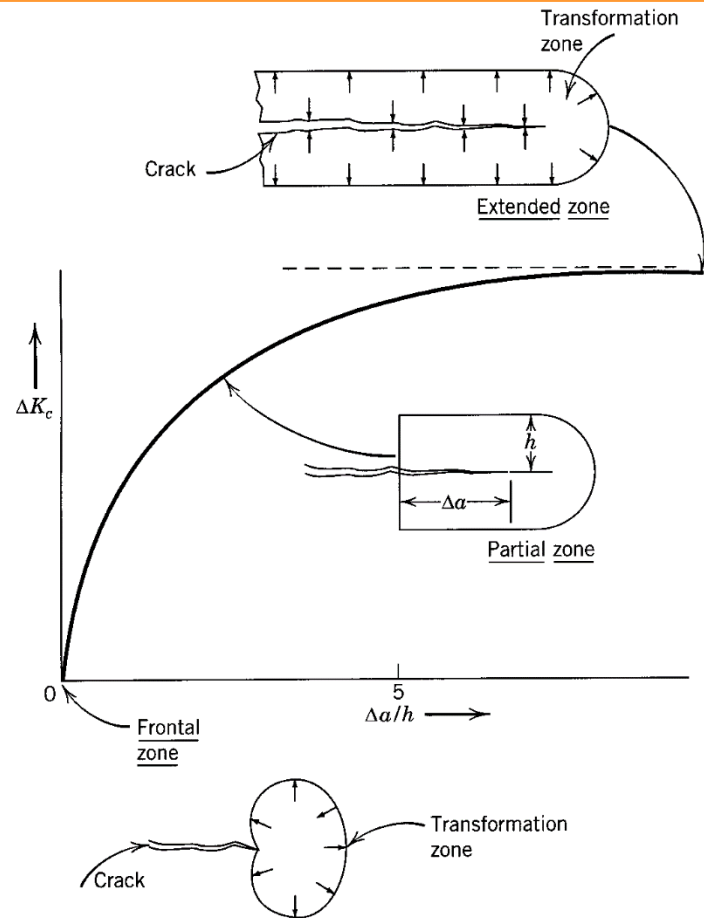
3. Crack deflection due to volume stresses: The deflection of cracks increases the crack surface. The stress release per unit penetration is, therefore, larger than for an inclusion-free zirconia.

Transformation toughening III



Initially tetragonal zirconia inclusion in a cubic zirconia matrix, which are completely transformed to the monoclinic structure. The bands within the inclusions are twin lamellae.

Crack Resistance R-Curve



Predicted crack resistance R -curve for transformation toughening showing asymptotic approach to a limiting K_c as the crack extension.

The R -curve describes the crack resistance as a function of crack extension, $K_R(\Delta c)$, and increases approximately as $\tan^{-1}(\Delta c/h)$, where h is the width of the transformation zone.

The influence R-Curve on fracture

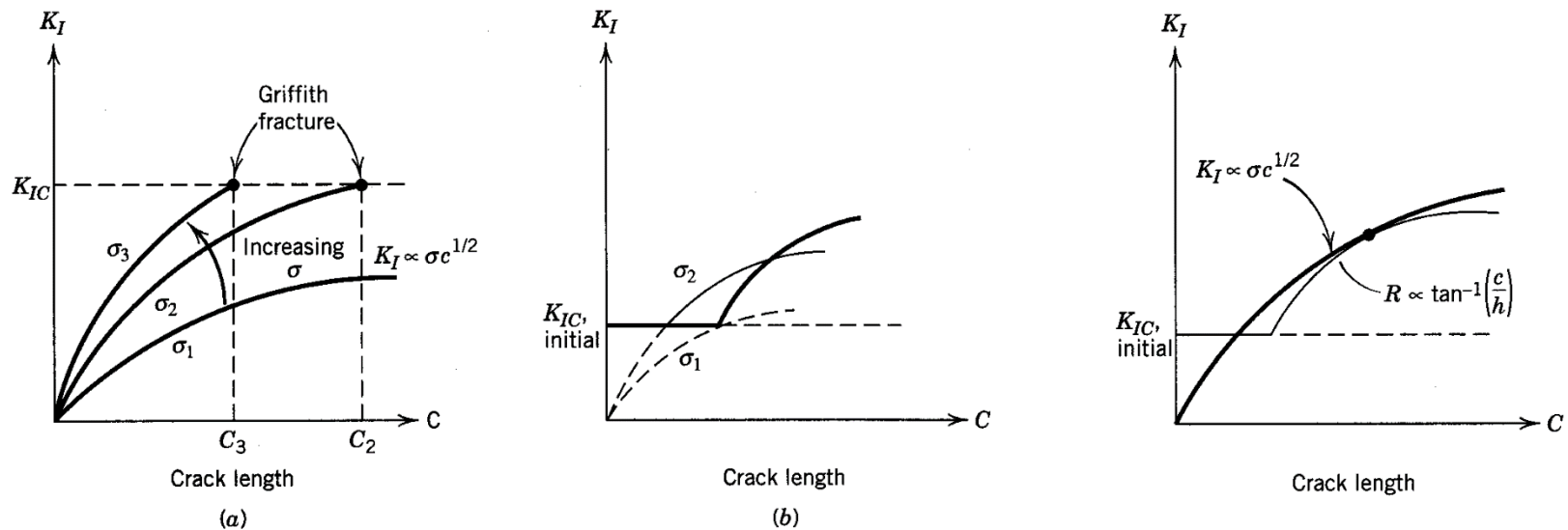
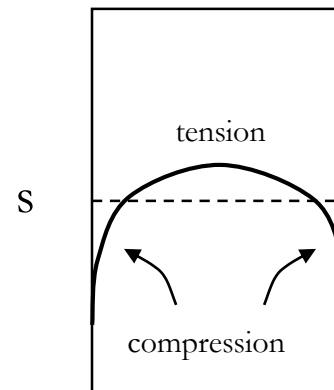
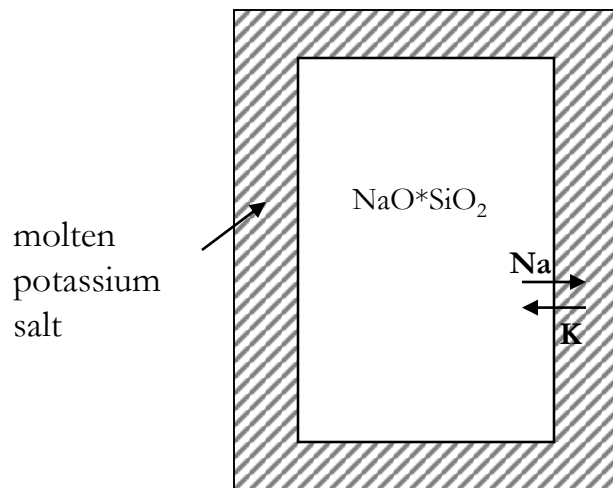
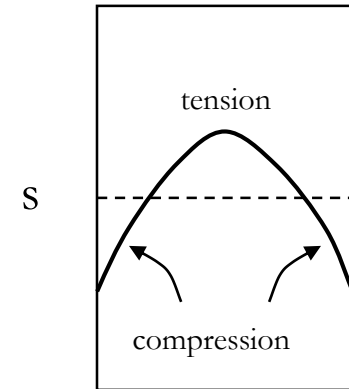


Figure 1. (a) Griffith fracture occurs when the applied K_I ($\propto \sigma c^{1/2}$) reaches K_{IC} ; this corresponds to reaching a critical stress for the flaw size present. Curves represent K_I for increasing values of stress σ . (b) With an R -curve being present, the crack resistance increases beyond its initial value as the crack grows. If the crack resistance rises faster with crack extension than does the applied K_I , the crack can extend stably without failure occurring. (c) Eventually the R -curve flattens, and crack resistance cannot keep up with the applied K_I . The point of tangency between the two curves shown defines the failure criterion.

Strengthening of Ceramics

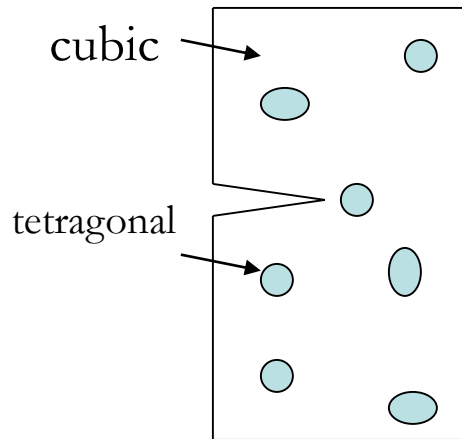
- ✚ Process to eliminate cracks (internal)
- ✚ Polish to eliminate surface cracks
- ✚ Blunt crack tip
- ✚ Anneal (heat treat) to eliminate randomly distributed internal stresses
- ✚ Quench (a silicate glass) to induce compressive stress on surface
- ✚ Ion exchange to induce surface compressive stress



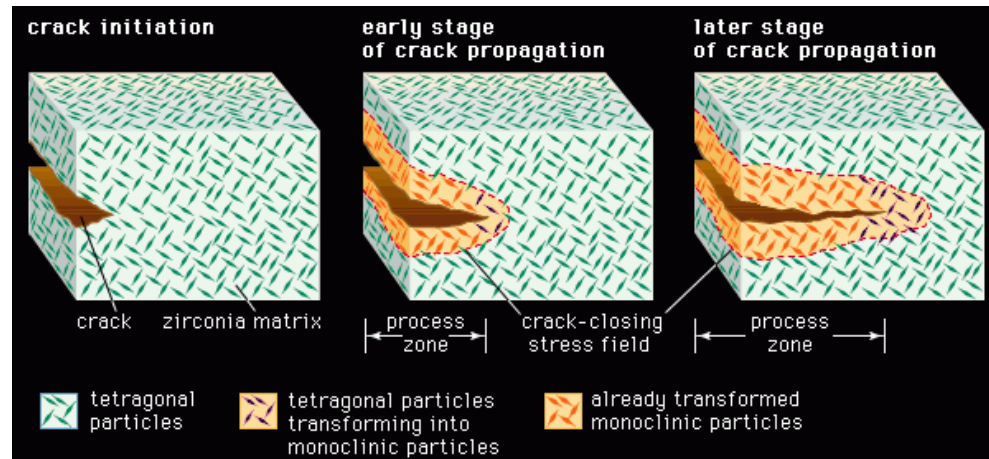
once crack penetrates
compressive region,
material shatters
explosively

Strengthening of Ceramics

- Transformation toughening
- Cool ZrO_2 : cubic $\rightarrow$ tetragonal $\rightarrow$ monoclinic
- Modify with CaO:
cubic $\rightarrow$ tetragonal $\rightarrow$ monoclinic + cubic
- Rapid cooling: tetragonal $\rightarrow$ monoclinic is slow
obtain tetragonal + cubic



crack catalyzes tetragonal $\rightarrow$ monoclinic transition
increase in volume upon transition
DV places compressive stress on crack (closes it)



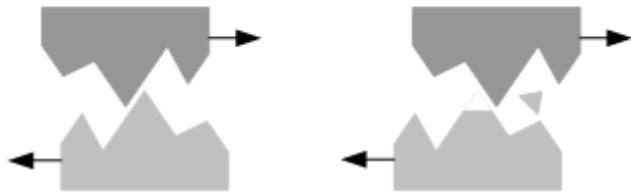
Friction, Wear and Abrasion

Ceramic wear is more of the abrasive type and is characterized by the generation of debris giving rise to propagation of cracks in type I and II mixed (opening and straight sliding).

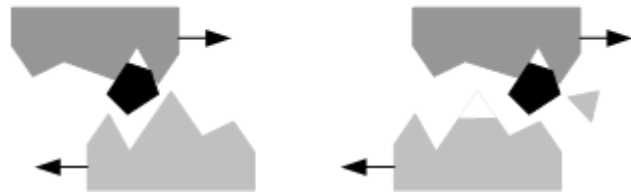
$$F = \mu N$$

Abrasive wear

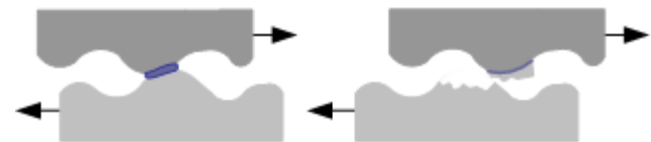
Two body abrasive wear



Three body abrasive wear



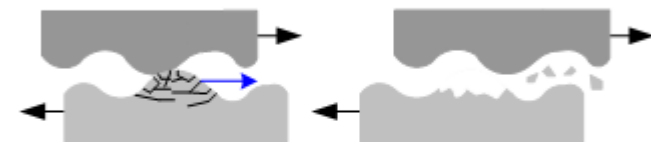
Adhesive wear



Fatigue wear

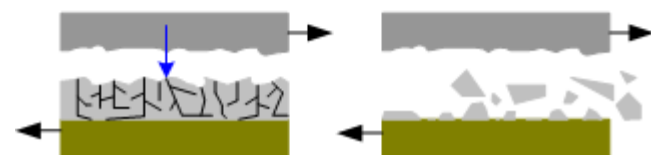
Fretting fatigue wear

Tangential cycling load



Fatigue wear of an overlay

Normal cycling load



Abrasion is the process of scuffing, scratching, wearing down, marring, or rubbing away.

Wear Resistance

Wear is the removal of the material from the surface of a solid body as a result of mechanical action of the counterbody.

Sliding wear is the wear mode when two ceramic surfaces slide over each other, and no abrasive particles are in the system.

Archard's Equation:

$$Q = \frac{KWL}{H}$$

where:

Q is the total volume of wear debris produced K is a dimensionless constant W is the total normal load L is the sliding distance H is the [hardness](#) of the softest contacting surfaces



The wear mechanisms and wear rates are not intrinsic material properties, but they depend on the material's triboenvironment.

Mechanical Properties of Ceramics_Summary

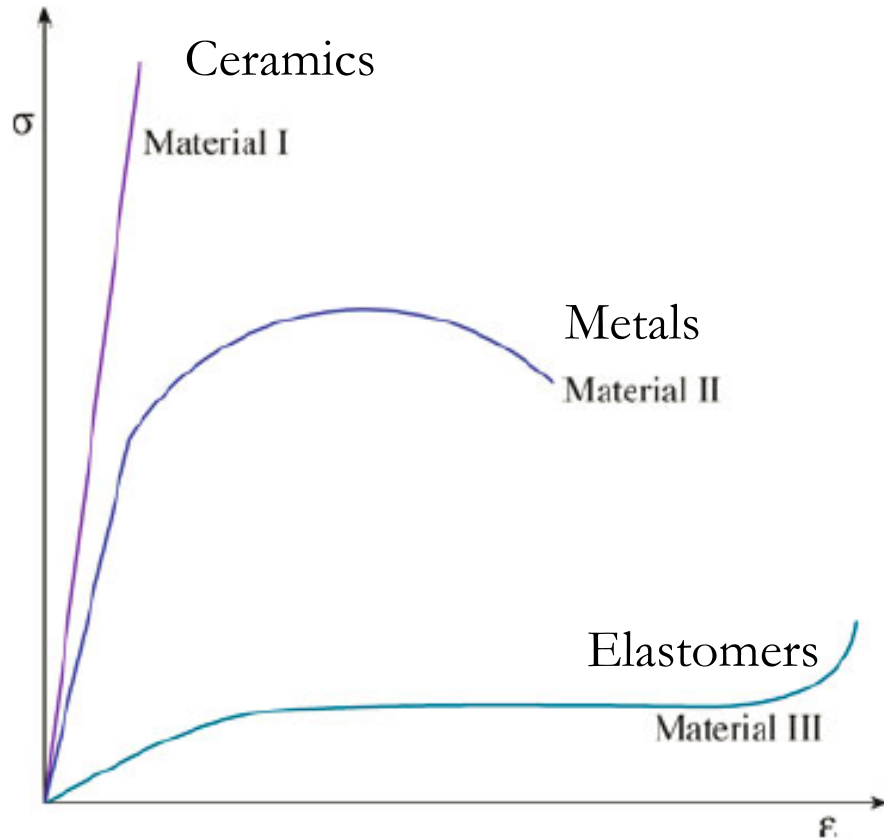
- ✦ Low ductility
 - Very brittle
 - High elastic modulus
 - Dimensional stability

- ✦ High strength at elevated temperatures

- ✦ Extreme hardness
 - High wear resistance
 - Extreme hardness can reduce wear caused by friction

- ✦ Low toughness
 - Low fracture toughness
 - Indicates the ability of a crack or flaw to produce a catastrophic failure

Stress-strain Curves for Different Materials



Material I: high Young's modulus, high Elastic Modulus and Hardness, low ductility, low toughness, and fractures without significant plastic deformation;

Material II: Moderate strength, moderate ductility, deforms plastically prior to failure, toughest;

Material III: Low Young's modulus, very ductile, low ultimate tensile strength, and limited toughness.